

# TL07xx Low-Noise, FET-Input Operational Amplifiers

## 1 Features

- High slew rate: 20V/μs (TL07xH, typ)
- Low offset voltage: 1mV (TL07xH, typ)
- Low offset voltage drift: 2μV/°C
- Low power consumption: 940μA/ch (TL07xH, typ)
- Wide common-mode and differential voltage ranges
  - Common-mode input voltage range includes V<sub>CC+</sub>
- Low input bias and offset currents
- Low noise:
  - V<sub>n</sub> = 37nV/√Hz (typ) at f = 1kHz
- Output short-circuit protection
- Low total harmonic distortion: 0.003% (typ)
- Wide supply voltage:
  - ±2.25V to ±20V, 4.5V to 40V

## 2 Applications

- [Solar energy: string and central inverter](#)
- [Motor drives: ac and servo drive control and power-stage modules](#)
- [Single-phase online UPS](#)
- [Three-phase UPS](#)
- [Pro audio mixers](#)
- [Battery test equipment](#)

## 3 Description

The TL071H, TL072H, and TL074H (TL07xH) family of devices are next-generation versions of the industry-standard TL071, TL072, and TL074 (TL07x) devices. These devices provide outstanding value for cost-sensitive applications, with features including low offset (1mV, typical), high slew rate (20V/μs), and common-mode input to the positive supply.

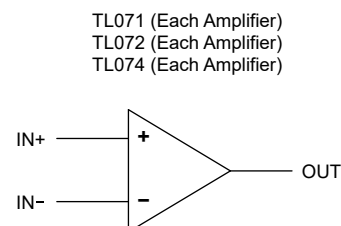
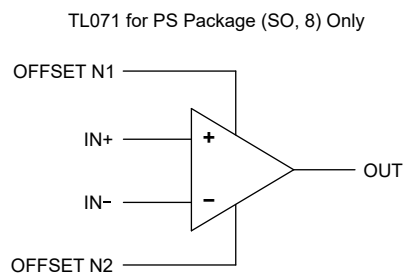
High ESD (1.5kV, HBM), integrated EMI and RF filters, and operation across the full –40°C to +125°C range enable the TL07xH devices for use in the most rugged and demanding applications.

### Device Information

| PART NUMBER <sup>(1)</sup> | CHANNEL COUNT | PACKAGE               |
|----------------------------|---------------|-----------------------|
| TL071x                     | Single        | D (SOIC, 8)           |
|                            |               | DBV (SOT-23, 5)       |
|                            |               | DCK (SC70, 5)         |
|                            |               | P (PDIP, 8)           |
|                            |               | PS (SO, 8)            |
| TL072x                     | Dual          | D (SOIC, 8)           |
|                            |               | DDF (SOT-23-THIN, 8)  |
|                            |               | P (PDIP, 8)           |
|                            |               | PS (SO, 8)            |
|                            |               | PW (TSSOP, 8)         |
| TL072M <sup>(2)</sup>      | Dual          | FK (LCCC, 20)         |
|                            |               | JG (CDIP, 8)          |
|                            |               | U (CFP, 10)           |
| TL074x                     | Quad          | D (SOIC, 14)          |
|                            |               | DB (SSOP, 14)         |
|                            |               | DYY (SOT-23-THIN, 14) |
|                            |               | N (PDIP, 14)          |
|                            |               | NS (SOP, 14)          |
|                            |               | PW (TSSOP, 14)        |
| TL074M <sup>(2)</sup>      | Quad          | FK (LCCC, 20)         |
|                            |               | J (CDIP, 14)          |
|                            |               | W (CFP, 14)           |

(1) For more information, see [Section 11](#).

(2) Devices with M suffix have an extended temperature range of –55°C to +125°C.



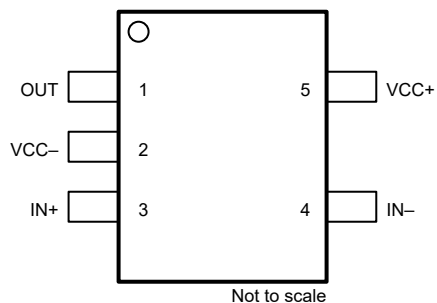
Logic Symbols



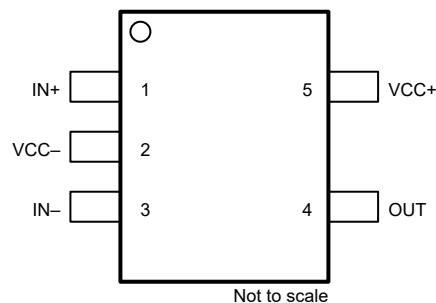
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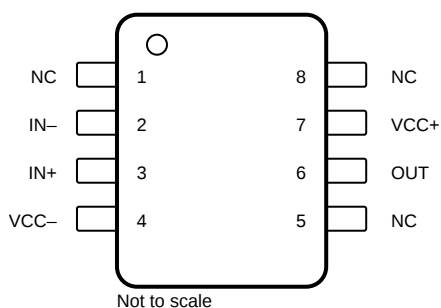
## 4 Pin Configuration and Functions



**Figure 4-1. TL071H DBV Package, 5-Pin SOT-23 (Top View)**



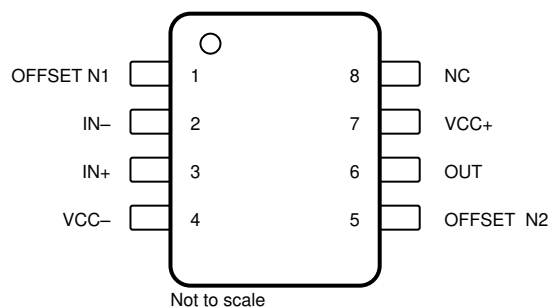
**Figure 4-2. TL071H DCK Package, 5-Pin SC70 (Top View)**



**Figure 4-3. TL071x D Package, 8-Pin SOIC and P Package, 8-pin PDIP (Top View)**

**Table 4-1. Pin Functions: TL071x**

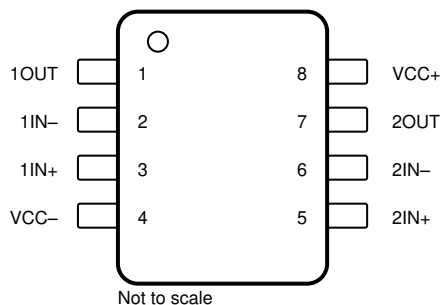
| NAME | PIN NO.      |            |          |          | TYPE   | DESCRIPTION        |
|------|--------------|------------|----------|----------|--------|--------------------|
|      | DBV (SOT-23) | DCK (SC70) | D (SOIC) | P (PDIP) |        |                    |
| IN–  | 4            | 3          | 2        | 2        | Input  | Inverting input    |
| IN+  | 3            | 1          | 3        | 3        | Input  | Noninverting input |
| NC   | —            | —          | 8        | 8        | —      | Do not connect     |
| NC   | —            | —          | 1        | 1        | —      | Do not connect     |
| NC   | —            | —          | 5        | 5        | —      | Do not connect     |
| OUT  | 1            | 4          | 6        | 6        | Output | Output             |
| VCC– | 2            | 2          | 4        | 4        | —      | Power supply       |
| VCC+ | 5            | 5          | 7        | 7        | —      | Power supply       |



**Figure 4-4. TL071C PS Package, 8-Pin SO  
(Top View)**

**Table 4-2. Pin Functions: TL071C**

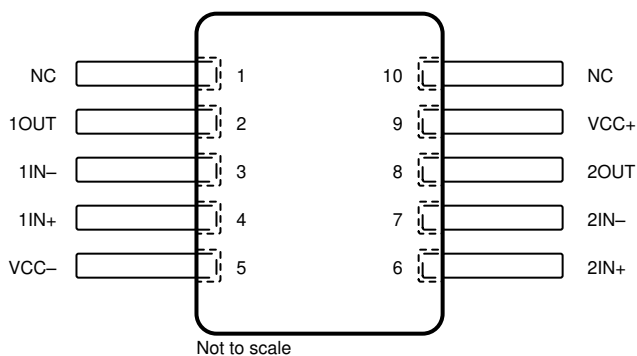
| PIN       |     | TYPE   | DESCRIPTION             |
|-----------|-----|--------|-------------------------|
| NAME      | NO. |        |                         |
| IN–       | 2   | Input  | Inverting input         |
| IN+       | 3   | Input  | Noninverting input      |
| NC        | 8   | —      | Do not connect          |
| OFFSET N1 | 1   | —      | Input offset adjustment |
| OFFSET N2 | 5   | —      | Input offset adjustment |
| OUT       | 6   | Output | Output                  |
| VCC–      | 4   | —      | Power supply            |
| VCC+      | 7   | —      | Power supply            |



**Figure 4-5. TL072x D, DDF, JG, P, PS, and PW Packages,  
8-Pin SOIC, SOT-23-THIN, CDIP, PDIP, SO, and TSSOP  
(Top View)**

**Table 4-3. Pin Functions: TL072x**

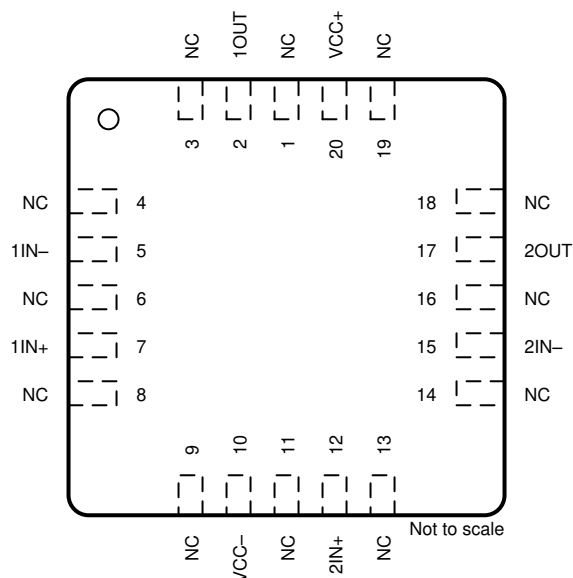
| PIN  |     | TYPE   | DESCRIPTION        |
|------|-----|--------|--------------------|
| NAME | NO. |        |                    |
| 1IN- | 2   | Input  | Inverting input    |
| 1IN+ | 3   | Input  | Noninverting input |
| 1OUT | 1   | Output | Output             |
| 2IN- | 6   | Input  | Inverting input    |
| 2IN+ | 5   | Input  | Noninverting input |
| 2OUT | 7   | Output | Output             |
| VCC- | 4   | —      | Power supply       |
| VCC+ | 8   | —      | Power supply       |



**Figure 4-6. TL072M U Package, 10-Pin CFP  
(Top View)**

**Table 4-4. Pin Functions: TL072M**

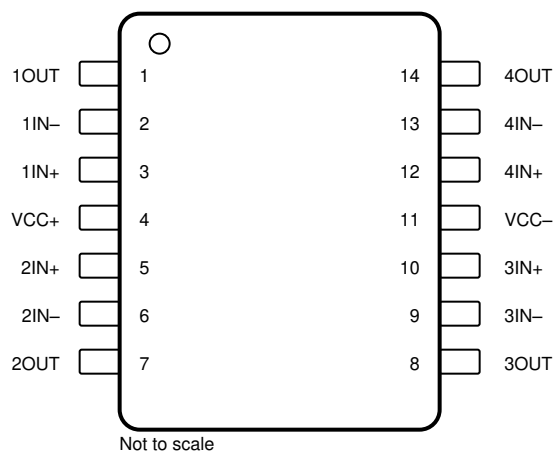
| PIN  |       | TYPE   | DESCRIPTION        |
|------|-------|--------|--------------------|
| NAME | NO.   |        |                    |
| 1IN- | 3     | Input  | Inverting input    |
| 1IN+ | 4     | Input  | Noninverting input |
| 1OUT | 2     | Output | Output             |
| 2IN- | 7     | Input  | Inverting input    |
| 2IN+ | 6     | Input  | Noninverting input |
| 2OUT | 8     | Output | Output             |
| NC   | 1, 10 | —      | Do not connect     |
| VCC- | 5     | —      | Power supply       |
| VCC+ | 9     | —      | Power supply       |



**Figure 4-7. TL072M FK Package, 20-Pin LCCC  
(Top View)**

**Table 4-5. Pin Functions: TL072M**

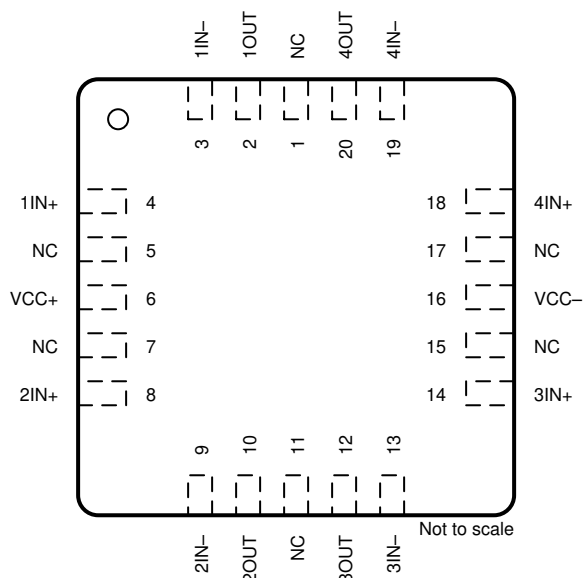
| PIN  |                                                | TYPE   | DESCRIPTION        |
|------|------------------------------------------------|--------|--------------------|
| NAME | NO.                                            |        |                    |
| 1IN– | 5                                              | Input  | Inverting input    |
| 1IN+ | 7                                              | Input  | Noninverting input |
| 1OUT | 2                                              | Output | Output             |
| 2IN– | 15                                             | Input  | Inverting input    |
| 2IN+ | 12                                             | Input  | Noninverting input |
| 2OUT | 17                                             | Output | Output             |
| NC   | 1, 3, 4, 6, 8,<br>9, 11, 13, 14,<br>16, 18, 19 | —      | Do not connect     |
| VCC– | 10                                             | —      | Power supply       |
| VCC+ | 20                                             | —      | Power supply       |



**Figure 4-8. TL074x D, DYY, J, N, NS, PW and W Packages,  
 14-Pin SOIC, SOT-23-THIN, CDIP, PDIP, SOP, TSSOP, and CFP  
 (Top View)**

**Table 4-6. Pin Functions: TL074x**

| PIN  |     | TYPE   | DESCRIPTION        |
|------|-----|--------|--------------------|
| NAME | NO. |        |                    |
| 1IN– | 2   | Input  | Inverting input    |
| 1IN+ | 3   | Input  | Noninverting input |
| 1OUT | 1   | Output | Output             |
| 2IN– | 6   | Input  | Inverting input    |
| 2IN+ | 5   | Input  | Noninverting input |
| 2OUT | 7   | Output | Output             |
| 3IN– | 9   | Input  | Inverting input    |
| 3IN+ | 10  | Input  | Noninverting input |
| 3OUT | 8   | Output | Output             |
| 4IN– | 13  | Input  | Inverting input    |
| 4IN+ | 12  | Input  | Noninverting input |
| 4OUT | 14  | Output | Output             |
| VCC– | 11  | —      | Power supply       |
| VCC+ | 4   | —      | Power supply       |



**Figure 4-9. TL074M FK Package, 20-Pin LCCC  
(Top View)**

**Table 4-7. Pin Functions: TL074M**

| PIN  |                     | TYPE   | DESCRIPTION        |
|------|---------------------|--------|--------------------|
| NAME | NO.                 |        |                    |
| 1IN- | 3                   | Input  | Inverting input    |
| 1IN+ | 4                   | Input  | Noninverting input |
| 1OUT | 2                   | Output | Output             |
| 2IN- | 9                   | Input  | Inverting input    |
| 2IN+ | 8                   | Input  | Noninverting input |
| 2OUT | 10                  | Output | Output             |
| 3IN- | 13                  | Input  | Inverting input    |
| 3IN+ | 14                  | Input  | Noninverting input |
| 3OUT | 12                  | Output | Output             |
| 4IN- | 19                  | Input  | Inverting input    |
| 4IN+ | 18                  | Input  | Noninverting input |
| 4OUT | 20                  | Output | Output             |
| NC   | 1, 5, 7, 11, 15, 17 | —      | Do not connect     |
| VCC- | 16                  | —      | Power supply       |
| VCC+ | 6                   | —      | Power supply       |

## 5 Specifications

### Note

The TLV07xx series has transitioned new die fabrication into a modern process.

This new die is available with an H suffix.

A die with a different suffix is either older or newer; see also [Section 9.1.1](#).

[Section 5.7](#) and [Section 5.10](#) describe the performance of the new die.

[Section 5.8](#), [Section 5.9](#), and [Section 5.11](#) describe the performance of the old die.

### 5.1 Absolute Maximum Ratings

over operating ambient temperature range (unless otherwise noted)<sup>(1)</sup>

|                                                              |                                     |                                                           | MIN               | MAX               | UNIT |
|--------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------|-------------------|-------------------|------|
| Supply voltage, $V_S = (V+) - (V-)$                          |                                     | All NS and PS packages; All TL07xM devices                | −0.3              | 36                | V    |
|                                                              |                                     | All other devices                                         | 0                 | 42                |      |
| Signal input pins                                            | Common-mode voltage <sup>(2)</sup>  | All NS and PS packages; All TL07xM devices                | $(V_{CC-}) - 0.3$ | $(V_{CC-}) + 36$  | V    |
|                                                              |                                     | All other devices                                         | $(V_{CC-}) - 0.5$ | $(V_{CC+}) + 0.5$ |      |
|                                                              | Differential voltage <sup>(2)</sup> | All NS and PS packages; All TL07xM devices <sup>(3)</sup> | $(V_{CC-}) - 0.3$ | $(V_{CC-}) + 36$  | V    |
|                                                              |                                     | All other devices                                         | $V_S + 0.2$       |                   |      |
|                                                              | Current <sup>(2)</sup>              | All NS and PS packages; All TL07xM devices                | 50                |                   | mA   |
|                                                              |                                     | All other devices                                         | −10 10            |                   |      |
| Output short-circuit <sup>(4)</sup>                          |                                     |                                                           | Continuous        |                   |      |
| Operating ambient temperature, $T_A$                         |                                     |                                                           | −55               | 150               | °C   |
| Junction temperature, $T_J$                                  |                                     |                                                           |                   | 150               | °C   |
| Case temperature for 60 seconds - FK package                 |                                     |                                                           |                   | 260               | °C   |
| Lead temperature 1.8 mm (1/16 inch) from case for 10 seconds |                                     |                                                           |                   | 300               | °C   |
| Storage temperature, $T_{stg}$                               |                                     |                                                           | −65               | 150               | °C   |

- (1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) Input pins are diode-clamped to both power-supply rails on all new die. Current limit input signals that swing more than 0.5 V beyond the supply rails to 10 mA or less.
- (3) Differential voltage only limited by input voltage.
- (4) Short-circuit to ground, one amplifier per package.

### 5.2 ESD Ratings

|             |                         |                                                                                | VALUE | UNIT |
|-------------|-------------------------|--------------------------------------------------------------------------------|-------|------|
| $V_{(ESD)}$ | Electrostatic discharge | Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001 <sup>(1)</sup>              | ±2000 | V    |
|             |                         | Charged-device model (CDM), per JEDEC specification JESD22-C101 <sup>(2)</sup> | ±1000 |      |

- (1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process.
- (2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process.

## 5.3 Recommended Operating Conditions

over operating ambient temperature range (unless otherwise noted)

|                |                                                           |                                                           | MIN                     | MAX                       | UNIT |
|----------------|-----------------------------------------------------------|-----------------------------------------------------------|-------------------------|---------------------------|------|
| V <sub>S</sub> | Supply voltage, (V <sub>CC+</sub> ) – (V <sub>CC–</sub> ) | All NS and PS packages; All TL07xM devices <sup>(1)</sup> | 10                      | 30                        | V    |
|                |                                                           | All other devices                                         | 4.5                     | 40                        |      |
| V <sub>I</sub> | Input voltage                                             | All NS and PS packages; All TL07xM devices                | (V <sub>CC–</sub> ) + 2 | (V <sub>CC+</sub> ) + 0.1 | V    |
|                |                                                           | All other devices                                         | (V <sub>CC–</sub> ) + 4 | (V <sub>CC+</sub> ) + 0.1 |      |
| T <sub>A</sub> | Specified temperature <sup>(2)</sup>                      | TL07xM                                                    | –55                     | 125                       | °C   |
|                |                                                           | TL07xH                                                    | –40                     | 125                       |      |
|                |                                                           | TL07xI                                                    | –40                     | 85                        |      |
|                |                                                           | TL07xC                                                    | 0                       | 70                        |      |

(1) V<sub>CC+</sub> and V<sub>CC–</sub> are not required to be of equal magnitude, provided that the total V<sub>S</sub> ((V<sub>CC+</sub>) – (V<sub>CC–</sub>)) is between 10 V and 30 V.

(2) See also [Section 9.1.1](#).

## 5.4 Thermal Information for Single Channel

| THERMAL METRIC <sup>(1)</sup> |                                              | TL071xx     |               |                 |             |            | UNIT |
|-------------------------------|----------------------------------------------|-------------|---------------|-----------------|-------------|------------|------|
|                               |                                              | D<br>(SOIC) | DCK<br>(SC70) | DBV<br>(SOT-23) | P<br>(PDIP) | PS<br>(SO) |      |
|                               |                                              | 8 PINS      | 5 PINS        | 5 PINS          | 8 PINS      | 8 PINS     |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 158.8       | 217.5         | 212.2           | 85          | 95         | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 98.6        | 113.1         | 111.1           | –           | –          | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 102.3       | 63.8          | 79.4            | –           | –          | °C/W |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 45.8        | 34.8          | 51.8            | –           | –          | °C/W |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 101.5       | 63.5          | 79.0            | –           | –          | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A         | N/A           | N/A             | N/A         | N/A        | °C/W |

(1) For information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

## 5.5 Thermal Information for Dual Channel

| THERMAL METRIC <sup>(1)</sup> |                                              | TL072xx     |                 |              |              |             |            |               |            | UNIT |
|-------------------------------|----------------------------------------------|-------------|-----------------|--------------|--------------|-------------|------------|---------------|------------|------|
|                               |                                              | D<br>(SOIC) | DDF<br>(SOT-23) | FK<br>(LCCC) | JG<br>(CDIP) | P<br>(PDIP) | PS<br>(SO) | PW<br>(TSSOP) | U<br>(CFP) |      |
|                               |                                              | 8 PINS      | 8 PINS          | 20 PINS      | 8 PINS       | 8 PINS      | 8 PINS     | 8 PINS        | 10 PINS    |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 147.8       | 181.5           | –            | –            | 85          | 95         | 200.3         | 169.8      | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 88.2        | 112.5           | 5.61         | 15.05        | –           | –          | 89.4          | 62.1       | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 91.4        | 98.2            | –            | –            | –           | –          | 131.0         | 176.2      | °C/W |
| ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 36.8        | 17.2            | –            | –            | –           | –          | 22.2          | 48.4       | °C/W |
| ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 90.6        | 97.6            | –            | –            | –           | –          | 129.3         | 144.1      | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A         | N/A             | –            | –            | –           | –          | N/A           | 5.4        | °C/W |

(1) For information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

## 5.6 Thermal Information for Quad Channel

| THERMAL METRIC <sup>(1)</sup> |                                              | TL074xx     |                 |               |              |              |               |               |              | UNIT |
|-------------------------------|----------------------------------------------|-------------|-----------------|---------------|--------------|--------------|---------------|---------------|--------------|------|
|                               |                                              | D<br>(SOIC) | DYY<br>(SOT-23) | FK<br>(TSSOP) | J<br>(TSSOP) | N<br>(TSSOP) | NS<br>(TSSOP) | PW<br>(TSSOP) | W<br>(TSSOP) |      |
|                               |                                              | 14 PINS     | 14 PINS         | 20 PINS       | 14 PINS      | 14 PINS      | 14 PINS       | 14 PINS       | 14 PINS      |      |
| R <sub>θJA</sub>              | Junction-to-ambient thermal resistance       | 114.2       | 153.2           | –             | –            | 80           | 76            | –             | 128.8        | °C/W |
| R <sub>θJC(top)</sub>         | Junction-to-case (top) thermal resistance    | 70.3        | 88.7            | 5.61          | 14.5         | –            | –             | 14.5          | 56.1         | °C/W |
| R <sub>θJB</sub>              | Junction-to-board thermal resistance         | 70.2        | 65.4            | –             | –            | –            | –             | –             | 127.6        | °C/W |
| Ψ <sub>JT</sub>               | Junction-to-top characterization parameter   | 28.8        | 9.5             | –             | –            | –            | –             | –             | 29           | °C/W |
| Ψ <sub>JB</sub>               | Junction-to-board characterization parameter | 69.8        | 65.0            | –             | –            | –            | –             | –             | 106.1        | °C/W |
| R <sub>θJC(bot)</sub>         | Junction-to-case (bottom) thermal resistance | N/A         | N/A             | –             | –            | –            | –             | –             | 0.5          | °C/W |

(1) For information about traditional and new thermal metrics, see the [Semiconductor and IC Package Thermal Metrics](#) application report.

## 5.7 Electrical Characteristics for TL07xH

at  $V_S = (V_{CC+}) - (V_{CC-}) = 4.5\text{ V to }40\text{ V}$  ( $\pm 2.25\text{ V to } \pm 20\text{ V}$ ),  $T_A = 25^\circ\text{C}$ ,  $R_L = 10\text{ k}\Omega$  connected to  $V_S / 2$ ,  $V_{CM} = V_S / 2$ , and  $V_{OUT} = V_S / 2$  (unless otherwise noted)

| PARAMETER            |                                          | TEST CONDITIONS                                                                                                                                                |                                                 | MIN                       | TYP                 | MAX | UNIT              |
|----------------------|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|---------------------------|---------------------|-----|-------------------|
| OFFSET VOLTAGE       |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| V <sub>OS</sub>      | Input offset voltage                     |                                                                                                                                                                |                                                 | ±1                        | ±4                  |     | mV                |
|                      |                                          |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C                |                           | ±5                  |     |                   |
| dV <sub>OS</sub> /dT | Input offset voltage drift               |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C                | ±2                        |                     |     | μV/°C             |
| PSRR                 | Input offset voltage versus power supply | V <sub>S</sub> = 5 V to 40 V, V <sub>CM</sub> = V <sub>S</sub> / 2                                                                                             | T <sub>A</sub> = −40°C to +125°C                | ±1                        | ±10                 |     | μV/V              |
|                      | Channel separation                       | f = 0 Hz                                                                                                                                                       |                                                 | 10                        |                     |     | μV/V              |
| INPUT BIAS CURRENT   |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| I <sub>B</sub>       | Input bias current                       |                                                                                                                                                                |                                                 | ±1                        | ±120                |     | pA                |
|                      |                                          |                                                                                                                                                                | DCK and DBV packages                            | ±1                        | ±300                |     | pA                |
|                      |                                          |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C <sup>(1)</sup> |                           | ±5                  |     | nA                |
| I <sub>OS</sub>      | Input offset current                     |                                                                                                                                                                |                                                 | ±0.5                      | ±120                |     | pA                |
|                      |                                          |                                                                                                                                                                | DCK and DBV packages                            | ±0.5                      | ±250                |     | pA                |
|                      |                                          |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C <sup>(1)</sup> |                           | ±5                  |     | nA                |
| NOISE                |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| E <sub>N</sub>       | Input voltage noise                      | f = 0.1 Hz to 10 Hz                                                                                                                                            |                                                 | 9.2                       |                     |     | μV <sub>PP</sub>  |
|                      |                                          |                                                                                                                                                                |                                                 | 1.4                       |                     |     | μV <sub>RMS</sub> |
| e <sub>N</sub>       | Input voltage noise density              | f = 1 kHz                                                                                                                                                      |                                                 | 37                        |                     |     | nV/√Hz            |
|                      |                                          | f = 10 kHz                                                                                                                                                     |                                                 | 21                        |                     |     |                   |
| i <sub>N</sub>       | Input current noise                      | f = 1 kHz                                                                                                                                                      |                                                 | 80                        |                     |     | fA/√Hz            |
| INPUT VOLTAGE RANGE  |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| V <sub>CM</sub>      | Common-mode voltage                      |                                                                                                                                                                |                                                 | (V <sub>CC−</sub> ) + 1.5 | (V <sub>CC+</sub> ) |     | V                 |
| CMRR                 | Common-mode rejection ratio              | V <sub>S</sub> = 40 V, (V <sub>CC−</sub> ) + 2.5 V < V <sub>CM</sub> < (V <sub>CC+</sub> ) − 1.5 V                                                             |                                                 | 100                       | 105                 |     | dB                |
|                      |                                          |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C                | 95                        |                     |     | dB                |
|                      |                                          | V <sub>S</sub> = 40 V, (V <sub>CC−</sub> ) + 2.5 V < V <sub>CM</sub> < (V <sub>CC+</sub> )                                                                     |                                                 | 90                        | 105                 |     | dB                |
|                      |                                          |                                                                                                                                                                | T <sub>A</sub> = −40°C to +125°C                | 80                        |                     |     | dB                |
| INPUT CAPACITANCE    |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| Z <sub>ID</sub>      | Differential                             |                                                                                                                                                                |                                                 | 100    2                  |                     |     | MΩ    pF          |
| Z <sub>ICM</sub>     | Common-mode                              |                                                                                                                                                                |                                                 | 6    1                    |                     |     | TΩ    pF          |
| OPEN-LOOP GAIN       |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| A <sub>OL</sub>      | Open-loop voltage gain                   | V <sub>S</sub> = 40 V, V <sub>CM</sub> = V <sub>S</sub> / 2, (V <sub>CC−</sub> ) + 0.3 V < V <sub>O</sub> < (V <sub>CC+</sub> ) − 0.3 V                        | T <sub>A</sub> = −40°C to +125°C                | 118                       | 125                 |     | dB                |
| A <sub>OL</sub>      | Open-loop voltage gain                   | V <sub>S</sub> = 40 V, V <sub>CM</sub> = V <sub>S</sub> / 2, R <sub>L</sub> = 2 kΩ, (V <sub>CC−</sub> ) + 1.2 V < V <sub>O</sub> < (V <sub>CC+</sub> ) − 1.2 V | T <sub>A</sub> = −40°C to +125°C                | 115                       | 120                 |     | dB                |
| FREQUENCY RESPONSE   |                                          |                                                                                                                                                                |                                                 |                           |                     |     |                   |
| GBW                  | Gain-bandwidth product                   |                                                                                                                                                                |                                                 | 5.25                      |                     |     | MHz               |
| SR                   | Slew rate                                | V <sub>S</sub> = 40 V, G = +1, C <sub>L</sub> = 20 pF                                                                                                          |                                                 | 20                        |                     |     | V/μs              |
| t <sub>S</sub>       | Settling time                            | To 0.1%, V <sub>S</sub> = 40 V, V <sub>STEP</sub> = 10 V , G = +1, C <sub>L</sub> = 20 pF                                                                      |                                                 | 0.63                      |                     |     | μs                |
|                      |                                          | To 0.1%, V <sub>S</sub> = 40 V, V <sub>STEP</sub> = 2 V , G = +1, C <sub>L</sub> = 20 pF                                                                       |                                                 | 0.56                      |                     |     |                   |
|                      |                                          | To 0.01%, V <sub>S</sub> = 40 V, V <sub>STEP</sub> = 10 V , G = +1, C <sub>L</sub> = 20 pF                                                                     |                                                 | 0.91                      |                     |     |                   |
|                      |                                          | To 0.01%, V <sub>S</sub> = 40 V, V <sub>STEP</sub> = 2 V , G = +1, C <sub>L</sub> = 20 pF                                                                      |                                                 | 0.48                      |                     |     |                   |
|                      | Phase margin                             | G = +1, R <sub>L</sub> = 10kΩ, C <sub>L</sub> = 20 pF                                                                                                          |                                                 | 56                        |                     |     | °                 |
|                      | Overload recovery time                   | V <sub>IN</sub> × gain > V <sub>S</sub>                                                                                                                        |                                                 | 300                       |                     |     | ns                |
| THD+N                | Total harmonic distortion + noise        | V <sub>S</sub> = 40 V, V <sub>O</sub> = 6 V <sub>RMS</sub> , G = +1, f = 1 kHz                                                                                 |                                                 | 0.00012                   |                     |     | %                 |
| EMIRR                | EMI rejection ratio                      | f = 1 GHz                                                                                                                                                      |                                                 | 53                        |                     |     | dB                |

## 5.7 Electrical Characteristics for TL07xH (continued)

at  $V_S = (V_{CC+}) - (V_{CC-}) = 4.5 \text{ V to } 40 \text{ V}$  ( $\pm 2.25 \text{ V to } \pm 20 \text{ V}$ ),  $T_A = 25^\circ\text{C}$ ,  $R_L = 10 \text{ k}\Omega$  connected to  $V_S / 2$ ,  $V_{CM} = V_S / 2$ , and  $V_{OUT} = V_S / 2$  (unless otherwise noted)

| PARAMETER         |                                 | TEST CONDITIONS                                                                      |                                               | MIN | TYP   | MAX  | UNIT |
|-------------------|---------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------|-----|-------|------|------|
| OUTPUT            |                                 |                                                                                      |                                               |     |       |      |      |
|                   | Voltage output swing from rail  | Positive rail headroom                                                               | V <sub>S</sub> = 40 V, R <sub>L</sub> = 10 kΩ |     | 115   | 210  | mV   |
|                   |                                 |                                                                                      | V <sub>S</sub> = 40 V, R <sub>L</sub> = 2 kΩ  |     | 520   | 965  |      |
|                   |                                 | Negative rail headroom                                                               | V <sub>S</sub> = 40 V, R <sub>L</sub> = 10 kΩ |     | 105   | 215  |      |
|                   |                                 |                                                                                      | V <sub>S</sub> = 40 V, R <sub>L</sub> = 2 kΩ  |     | 500   | 1030 |      |
| I <sub>SC</sub>   | Short-circuit current           |                                                                                      |                                               |     | ±26   |      | mA   |
| C <sub>LOAD</sub> | Capacitive load drive           |                                                                                      |                                               |     | 300   |      | pF   |
| Z <sub>O</sub>    | Open-loop output impedance      | f = 1 MHz, I <sub>O</sub> = 0 A                                                      |                                               |     | 125   |      | Ω    |
| POWER SUPPLY      |                                 |                                                                                      |                                               |     |       |      |      |
| I <sub>Q</sub>    | Quiescent current per amplifier | I <sub>O</sub> = 0 A                                                                 | T <sub>A</sub> = −40°C to +125°C              |     | 937.5 | 1125 | μA   |
|                   |                                 | I <sub>O</sub> = 0 A, (TL071H)                                                       |                                               |     | 960   | 1156 |      |
|                   |                                 | I <sub>O</sub> = 0 A                                                                 |                                               |     |       | 1130 |      |
|                   |                                 | I <sub>O</sub> = 0 A, (TL072H)                                                       |                                               |     |       | 1143 |      |
|                   |                                 | I <sub>O</sub> = 0 A, (TL071H)                                                       |                                               |     |       | 1160 |      |
|                   | Turn-on time                    | At T <sub>A</sub> = 25°C, V <sub>S</sub> = 40 V, V <sub>S</sub> ramp rate > 0.3 V/μs |                                               |     | 60    |      | μs   |

(1) Maximum  $I_B$  and  $I_{OS}$  data are specified based on characterization results.

## 5.8 Electrical Characteristics (DC) for TL07xC, TL07xAC, TL07xBC, TL07xI, TL07xM

at  $V_S = (V_{CC+}) - (V_{CC-}) = \pm 15\text{ V}$  and  $T_A = 25^\circ\text{C}$  (unless otherwise noted)

| PARAMETER            |                                          | TEST CONDITIONS <sup>(1) (2)</sup>                                                       |                                  |                             | MIN   | TYP       | MAX   | UNIT |
|----------------------|------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------|-----------------------------|-------|-----------|-------|------|
| DC                   |                                          |                                                                                          |                                  |                             |       |           |       |      |
| V <sub>OS</sub>      | Input offset voltage                     | V <sub>O</sub> = 0 V<br>R <sub>S</sub> = 50 Ω                                            | TL07xC                           |                             | 3     | 10        | mV    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 13    |           |       |      |
|                      |                                          |                                                                                          | TL07xAC                          |                             | 3     | 6         |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 7.5   |           |       |      |
|                      |                                          |                                                                                          | TL07xBC                          |                             | 2     | 3         |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 5     |           |       |      |
|                      |                                          |                                                                                          | TL07xI                           |                             | 3     | 6         |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 8     |           |       |      |
|                      |                                          |                                                                                          | TL071M, TL072M                   |                             | 3     | 6         |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 9     |           |       |      |
| TL074M               |                                          | 3                                                                                        | 9                                |                             |       |           |       |      |
|                      | T <sub>A</sub> = Full range              | 15                                                                                       |                                  |                             |       |           |       |      |
| dV <sub>OS</sub> /dT | Input offset voltage drift               | V <sub>O</sub> = 0 V, R <sub>S</sub> = 50 Ω                                              | T <sub>A</sub> = Full range      |                             | ±18   |           | μV/°C |      |
| I <sub>OS</sub>      | Input offset current                     | V <sub>O</sub> = 0 V                                                                     | TL07xC                           |                             | 5     | 100       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 10    | nA        |       |      |
|                      |                                          |                                                                                          | TL07xAC, TL07xBC, TL07xI         |                             | 5     | 100       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 2     | nA        |       |      |
|                      |                                          |                                                                                          | TL07xM                           |                             | 5     | 100       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 20    | nA        |       |      |
| I <sub>B</sub>       | Input bias current                       | V <sub>O</sub> = 0 V                                                                     | TL07xC, TL07xAC, TL07xBC, TL07xI |                             | 65    | 200       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 7     | nA        |       |      |
|                      |                                          |                                                                                          | TL071M, TL072M                   |                             | 65    | 200       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 50    | nA        |       |      |
|                      |                                          |                                                                                          | TL074M                           |                             | 65    | 200       | pA    |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 20    | nA        |       |      |
| V <sub>CM</sub>      | Common-mode voltage                      |                                                                                          |                                  |                             | ±11   | −12 to 15 | V     |      |
| V <sub>OM</sub>      | Maximum peak output voltage swing        | R <sub>L</sub> = 10 kΩ                                                                   | T <sub>A</sub> = Full range      | ±12                         | ±13.5 | V         |       |      |
|                      |                                          | R <sub>L</sub> ≥ 10 kΩ                                                                   |                                  | ±12                         |       |           |       |      |
|                      |                                          | R <sub>L</sub> ≥ 2 kΩ                                                                    |                                  | ±10                         |       |           |       |      |
| A <sub>OL</sub>      | Open-loop voltage gain                   | V <sub>O</sub> = 0 V                                                                     | TL07xC                           |                             | 25    | 200       | V/mV  |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 15    |           |       |      |
|                      |                                          |                                                                                          | TL07xAC, TL07xBC, TL07xI         |                             | 50    | 200       |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 25    |           |       |      |
|                      |                                          |                                                                                          | TL07xM                           |                             | 35    | 200       |       |      |
|                      |                                          |                                                                                          |                                  | T <sub>A</sub> = Full range | 15    |           |       |      |
| GBW                  | Gain-bandwidth product                   | All NS and PS packages; All TL07xM devices                                               |                                  |                             | 3     | MHz       |       |      |
|                      |                                          | All other devices                                                                        |                                  |                             | 5.25  |           |       |      |
| R <sub>ID</sub>      | Common-mode input resistance             |                                                                                          |                                  |                             | 1     | TΩ        |       |      |
| CMRR                 | Common-mode rejection ratio              | V <sub>IC</sub> = V <sub>ICR(min)</sub><br>V <sub>O</sub> = 0 V<br>R <sub>S</sub> = 50 Ω | TL07xC                           | 70                          | 100   | dB        |       |      |
|                      |                                          |                                                                                          | TL07xAC, TL07xBC, TL07xI         | 75                          | 100   |           |       |      |
|                      |                                          |                                                                                          | TL07xM                           | 80                          | 86    |           |       |      |
| PSRR                 | Input offset voltage versus power supply | V <sub>S</sub> = ±9 V to ± 18 V<br>V <sub>O</sub> = 0 V<br>R <sub>S</sub> = 50 Ω         | TL07xC                           | 70                          | 100   | dB        |       |      |
|                      |                                          |                                                                                          | TL07xAC, TL07xBC, TL07xI         | 80                          | 100   |           |       |      |
|                      |                                          |                                                                                          | TL07xM                           | 80                          | 86    |           |       |      |

## 5.8 Electrical Characteristics (DC) for TL07xC, TL07xAC, TL07xBC, TL07xI, TL07xM (continued)

at  $V_S = (V_{CC+}) - (V_{CC-}) = \pm 15\text{ V}$  and  $T_A = 25^\circ\text{C}$  (unless otherwise noted)

| PARAMETER |                                 | TEST CONDITIONS <sup>(1) (2)</sup> | MIN | TYP | MAX | UNIT            |
|-----------|---------------------------------|------------------------------------|-----|-----|-----|-----------------|
| $I_Q$     | Quiescent current per amplifier | $V_O = 0\text{ V}$ , no load       |     | 1.4 | 2.5 | mA              |
|           | Channel separation              | $f = 0\text{ Hz}$                  |     | 1   |     | $\mu\text{V/V}$ |

(1) All characteristics are measured under open-loop conditions with zero common-mode voltage, unless otherwise specified.

(2) Full range is  $T_A = 0^\circ\text{C}$  to  $70^\circ\text{C}$  for the TL07xC, TL07xAC, and TL07xBC;  $T_A = -40^\circ\text{C}$  to  $+85^\circ\text{C}$  for the TL07xI; and  $T_A = -55^\circ\text{C}$  to  $+125^\circ\text{C}$  for the TL07xM.

## 5.9 Electrical Characteristics (AC) for TL07xC, TL07xAC, TL07xBC, TL07xI, TL07xM

at  $V_S = (V_{CC+}) - (V_{CC-}) = \pm 15\text{ V}$  and  $T_A = 25^\circ\text{C}$  (unless otherwise noted)

| PARAMETER      |                                   | TEST CONDITIONS                                                          |                                                                                                          | MIN     | TYP | MAX    | UNIT              |
|----------------|-----------------------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------|-----|--------|-------------------|
| AC             |                                   |                                                                          |                                                                                                          |         |     |        |                   |
| SR             | Slew rate                         | V <sub>I</sub> = 10 V, C <sub>L</sub> = 100 pF,<br>R <sub>L</sub> = 2 kΩ | TL07xM                                                                                                   | 5       | 20  |        | V/μs              |
|                |                                   |                                                                          | TL07xC, TL07xAC,<br>TL07xBC, TL07xI                                                                      | 8       | 20  |        | V/μs              |
| t <sub>S</sub> | Settling time                     | V <sub>I</sub> = 20 V, C <sub>L</sub> = 100 pF, R <sub>L</sub> = 2 kΩ    |                                                                                                          | 0.1     |     |        | μs                |
|                |                                   |                                                                          |                                                                                                          | 20%     |     |        |                   |
| e <sub>N</sub> | Input voltage noise density       | All PS and NS packages,<br>all TL07xM devices                            | R <sub>S</sub> = 20 Ω, f = 1 kHz                                                                         | 18      |     |        | nV/√Hz            |
|                |                                   |                                                                          | f = 1 kHz                                                                                                | 37      |     | nV/√Hz |                   |
|                |                                   | All other devices                                                        | f = 10 kHz                                                                                               | 21      |     |        |                   |
| E <sub>N</sub> | Input voltage noise               | All PS and NS packages,<br>all TL07xM devices                            | R <sub>S</sub> = 20 Ω,<br>f = 10 Hz to 10 kHz                                                            | 4       |     |        | μV <sub>RMS</sub> |
|                |                                   | All other devices                                                        | f = 0.1 Hz to 10 Hz                                                                                      | 1.4     |     |        | μV <sub>RMS</sub> |
| i <sub>N</sub> | Input current noise               | R <sub>S</sub> = 20 Ω, f = 1 kHz                                         |                                                                                                          | 10      |     |        | fA/√Hz            |
|                | Phase margin                      | TL07xC, TL07xAC,<br>TL07xBC, TL07xI                                      | G = +1, R <sub>L</sub> = 10kΩ,<br>C <sub>L</sub> = 20 pF                                                 | 56      |     |        | °                 |
|                | Overload recovery time            | V <sub>IN</sub> × gain > V <sub>S</sub>                                  |                                                                                                          | 300     |     |        | ns                |
| THD+N          | Total harmonic distortion + noise | All PS and NS packages,<br>all TL07xM devices                            | V <sub>O</sub> = 6 V <sub>RMS</sub> , R <sub>L</sub> ≥ 2 kΩ,<br>f = 1 kHz, G = +1, R <sub>S</sub> ≤ 1 kΩ | 0.003   |     |        | %                 |
|                |                                   | All other devices                                                        | V <sub>S</sub> = 40 V, V <sub>O</sub> = 6 V <sub>RMS</sub> ,<br>G = +1, f = 1 kHz                        | 0.00012 |     |        | %                 |
| EMIRR          | EMI rejection ratio               | TL07xC, TL07xAC,<br>TL07xBC, TL07xI                                      | f = 1 GHz                                                                                                | 53      |     |        | dB                |
| Z <sub>O</sub> | Open-loop output impedance        | TL07xC, TL07xAC,<br>TL07xBC, TL07xI                                      | f = 1 MHz, I <sub>O</sub> = 0 A                                                                          | 125     |     |        | Ω                 |

## 5.10 Typical Characteristics: TL07xH

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

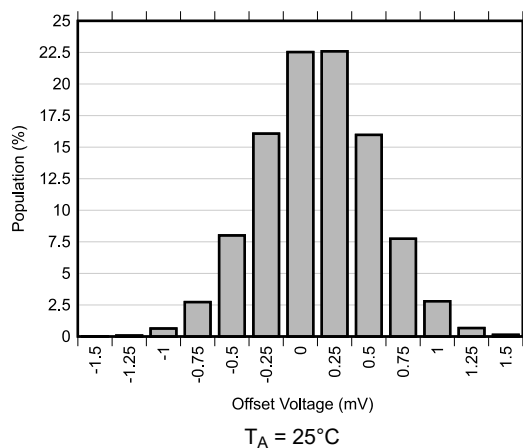


Figure 5-1. Offset Voltage Production Distribution

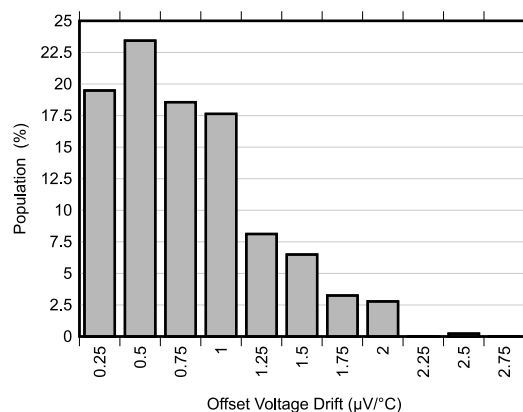


Figure 5-2. Offset Voltage Drift Distribution

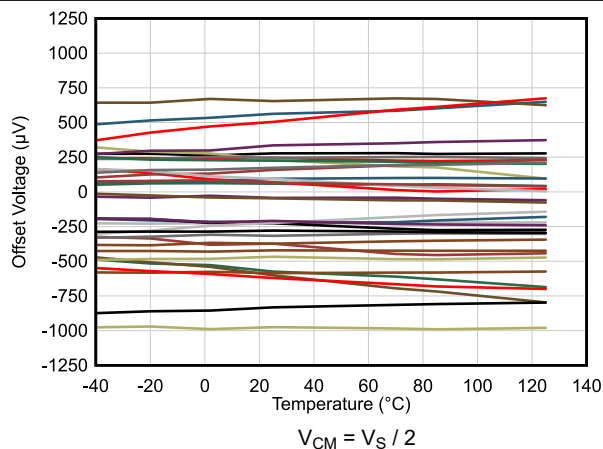


Figure 5-3. Offset Voltage vs Temperature

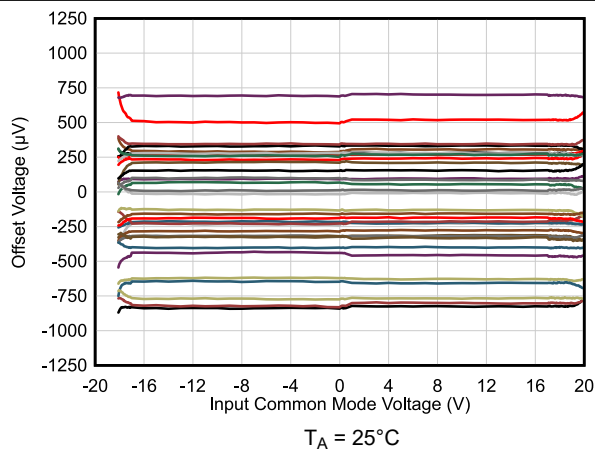


Figure 5-4. Offset Voltage vs Common-Mode Voltage

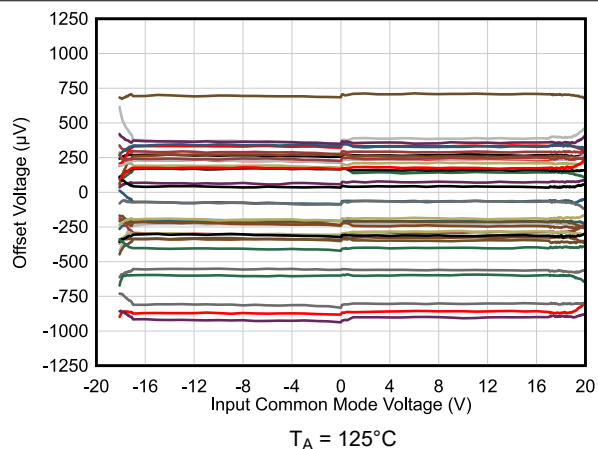


Figure 5-5. Offset Voltage vs Common-Mode Voltage

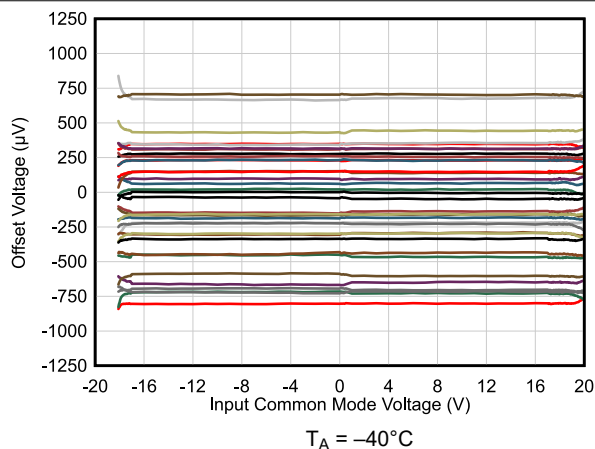


Figure 5-6. Offset Voltage vs Common-Mode Voltage

## 5.10 Typical Characteristics: TL07xH (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

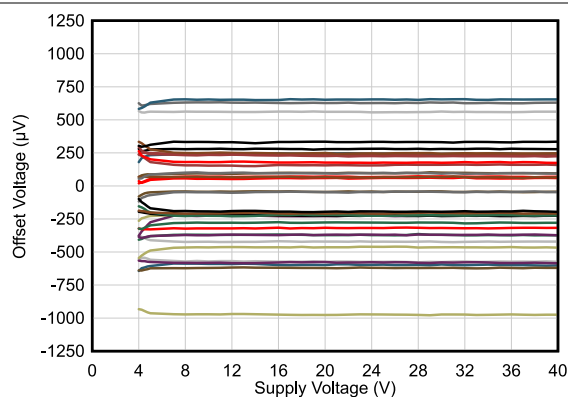


Figure 5-7. Offset Voltage vs Power Supply

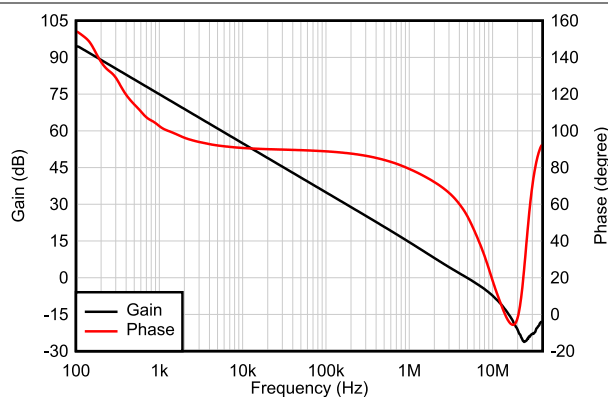


Figure 5-8. Open-Loop Gain and Phase vs Frequency

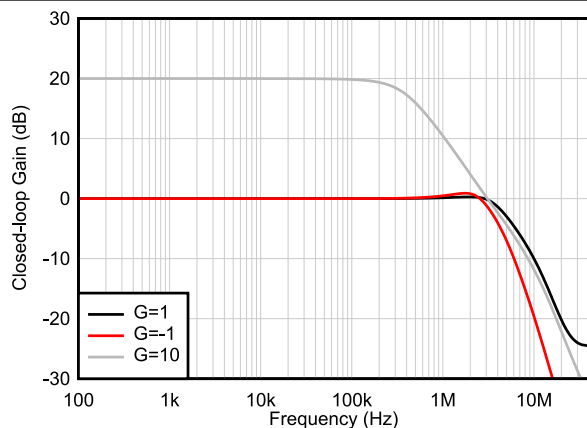


Figure 5-9. Closed-Loop Gain vs Frequency

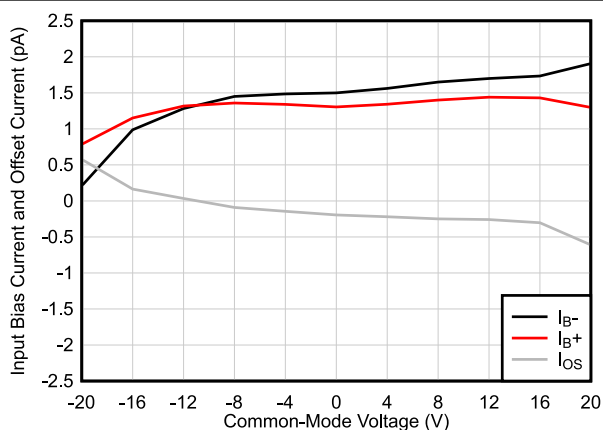


Figure 5-10. Input Bias Current vs Common-Mode Voltage

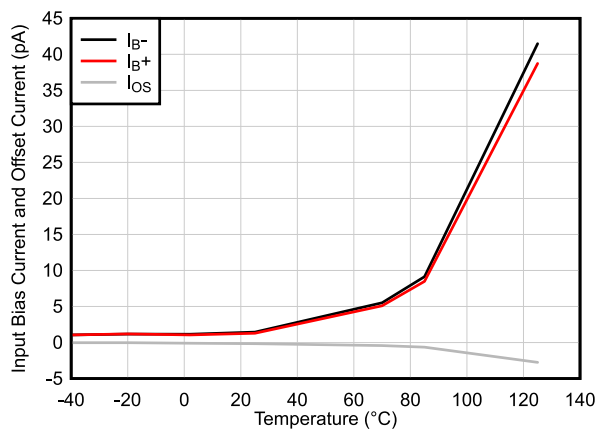


Figure 5-11. Input Bias Current vs Temperature

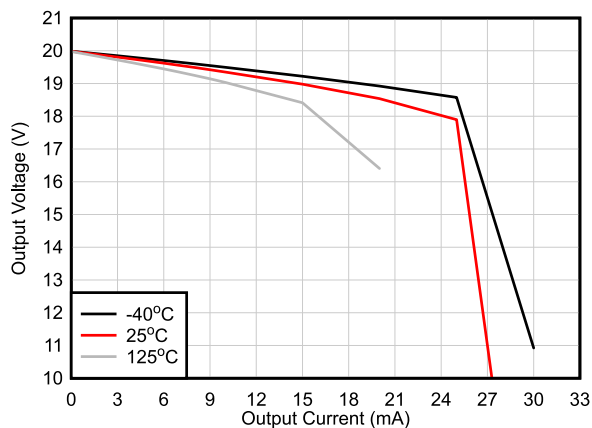


Figure 5-12. Output Voltage Swing vs Output Current (Sourcing)

## 5.10 Typical Characteristics: TL07xH (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

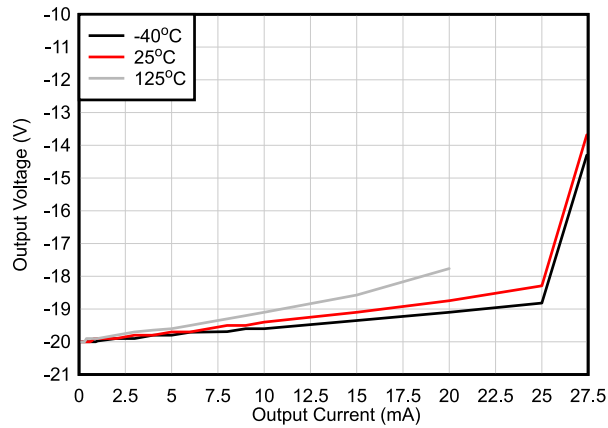


Figure 5-13. Output Voltage Swing vs Output Current (Sinking)

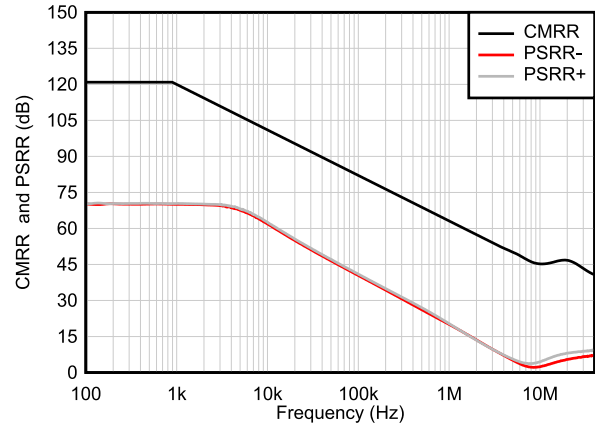


Figure 5-14. CMRR and PSRR vs Frequency

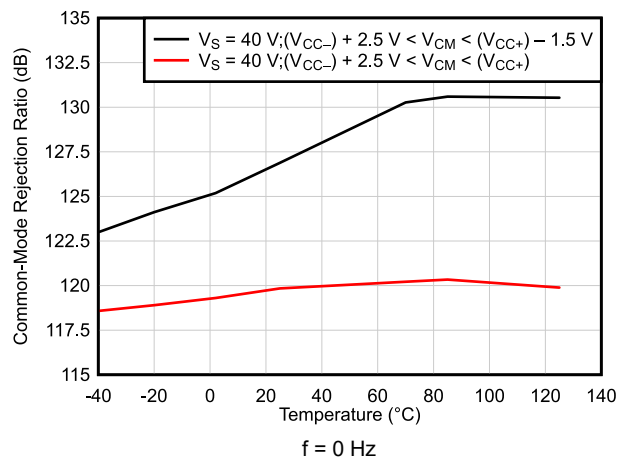


Figure 5-15. CMRR vs Temperature (dB)

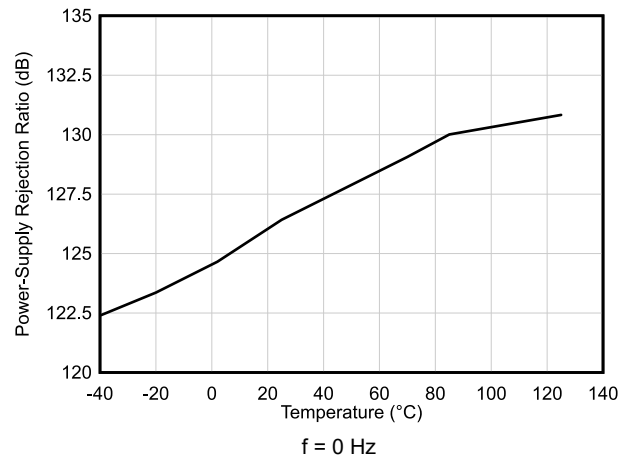


Figure 5-16. PSRR vs Temperature (dB)

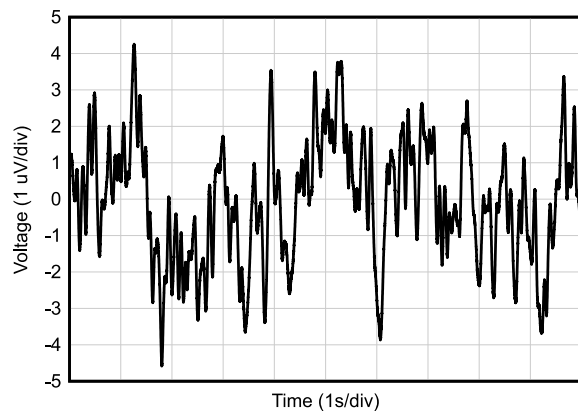


Figure 5-17. 0.1-Hz to 10-Hz Noise

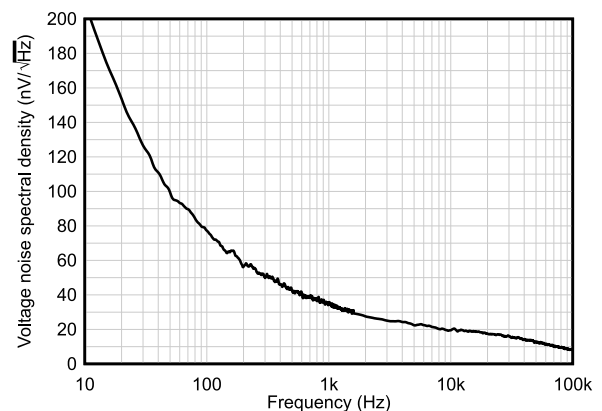


Figure 5-18. Input Voltage Noise Spectral Density vs Frequency

## 5.10 Typical Characteristics: TL07xH (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

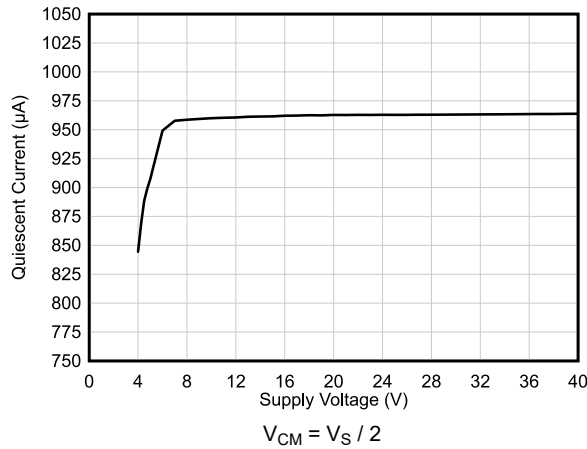


Figure 5-19. Quiescent Current vs Supply Voltage

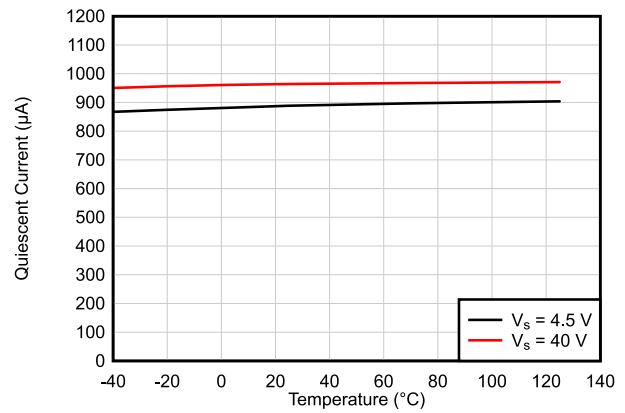


Figure 5-20. Quiescent Current vs Temperature

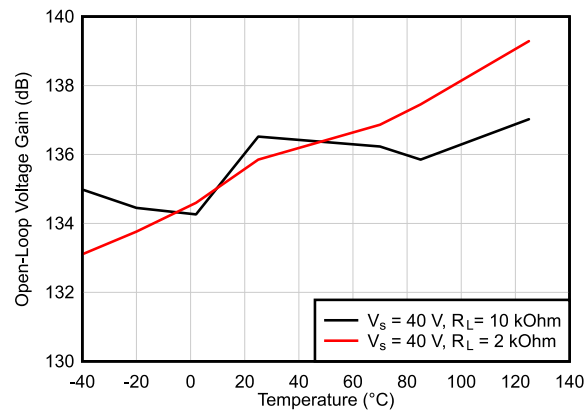


Figure 5-21. Open-Loop Voltage Gain vs Temperature

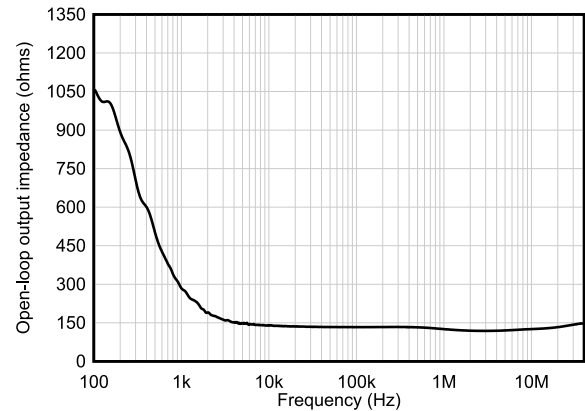


Figure 5-22. Open-Loop Output Impedance vs Frequency

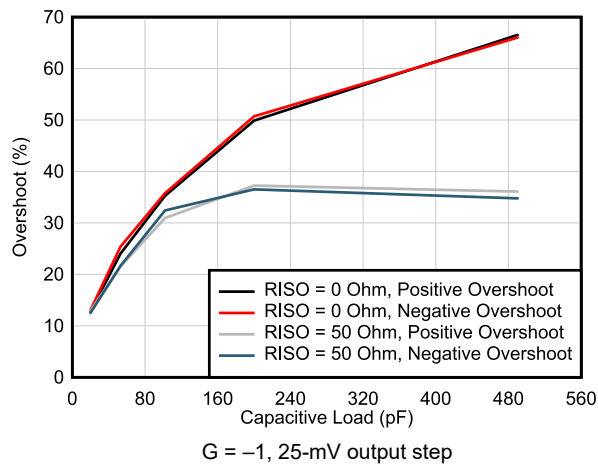


Figure 5-23. Small-Signal Overshoot vs Capacitive Load

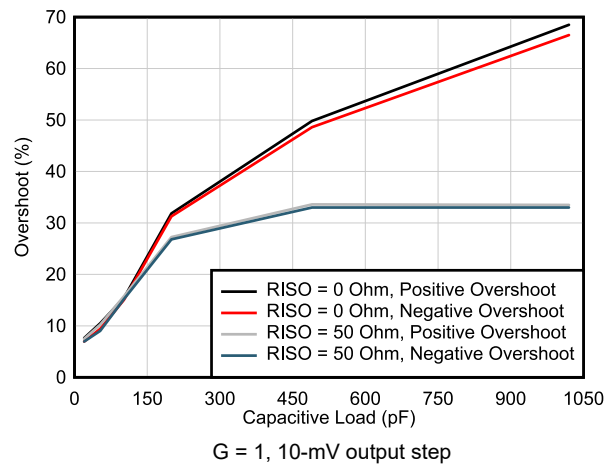


Figure 5-24. Small-Signal Overshoot vs Capacitive Load

## 5.10 Typical Characteristics: TL07xH (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

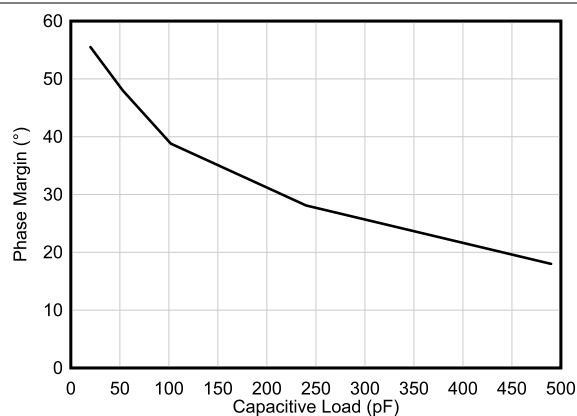


Figure 5-25. Phase Margin vs Capacitive Load

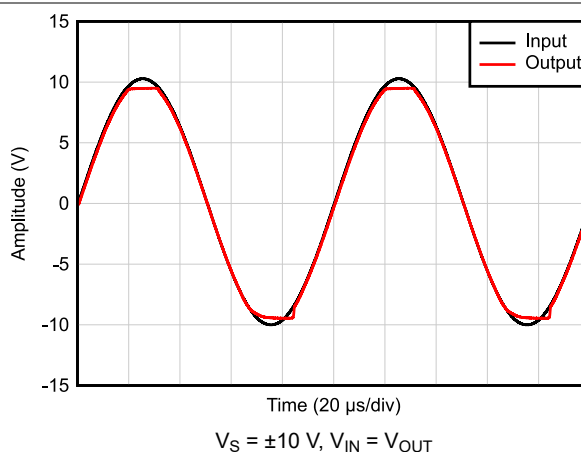


Figure 5-26. No Phase Reversal

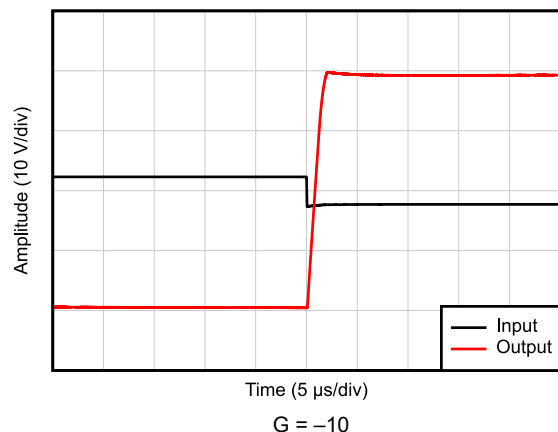


Figure 5-27. Positive Overload Recovery

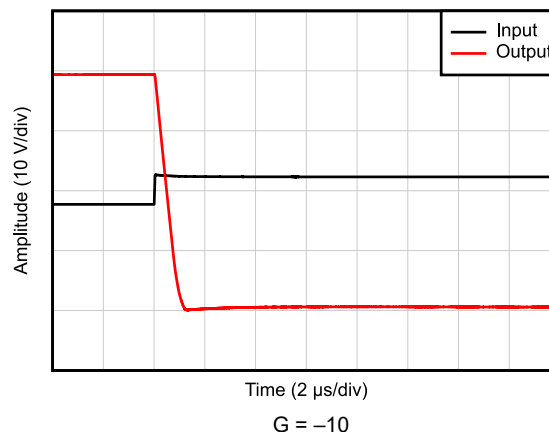


Figure 5-28. Negative Overload Recovery

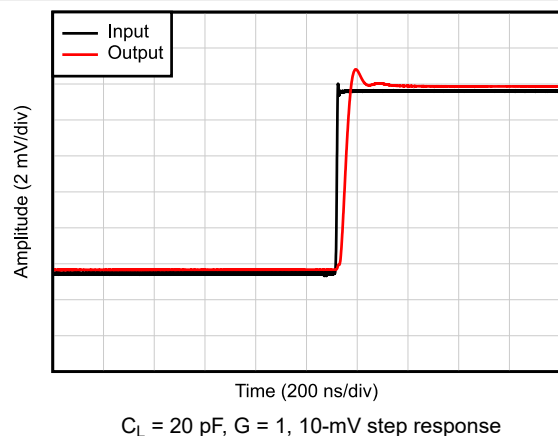


Figure 5-29. Small-Signal Step Response, Rising

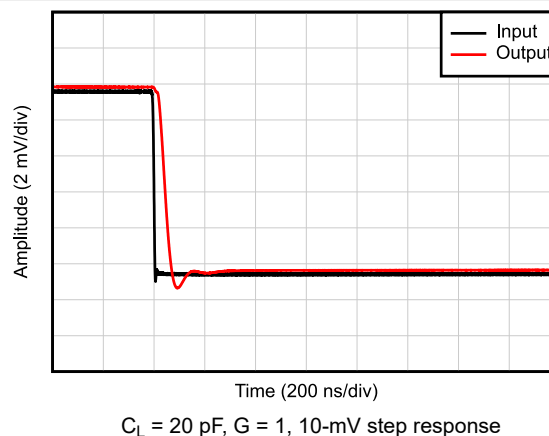
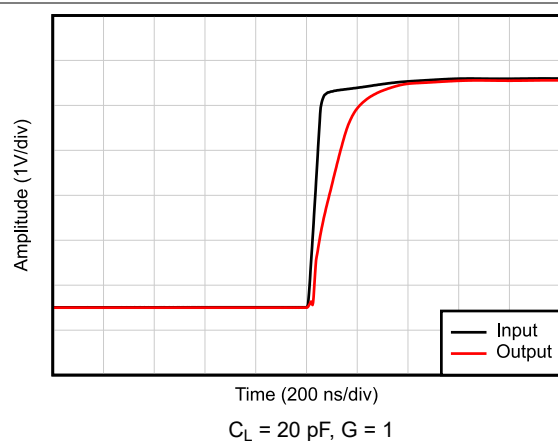


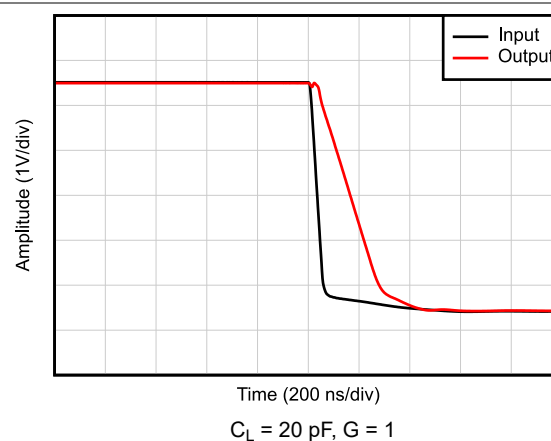
Figure 5-30. Small-Signal Step Response, Falling

## 5.10 Typical Characteristics: TL07xH (continued)

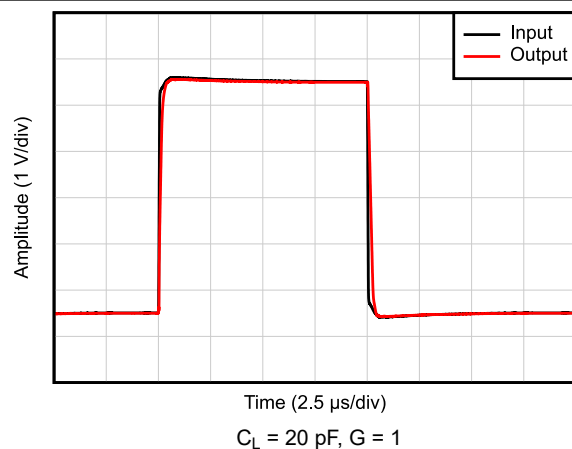
at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)



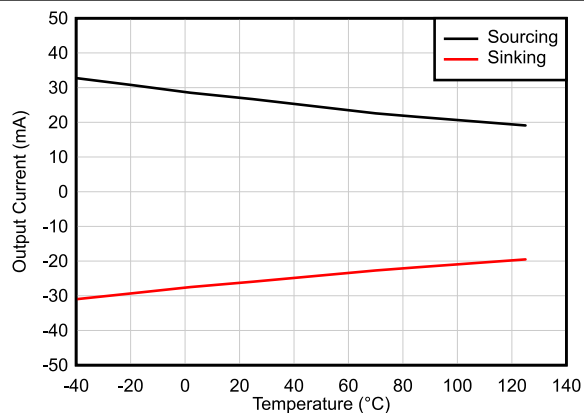
**Figure 5-31. Large-Signal Step Response (Rising)**



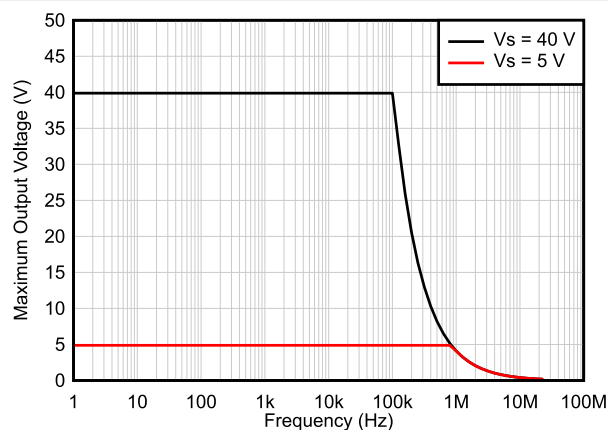
**Figure 5-32. Large-Signal Step Response (Falling)**



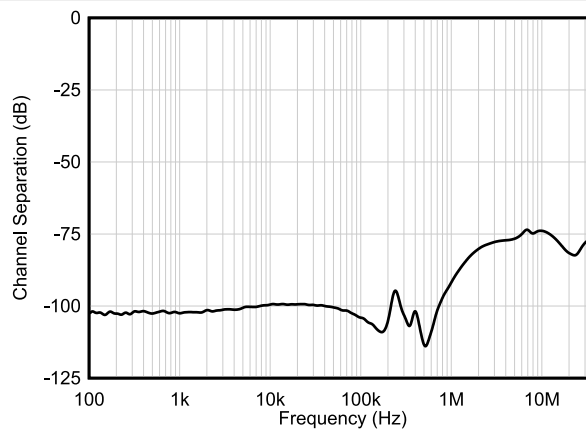
**Figure 5-33. Large-Signal Step Response**



**Figure 5-34. Short-Circuit Current vs Temperature**



**Figure 5-35. Maximum Output Voltage vs Frequency**



**Figure 5-36. Channel Separation vs Frequency**

## 5.10 Typical Characteristics: TL07xH (continued)

at  $T_A = 25^\circ\text{C}$ ,  $V_S = 40\text{ V}$  ( $\pm 20\text{ V}$ ),  $V_{CM} = V_S / 2$ ,  $R_{LOAD} = 10\text{ k}\Omega$  connected to  $V_S / 2$ , and  $C_L = 20\text{ pF}$  (unless otherwise noted)

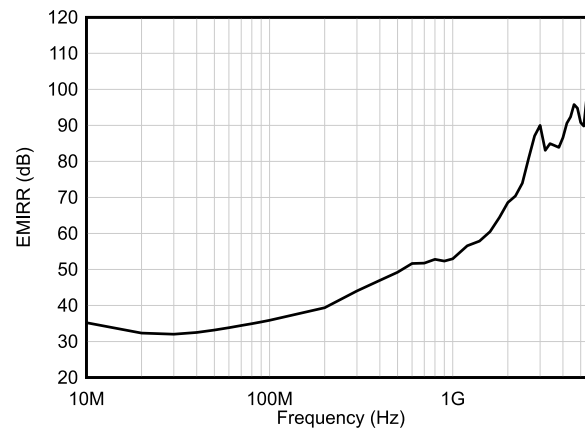


Figure 5-37. EMIRR (Electromagnetic Interference Rejection Ratio) vs Frequency

## 5.11 Typical Characteristics: All Devices Except TL07xH

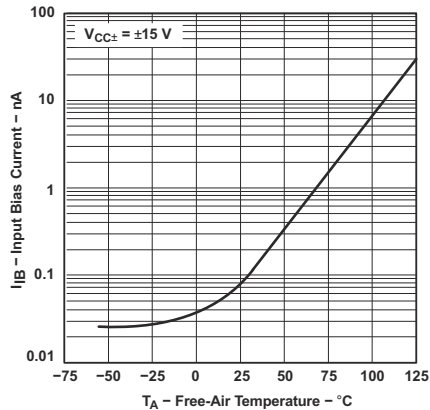


Figure 5-38. Input Bias Current vs Free-Air Temperature

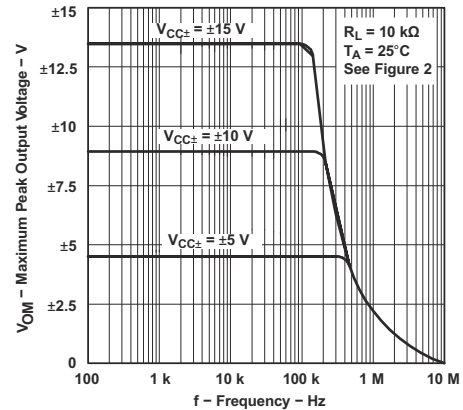


Figure 5-39. Maximum Peak Output Voltage vs Frequency

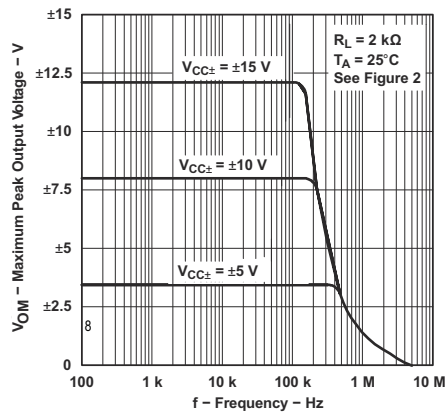


Figure 5-40. Maximum Peak Output Voltage vs Frequency

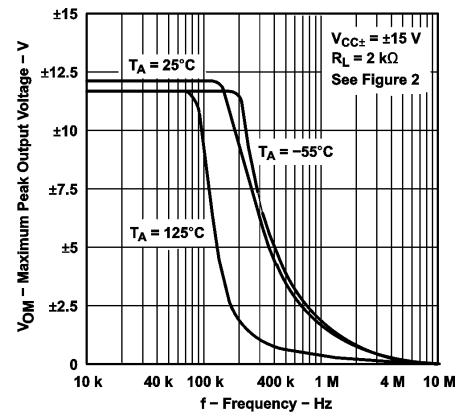


Figure 5-41. Maximum Peak Output Voltage vs Frequency

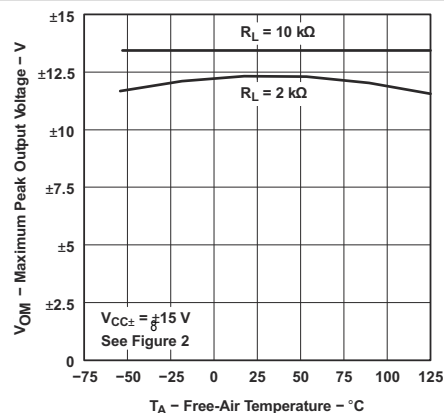


Figure 5-42. Maximum Peak Output Voltage vs Free-Air Temperature

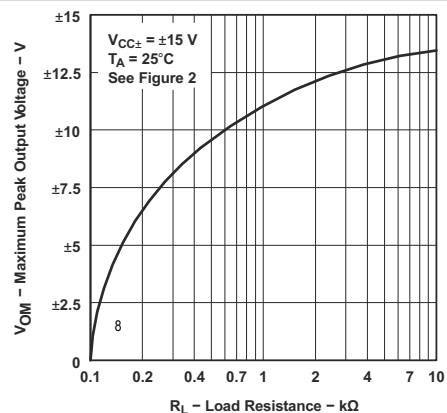


Figure 5-43. Maximum Peak Output Voltage vs Load Resistance

## 5.11 Typical Characteristics: All Devices Except TL07xH (continued)

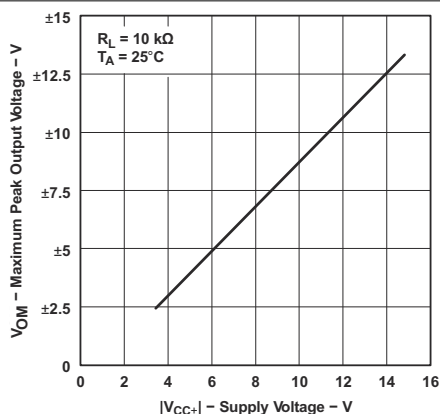


Figure 5-44. Maximum Peak Output Voltage vs Supply Voltage

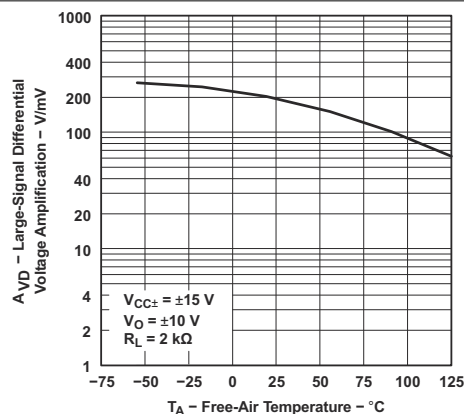


Figure 5-45. Large-Signal Differential Voltage Amplification vs Free-Air Temperature

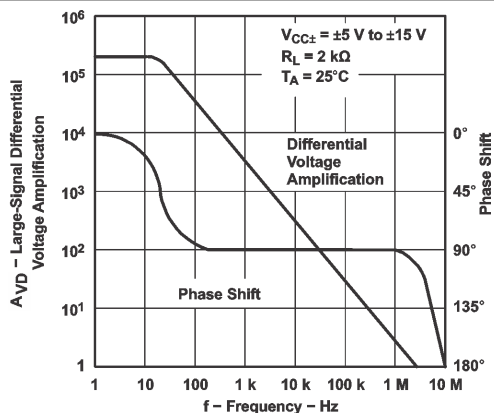


Figure 5-46. Large-Signal Differential Voltage Amplification and Phase Shift vs Frequency

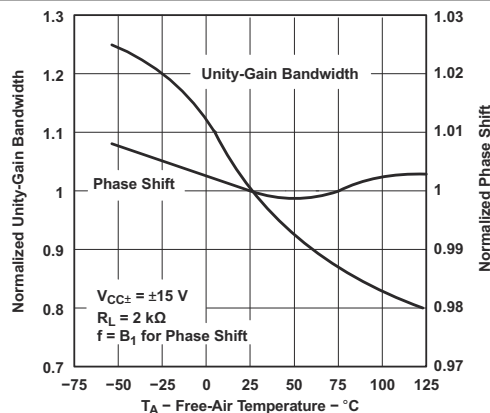


Figure 5-47. Normalized Unity-Gain Bandwidth and Phase Shift vs Free-Air Temperature

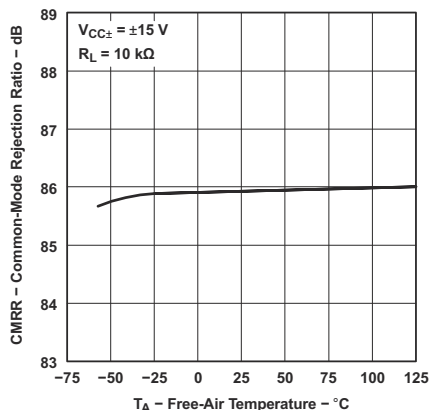


Figure 5-48. Common-Mode Rejection Ratio vs Free-Air Temperature

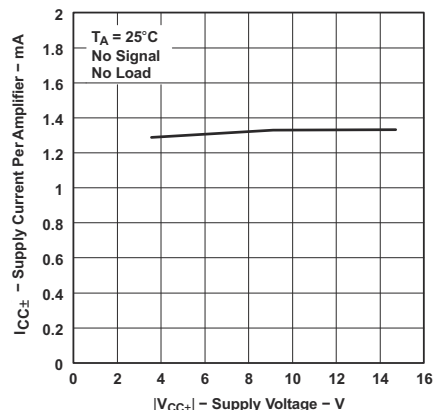


Figure 5-49. Supply Current Per Amplifier vs Supply Voltage

## 5.11 Typical Characteristics: All Devices Except TL07xH (continued)

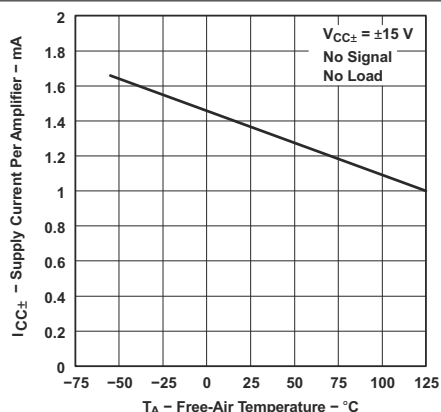


Figure 5-50. Supply Current Per Amplifier vs Free-Air Temperature

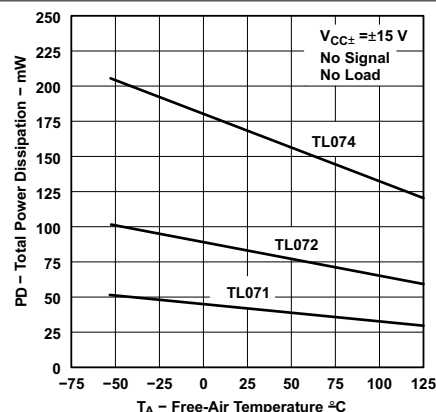


Figure 5-51. Total Power Dissipation vs Free-Air Temperature

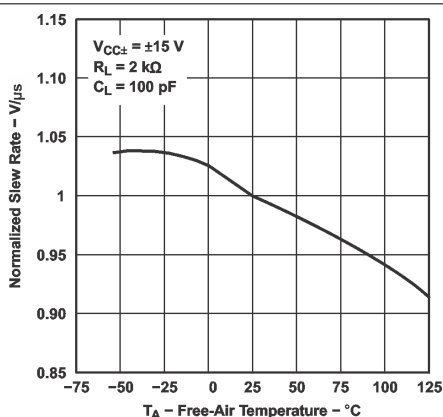


Figure 5-52. Normalized Slew Rate vs Free-Air Temperature

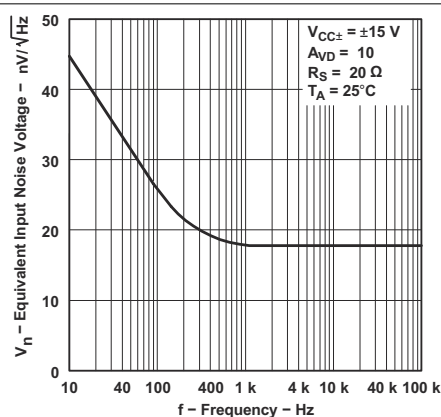


Figure 5-53. Equivalent Input Noise Voltage vs Frequency

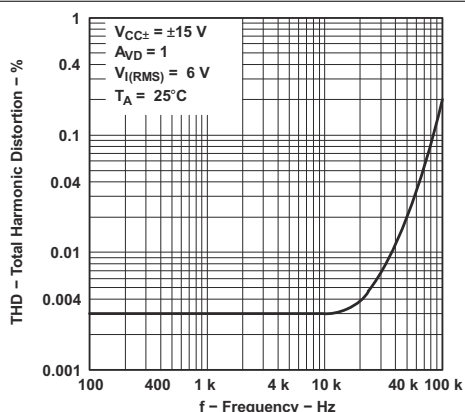


Figure 5-54. Total Harmonic Distortion vs Frequency

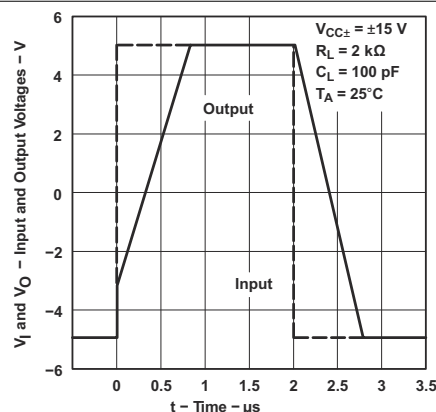


Figure 5-55. Voltage-Follower Large-Signal Pulse Response

## 5.11 Typical Characteristics: All Devices Except TL07xH (continued)

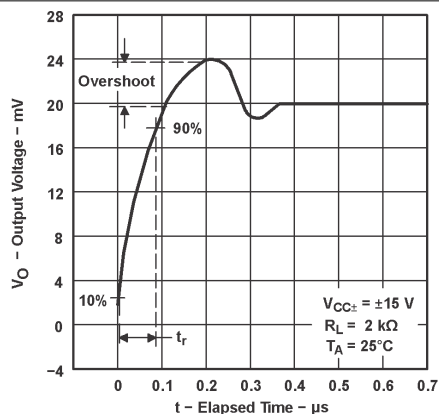


Figure 5-56. Output Voltage vs Elapsed Time

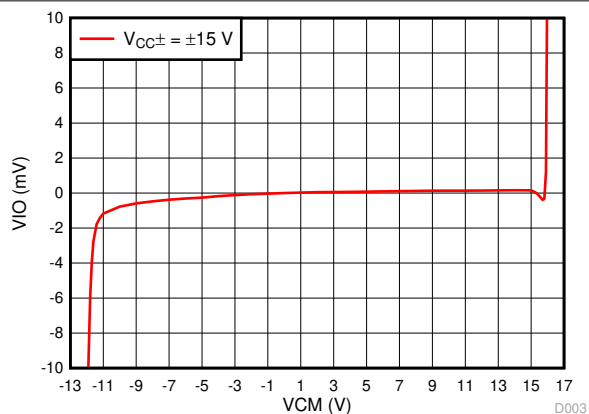
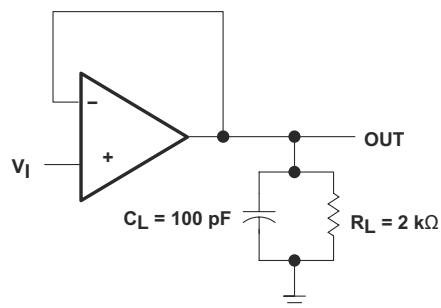
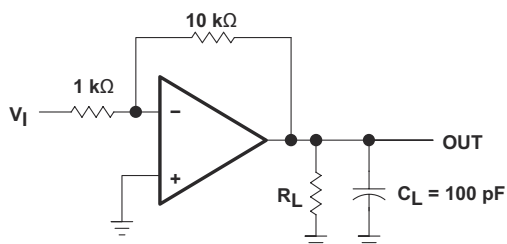


Figure 5-57.  $V_{IO}$  vs  $V_{CM}$

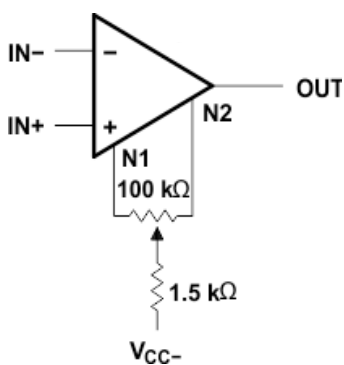
## 6 Parameter Measurement Information



**Figure 6-1. Unity-Gain Amplifier**



**Figure 6-2. Gain-of-10 Inverting Amplifier**



**Figure 6-3. Input Offset-Voltage Null Circuit  
for PS Package (SO, 8) Only**

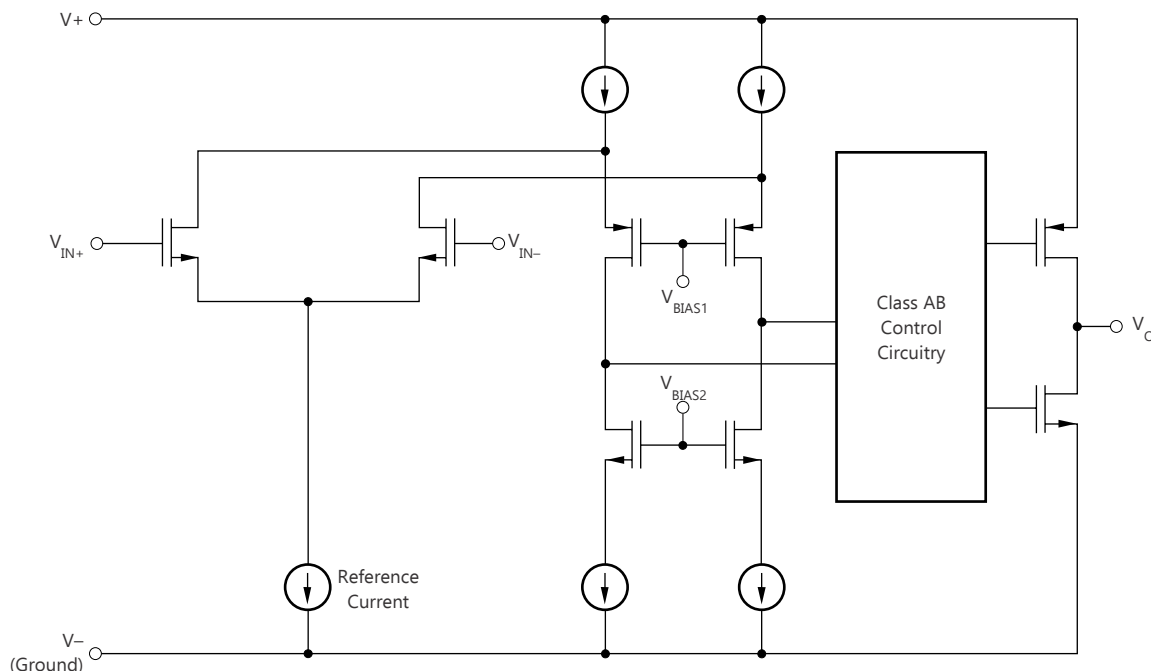
## 7 Detailed Description

### 7.1 Overview

The TL07xH (TL071H, TL072H, and TL074H) family of devices are the next-generation versions of the industry-standard TL07x (TL071, TL072, and TL074) devices. These devices provide outstanding value for cost-sensitive applications, with features including low offset (1 mV, typical), high slew rate (20 V/ $\mu$ s, typical), and common-mode input to the positive supply. High ESD (2 kV, HBM), integrated EMI and RF filters, and operation across the full  $-40^{\circ}\text{C}$  to  $125^{\circ}\text{C}$  enable the TL07xH devices to be used in the most rugged and demanding applications.

The C-suffix devices are characterized for operation from  $0^{\circ}\text{C}$  to  $70^{\circ}\text{C}$ . The I-suffix devices are characterized for operation from  $-40^{\circ}\text{C}$  to  $+85^{\circ}\text{C}$ . The M-suffix devices are characterized for operation over the full military temperature range of  $-55^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ .

### 7.2 Functional Block Diagram



### 7.3 Feature Description

The TL07xH family of devices improve many specifications as compared to the industry-standard TL07x family. Several comparisons of key specifications between these families are included in the following sections to show the advantages of the TL07xH family.

#### 7.3.1 Total Harmonic Distortion

Harmonic distortions to an audio signal are created by electronic components in a circuit. Total harmonic distortion (THD) is a measure of harmonic distortions accumulated by a signal in an audio system. These devices have a very low THD of 0.003% meaning that the TL07x device adds little harmonic distortion when used in audio signal applications.

#### 7.3.2 Slew Rate

The slew rate is the rate at which an operational amplifier can change the output when there is a change on the input. These devices have a 20-V/ $\mu$ s slew rate.

### 7.4 Device Functional Modes

These devices are powered on when the supply is connected. These devices can be operated as a single-supply operational amplifier or dual-supply amplifier depending on the application.

## 8 Application and Implementation

### Note

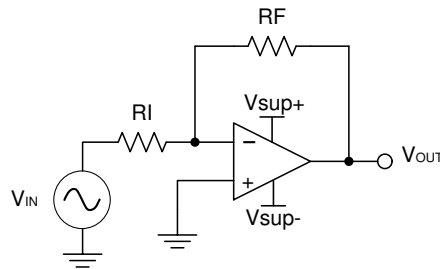
Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

### 8.1 Application Information

A typical application for an operational amplifier is an inverting amplifier. This amplifier takes a positive voltage on the input, and makes the voltage a negative voltage. In the same manner, the amplifier makes negative voltages positive.

### 8.2 Typical Applications

#### 8.2.1 Inverting Amplifier



**Figure 8-1. Inverting Amplifier**

##### 8.2.1.1 Design Requirements

The supply voltage must be selected so the supply voltage is larger than the input voltage range and output range. For instance, this application scales a signal of  $\pm 0.5$  V to  $\pm 1.8$  V. Setting the supply at  $\pm 12$  V is sufficient to accommodate this application.

##### 8.2.1.2 Detailed Design Procedure

Determine the gain required by the inverting amplifier:

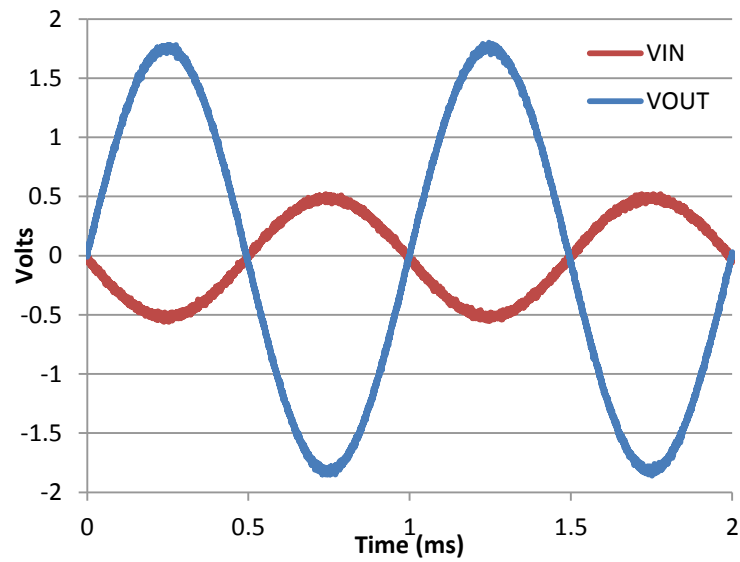
$$A_V = \frac{V_{OUT}}{V_{IN}} \quad (1)$$

$$A_V = \frac{1.8}{-0.5} = -3.6 \quad (2)$$

After the desired gain is determined, select a value for  $R_I$  or  $R_F$ . Selecting a value in the kilohm range is desirable because the amplifier circuit uses currents in the milliamp range. This example uses 10 k $\Omega$  for  $R_I$ , which means 36 k $\Omega$  is used for  $R_F$ . The gain is determined by [Equation 3](#).

$$A_V = -\frac{R_F}{R_I} \quad (3)$$

### 8.2.1.3 Application Curve



**Figure 8-2. Input and Output Voltages of the Inverting Amplifier**

## 8.3 Power Supply Recommendations

### CAUTION

Supply voltages larger than 36 V for a single-supply or outside the range of  $\pm 18$  V for a dual-supply can permanently damage the device (see [Section 5.1](#)).

Place 0.1- $\mu$ F bypass capacitors close to the power-supply pins to reduce errors coupling in from noisy or high-impedance power supplies. For more detailed information on bypass capacitor placement, see [Section 8.4](#).

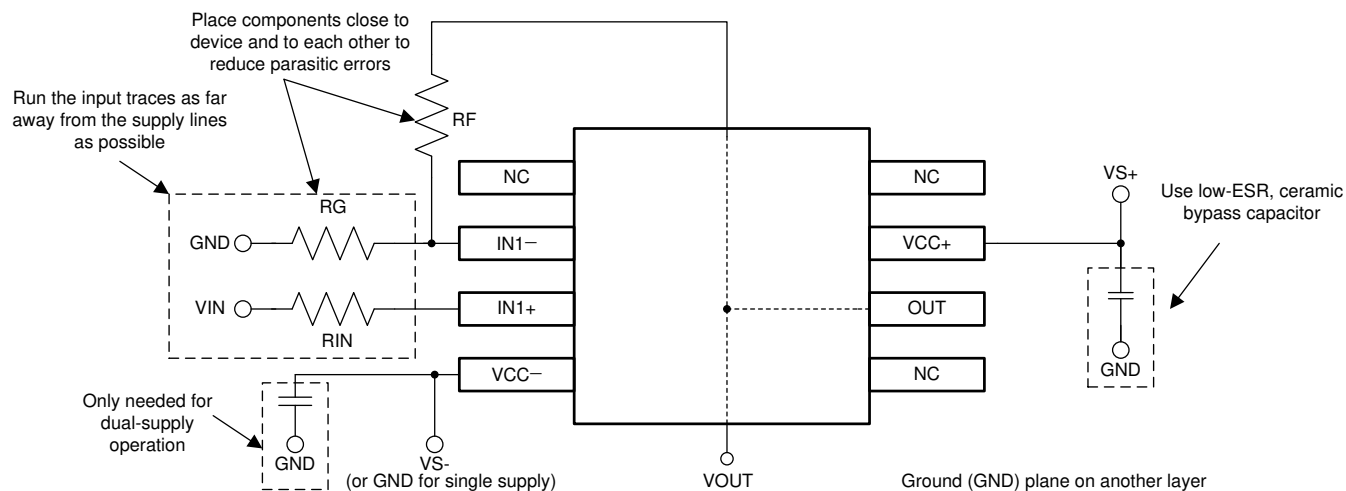
## 8.4 Layout

### 8.4.1 Layout Guidelines

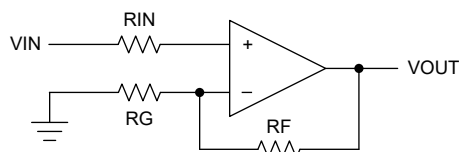
For best operational performance of the device, use good PCB layout practices, including:

- Noise can propagate into analog circuitry through the power pins of the circuit as a whole, as well as the operational amplifier. Bypass capacitors are used to reduce the coupled noise by providing low impedance power sources local to the analog circuitry.
  - Connect low-ESR, 0.1- $\mu$ F ceramic bypass capacitors between each supply pin and ground, placed as close to the device as possible. A single bypass capacitor from  $V_{CC+}$  to ground is applicable for single-supply applications.
- Separate grounding for analog and digital portions of circuitry is one of the simplest and most-effective methods of noise suppression. One or more layers on multilayer PCBs are typically devoted to ground planes. A ground plane helps distribute heat and reduces EMI noise pickup. Take care to physically separate digital and analog grounds, paying attention to the flow of the ground current.
- To reduce parasitic coupling, run the input traces as far away from the supply or output traces as possible. If not possible, then better to cross the sensitive trace perpendicular as opposed to in parallel with the noisy trace.
- Place the external components as close to the device as possible. Keeping RF and RG close to the inverting input minimizes parasitic capacitance; see also [Section 8.4.2](#).
- Keep the length of input traces as short as possible. Always remember that the input traces are the most sensitive part of the circuit.
- Consider a driven, low-impedance guard ring around the critical traces. A guard ring can significantly reduce leakage currents from nearby traces that are at different potentials.

### 8.4.2 Layout Example



**Figure 8-3. Operational Amplifier Board Layout for Noninverting Configuration**



**Figure 8-4. Operational Amplifier Schematic for Noninverting Configuration**

## 9 Device and Documentation Support

### 9.1 Device Support

#### 9.1.1 Device Nomenclature

**Table 9-1. Device Nomenclature**

| PART NUMBER  | DEFINITION                                                                                                                                                                                                                                                                                                                   |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TL07xyzzzzzz | x is the channel count                                                                                                                                                                                                                                                                                                       |
|              | If y = H, the die is manufactured on the latest flow (CSO: RFB).<br>Section 5.7 and Section 5.10 describe the performance of the new die.                                                                                                                                                                                    |
|              | If y ≠ H and y ≠ M, the die is manufactured on the legacy flow (CSO: SFAB) or the latest flow (CSO: RFB).<br>Section 5.8, Section 5.9, and Section 5.11 describe the performance of the original die.<br>Section 5.7 and Section 5.10 describe the performance of the new die.                                               |
|              | If y = M, the device is specified for the extended temperature range of –55°C to +125°C. The die is manufactured on the legacy flow (CSO:SFAB).<br>The letters and numbers represented by z are grade-out and package options described in Section 5.8 and the <i>Package Option Addendum</i> at the end of this data sheet. |

### 9.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on [ti.com](https://www.ti.com). Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

### 9.3 Support Resources

[TI E2E™ support forums](#) are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

Linked content is provided "AS IS" by the respective contributors. They do not constitute TI specifications and do not necessarily reflect TI's views; see TI's [Terms of Use](#).

### 9.4 Trademarks

TI E2E™ is a trademark of Texas Instruments.

All trademarks are the property of their respective owners.

### 9.5 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

### 9.6 Glossary

[TI Glossary](#) This glossary lists and explains terms, acronyms, and definitions.

## 10 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision V (April 2023) to Revision W (July 2025)                                                 | Page |
|----------------------------------------------------------------------------------------------------------------|------|
| • Deleted references to trim function from all packages except PS (SO, 8) package.....                         | 1    |
| • Changed $V_n$ from $18\text{nV}/\sqrt{\text{Hz}}$ to $37\text{nV}/\sqrt{\text{Hz}}$ in <i>Features</i> ..... | 1    |
| • Updated <i>Device Information</i> table to match <i>Package Option Addendum</i> .....                        | 1    |
| • Updated front page image to show which device uses the PS package only.....                                  | 1    |
| • Updated <i>Pin Configuration and Functions</i> to show that only PS package (PDIP, 8) has trim function..... | 3    |

|                                                                                                                |    |
|----------------------------------------------------------------------------------------------------------------|----|
| • Added note regarding old and new dies.....                                                                   | 10 |
| • Deleted Figure 5-19, <i>THD+N Ratio vs Frequency</i> and Figure 5-20, <i>THD+N vs Output Amplitude</i> ..... | 17 |
| • Added "for PS Package (SO, 8) Only" to Figure 7-3 caption.....                                               | 28 |
| • Deleted <i>Unity Gain Buffer</i> and <i>System Examples</i> sections.....                                    | 30 |
| • Deleted Equation 1 from <i>Detailed Design Procedure</i> .....                                               | 30 |
| • Deleted "This ensures the part does not draw too much current." from <i>Detailed Design Procedure</i> .....  | 30 |
| • Added <i>Device Nomenclature</i> table.....                                                                  | 34 |

| <b>Changes from Revision U (December 2022) to Revision V (April 2023)</b>                                   | <b>Page</b> |
|-------------------------------------------------------------------------------------------------------------|-------------|
| • Updated <i>Overview</i> , <i>Functional Block Diagram</i> , and <i>Feature Description</i> sections ..... | 29          |

## 11 Mechanical, Packaging, and Orderable Information

The following pages include mechanical packaging and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser based versions of this data sheet, refer to the left hand navigation.

## PACKAGING INFORMATION

| Orderable part number            | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|----------------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| <a href="#">81023052A</a>        | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023052A<br>TL072MFKB |
| <a href="#">8102305HA</a>        | Active        | Production           | CFP (U)   10   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305HA<br>TL072M    |
| <a href="#">8102305PA</a>        | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305PA<br>TL072M    |
| <a href="#">81023062A</a>        | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023062A<br>TL074MFKB |
| <a href="#">8102306CA</a>        | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306CA<br>TL074MJB  |
| <a href="#">8102306DA</a>        | Active        | Production           | CFP (W)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306DA<br>TL074MWB  |
| <a href="#">JM38510/11905BPA</a> | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | JM38510<br>/11905BPA   |
| JM38510/11905BPA.A               | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | JM38510<br>/11905BPA   |
| <a href="#">M38510/11905BPA</a>  | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | JM38510<br>/11905BPA   |
| <a href="#">TL071ACDR</a>        | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 071AC                  |
| TL071ACDR.A                      | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 071AC                  |
| <a href="#">TL071ACP</a>         | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071ACP               |
| TL071ACP.A                       | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071ACP               |
| <a href="#">TL071BCDR</a>        | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 071BC                  |
| TL071BCDR.A                      | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 071BC                  |
| <a href="#">TL071BCP</a>         | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071BCP               |
| TL071BCP.A                       | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071BCP               |
| <a href="#">TL071CDR</a>         | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL071C                 |
| TL071CDR.A                       | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL071C                 |
| TL071CDRE4                       | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| TL071CDRG4                       | Active        | Production           | SOIC (D)   8   | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| <a href="#">TL071CP</a>          | Active        | Production           | PDIP (P)   8   | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071CP                |

| Orderable part number       | Status<br>(1) | Material type<br>(2) | Package   Pins   | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|-----------------------------|---------------|----------------------|------------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| TL071CP.A                   | Active        | Production           | PDIP (P)   8     | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL071CP             |
| TL071CPE4                   | Active        | Production           | PDIP (P)   8     | 50   TUBE             | -           | Call TI                              | Call TI                           | 0 to 70      |                     |
| <a href="#">TL071CPSR</a>   | Active        | Production           | SO (PS)   8      | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T071                |
| TL071CPSR.A                 | Active        | Production           | SO (PS)   8      | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T071                |
| <a href="#">TL071HIDBVR</a> | Active        | Production           | SOT-23 (DBV)   5 | 3000   LARGE T&R      | Yes         | NIPDAU   SN                          | Level-1-260C-UNLIM                | -40 to 125   | T71V                |
| TL071HIDBVR.A               | Active        | Production           | SOT-23 (DBV)   5 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T71V                |
| TL071HIDBVR.B               | Active        | Production           | SOT-23 (DBV)   5 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T71V                |
| TL071HIDBVRG4               | Active        | Production           | SOT-23 (DBV)   5 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T71V                |
| TL071HIDBVRG4.A             | Active        | Production           | SOT-23 (DBV)   5 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T71V                |
| <a href="#">TL071HIDCKR</a> | Active        | Production           | SC70 (DCK)   5   | 3000   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | -40 to 125   | 1IO                 |
| TL071HIDCKR.A               | Active        | Production           | SC70 (DCK)   5   | 3000   LARGE T&R      | Yes         | SN                                   | Level-1-260C-UNLIM                | -40 to 125   | 1IO                 |
| <a href="#">TL071HIDR</a>   | Active        | Production           | SOIC (D)   8     | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL071D              |
| TL071HIDR.A                 | Active        | Production           | SOIC (D)   8     | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL071D              |
| <a href="#">TL071IDR</a>    | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL071I              |
| TL071IDR.A                  | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL071I              |
| TL071IDR1G4                 | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL071I              |
| TL071IDR1G4.A               | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL071I              |
| TL071IDRG4                  | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 85    |                     |
| <a href="#">TL071IP</a>     | Active        | Production           | PDIP (P)   8     | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL071IP             |
| TL071IP.A                   | Active        | Production           | PDIP (P)   8     | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL071IP             |
| <a href="#">TL072ACDR</a>   | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072AC               |
| TL072ACDR.A                 | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072AC               |
| TL072ACDRE4                 | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072AC               |
| TL072ACDRG4                 | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072AC               |
| <a href="#">TL072ACP</a>    | Active        | Production           | PDIP (P)   8     | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072ACP            |
| TL072ACP.A                  | Active        | Production           | PDIP (P)   8     | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072ACP            |
| TL072ACPE4                  | Active        | Production           | PDIP (P)   8     | 50   TUBE             | -           | Call TI                              | Call TI                           | 0 to 70      |                     |
| TL072ACPS                   | Active        | Production           | SO (PS)   8      | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072A               |
| TL072ACPS.A                 | Active        | Production           | SO (PS)   8      | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072A               |
| <a href="#">TL072BCD</a>    | Obsolete      | Production           | SOIC (D)   8     | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | 072BC               |
| <a href="#">TL072BCDR</a>   | Active        | Production           | SOIC (D)   8     | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072BC               |

| Orderable part number       | Status<br>(1) | Material type<br>(2) | Package   Pins        | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|-----------------------------|---------------|----------------------|-----------------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| TL072BCDR.A                 | Active        | Production           | SOIC (D)   8          | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | 072BC                  |
| <a href="#">TL072BCP</a>    | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072BCP               |
| TL072BCP.A                  | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072BCP               |
| <a href="#">TL072CDR</a>    | Active        | Production           | SOIC (D)   8          | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL072C                 |
| TL072CDR.A                  | Active        | Production           | SOIC (D)   8          | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL072C                 |
| <a href="#">TL072CP</a>     | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072CP                |
| TL072CP.A                   | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL072CP                |
| <a href="#">TL072CPS</a>    | Active        | Production           | SO (PS)   8           | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| TL072CPS.A                  | Active        | Production           | SO (PS)   8           | 80   TUBE             | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| <a href="#">TL072CPSR</a>   | Active        | Production           | SO (PS)   8           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| TL072CPSR.A                 | Active        | Production           | SO (PS)   8           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| TL072CPSRG4                 | Active        | Production           | SO (PS)   8           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| <a href="#">TL072CPWR</a>   | Active        | Production           | TSSOP (PW)   8        | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| TL072CPWR.A                 | Active        | Production           | TSSOP (PW)   8        | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T072                   |
| TL072CPWRE4                 | Active        | Production           | TSSOP (PW)   8        | 2000   LARGE T&R      | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| TL072CPWRG4                 | Active        | Production           | TSSOP (PW)   8        | 2000   LARGE T&R      | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| <a href="#">TL072HIDDFR</a> | Active        | Production           | SOT-23-THIN (DDF)   8 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | O72F                   |
| TL072HIDDFR.A               | Active        | Production           | SOT-23-THIN (DDF)   8 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | O72F                   |
| TL072HIDDFR.B               | Active        | Production           | SOT-23-THIN (DDF)   8 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | O72F                   |
| <a href="#">TL072HIDR</a>   | Active        | Production           | SOIC (D)   8          | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL072D                 |
| TL072HIDR.A                 | Active        | Production           | SOIC (D)   8          | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL072D                 |
| <a href="#">TL072HIPWR</a>  | Active        | Production           | TSSOP (PW)   8        | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | 072HPW                 |
| TL072HIPWR.A                | Active        | Production           | TSSOP (PW)   8        | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | 072HPW                 |
| <a href="#">TL072IDR</a>    | Active        | Production           | SOIC (D)   8          | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL072I                 |
| TL072IDR.A                  | Active        | Production           | SOIC (D)   8          | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL072I                 |
| <a href="#">TL072IP</a>     | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL072IP                |
| TL072IP.A                   | Active        | Production           | PDIP (P)   8          | 50   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL072IP                |
| TL072IPE4                   | Active        | Production           | PDIP (P)   8          | 50   TUBE             | -           | Call TI                              | Call TI                           | -40 to 85    |                        |
| <a href="#">TL072MFKB</a>   | Active        | Production           | LCCC (FK)   20        | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023052A<br>TL072MFKB |

| Orderable part number      | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|----------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| TL072MFKB.A                | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023052A<br>TL072MFKB |
| <a href="#">TL072MJG</a>   | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL072MJG               |
| TL072MJG.A                 | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL072MJG               |
| <a href="#">TL072MJGB</a>  | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305PA<br>TL072M    |
| TL072MJGB.A                | Active        | Production           | CDIP (JG)   8  | 50   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305PA<br>TL072M    |
| <a href="#">TL072MUB</a>   | Active        | Production           | CFP (U)   10   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305HA<br>TL072M    |
| TL072MUB.A                 | Active        | Production           | CFP (U)   10   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102305HA<br>TL072M    |
| <a href="#">TL074ACDR</a>  | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074AC                |
| TL074ACDR.A                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074AC                |
| <a href="#">TL074ACN</a>   | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074ACN               |
| TL074ACN.A                 | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074ACN               |
| <a href="#">TL074ACNSR</a> | Active        | Production           | SOP (NS)   14  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074A                 |
| TL074ACNSR.A               | Active        | Production           | SOP (NS)   14  | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074A                 |
| <a href="#">TL074BCD</a>   | Obsolete      | Production           | SOIC (D)   14  | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL074BC                |
| <a href="#">TL074BCDR</a>  | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074BC                |
| TL074BCDR.A                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074BC                |
| TL074BCDR.B                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | 0 to 70      |                        |
| TL074BCDRE4                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074BC                |
| TL074BCDRG4                | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074BC                |
| <a href="#">TL074BCN</a>   | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074BCN               |
| TL074BCN.A                 | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074BCN               |
| <a href="#">TL074CD</a>    | Obsolete      | Production           | SOIC (D)   14  | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | TL074C                 |
| <a href="#">TL074CDBR</a>  | Active        | Production           | SSOP (DB)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                   |
| TL074CDBR.A                | Active        | Production           | SSOP (DB)   14 | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                   |
| <a href="#">TL074CDR</a>   | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074C                 |
| TL074CDR.A                 | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074C                 |
| TL074CDR.B                 | Active        | Production           | SOIC (D)   14  | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 85    |                        |

| Orderable part number       | Status<br>(1) | Material type<br>(2) | Package   Pins            | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6) |
|-----------------------------|---------------|----------------------|---------------------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|---------------------|
| <a href="#">TL074CDRG4</a>  | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074C              |
| TL074CDRG4.A                | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074C              |
| <a href="#">TL074CN</a>     | Active        | Production           | PDIP (N)   14             | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074CN             |
| TL074CN.A                   | Active        | Production           | PDIP (N)   14             | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | 0 to 70      | TL074CN             |
| <a href="#">TL074CNSR</a>   | Active        | Production           | SOP (NS)   14             | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074               |
| TL074CNSR.A                 | Active        | Production           | SOP (NS)   14             | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | TL074               |
| <a href="#">TL074CPW</a>    | Obsolete      | Production           | TSSOP (PW)   14           | -                     | -           | Call TI                              | Call TI                           | 0 to 70      | T074                |
| <a href="#">TL074CPWR</a>   | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                |
| TL074CPWR.A                 | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                |
| TL074CPWRE4                 | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                |
| TL074CPWRG4                 | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | 0 to 70      | T074                |
| <a href="#">TL074HIDR</a>   | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074HID            |
| TL074HIDR.A                 | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074HID            |
| TL074HIDR.B                 | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 125   |                     |
| TL074HIDRG4                 | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074HID            |
| TL074HIDRG4.A               | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074HID            |
| <a href="#">TL074HIDYYR</a> | Active        | Production           | SOT-23-THIN<br>(DYY)   14 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T074HDYY            |
| TL074HIDYYR.A               | Active        | Production           | SOT-23-THIN<br>(DYY)   14 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T074HDYY            |
| TL074HIDYYR.B               | Active        | Production           | SOT-23-THIN<br>(DYY)   14 | 3000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | T074HDYY            |
| <a href="#">TL074HIPWR</a>  | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074PW             |
| TL074HIPWR.A                | Active        | Production           | TSSOP (PW)   14           | 2000   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 125   | TL074PW             |
| <a href="#">TL074ID</a>     | Obsolete      | Production           | SOIC (D)   14             | -                     | -           | Call TI                              | Call TI                           | -40 to 85    | TL074I              |
| <a href="#">TL074IDR</a>    | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL074I              |
| TL074IDR.A                  | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL074I              |
| TL074IDR.B                  | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | -           | Call TI                              | Call TI                           | -40 to 85    |                     |
| TL074IDRE4                  | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL074I              |
| TL074IDRG4                  | Active        | Production           | SOIC (D)   14             | 2500   LARGE T&R      | Yes         | NIPDAU                               | Level-1-260C-UNLIM                | -40 to 85    | TL074I              |
| <a href="#">TL074IN</a>     | Active        | Production           | PDIP (N)   14             | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL074IN             |

| Orderable part number     | Status<br>(1) | Material type<br>(2) | Package   Pins | Package qty   Carrier | RoHS<br>(3) | Lead finish/<br>Ball material<br>(4) | MSL rating/<br>Peak reflow<br>(5) | Op temp (°C) | Part marking<br>(6)    |
|---------------------------|---------------|----------------------|----------------|-----------------------|-------------|--------------------------------------|-----------------------------------|--------------|------------------------|
| TL074IN.A                 | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL074IN                |
| TL074ING4                 | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL074IN                |
| TL074ING4.A               | Active        | Production           | PDIP (N)   14  | 25   TUBE             | Yes         | NIPDAU                               | N/A for Pkg Type                  | -40 to 85    | TL074IN                |
| <a href="#">TL074MFK</a>  | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL074MFK               |
| TL074MFK.A                | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL074MFK               |
| <a href="#">TL074MFKB</a> | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023062A<br>TL074MFKB |
| TL074MFKB.A               | Active        | Production           | LCCC (FK)   20 | 55   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 81023062A<br>TL074MFKB |
| <a href="#">TL074MJ</a>   | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL074MJ                |
| TL074MJ.A                 | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | TL074MJ                |
| <a href="#">TL074MJB</a>  | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306CA<br>TL074MJB  |
| TL074MJB.A                | Active        | Production           | CDIP (J)   14  | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306CA<br>TL074MJB  |
| <a href="#">TL074MWB</a>  | Active        | Production           | CFP (W)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306DA<br>TL074MWB  |
| TL074MWB.A                | Active        | Production           | CFP (W)   14   | 25   TUBE             | No          | SNPB                                 | N/A for Pkg Type                  | -55 to 125   | 8102306DA<br>TL074MWB  |

<sup>(1)</sup> **Status:** For more details on status, see our [product life cycle](#).

<sup>(2)</sup> **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

<sup>(3)</sup> **RoHS values:** Yes, No, RoHS Exempt. See the [TI RoHS Statement](#) for additional information and value definition.

<sup>(4)</sup> **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

<sup>(5)</sup> **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

**(6) Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

**Important Information and Disclaimer:** The information provided on this page represents TI's knowledge and belief as of the date that it is provided. TI bases its knowledge and belief on information provided by third parties, and makes no representation or warranty as to the accuracy of such information. Efforts are underway to better integrate information from third parties. TI has taken and continues to take reasonable steps to provide representative and accurate information but may not have conducted destructive testing or chemical analysis on incoming materials and chemicals. TI and TI suppliers consider certain information to be proprietary, and thus CAS numbers and other limited information may not be available for release.

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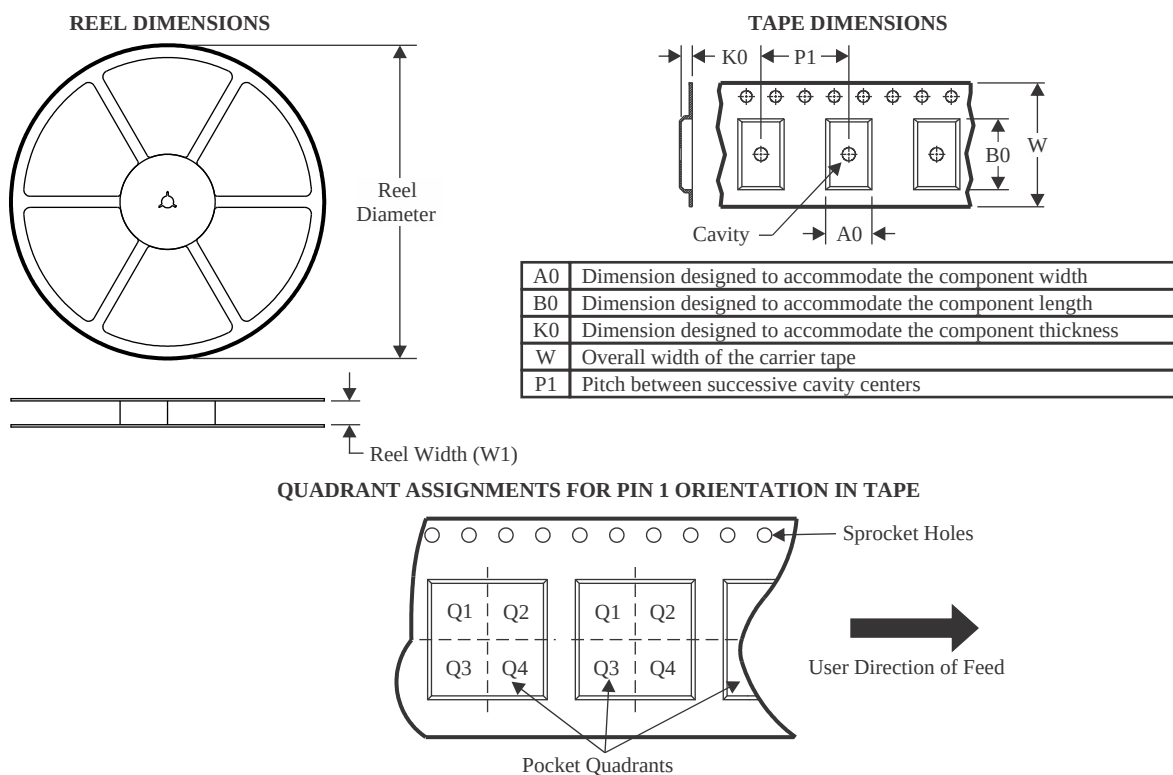
**OTHER QUALIFIED VERSIONS OF TL072, TL072M, TL074, TL074M :**

- Catalog : [TL072](#), [TL074](#)
- Enhanced Product : [TL072-EP](#), [TL072-EP](#), [TL074-EP](#), [TL074-EP](#)
- Military : [TL072M](#), [TL074M](#)

**NOTE: Qualified Version Definitions:**

- Catalog - TI's standard catalog product
- Enhanced Product - Supports Defense, Aerospace and Medical Applications
- Military - QML certified for Military and Defense Applications

## TAPE AND REEL INFORMATION

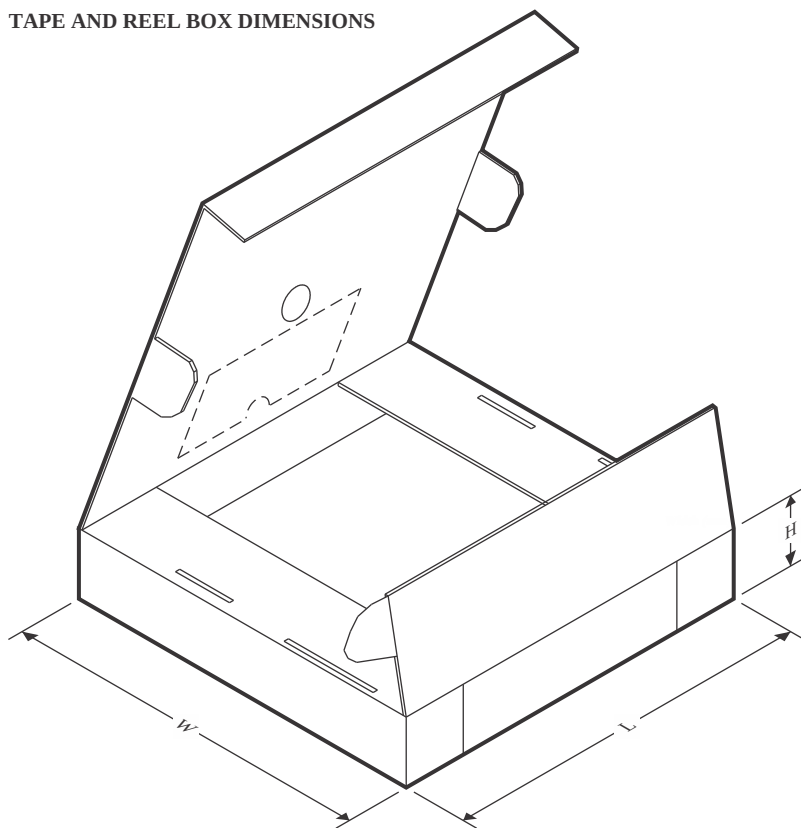


\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TL071ACDR     | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL071BCDR     | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL071CDR      | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL071CPSR     | SO           | PS              | 8    | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| TL071HIDBVR   | SOT-23       | DBV             | 5    | 3000 | 180.0              | 8.4                | 3.2     | 3.2     | 1.4     | 4.0     | 8.0    | Q3            |
| TL071HIDBVRG4 | SOT-23       | DBV             | 5    | 3000 | 180.0              | 8.4                | 3.2     | 3.2     | 1.4     | 4.0     | 8.0    | Q3            |
| TL071HIDCKR   | SC70         | DCK             | 5    | 3000 | 178.0              | 9.0                | 2.4     | 2.5     | 1.2     | 4.0     | 8.0    | Q3            |
| TL071HIDR     | SOIC         | D               | 8    | 3000 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL071IDR      | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL071IDR1G4   | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072ACDR     | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072ACDR     | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072BCDR     | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072CDR      | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072CDR      | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072CPSR     | SO           | PS              | 8    | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |

| Device      | Package Type | Package Drawing | Pins | SPQ  | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|-------------|--------------|-----------------|------|------|--------------------|--------------------|---------|---------|---------|---------|--------|---------------|
| TL072CPWR   | TSSOP        | PW              | 8    | 2000 | 330.0              | 12.4               | 7.0     | 3.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL072HIDDFR | SOT-23-THIN  | DDF             | 8    | 3000 | 180.0              | 8.4                | 3.2     | 3.2     | 1.4     | 4.0     | 8.0    | Q3            |
| TL072HIDR   | SOIC         | D               | 8    | 3000 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072HIPWR  | TSSOP        | PW              | 8    | 3000 | 330.0              | 12.4               | 7.0     | 3.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL072IDR    | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL072IDR    | SOIC         | D               | 8    | 2500 | 330.0              | 12.4               | 6.4     | 5.2     | 2.1     | 8.0     | 12.0   | Q1            |
| TL074ACDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074ACDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074ACNSR  | SOP          | NS              | 14   | 2000 | 330.0              | 16.4               | 8.1     | 10.4    | 2.5     | 12.0    | 16.0   | Q1            |
| TL074BCDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074BCDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074CDBR   | SSOP         | DB              | 14   | 2000 | 330.0              | 16.4               | 8.35    | 6.6     | 2.4     | 12.0    | 16.0   | Q1            |
| TL074CDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074CDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074CDRG4  | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074CDRG4  | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074CNSR   | SOP          | NS              | 14   | 2000 | 330.0              | 16.4               | 8.45    | 10.55   | 2.5     | 12.0    | 16.2   | Q1            |
| TL074CPWR   | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL074CPWR   | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL074HIDR   | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074HIDRG4 | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074HIDYYR | SOT-23-THIN  | DYY             | 14   | 3000 | 330.0              | 12.4               | 4.8     | 3.6     | 1.6     | 8.0     | 12.0   | Q3            |
| TL074HIPWR  | TSSOP        | PW              | 14   | 2000 | 330.0              | 12.4               | 6.9     | 5.6     | 1.6     | 8.0     | 12.0   | Q1            |
| TL074IDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |
| TL074IDR    | SOIC         | D               | 14   | 2500 | 330.0              | 16.4               | 6.5     | 9.0     | 2.1     | 8.0     | 16.0   | Q1            |

## TAPE AND REEL BOX DIMENSIONS

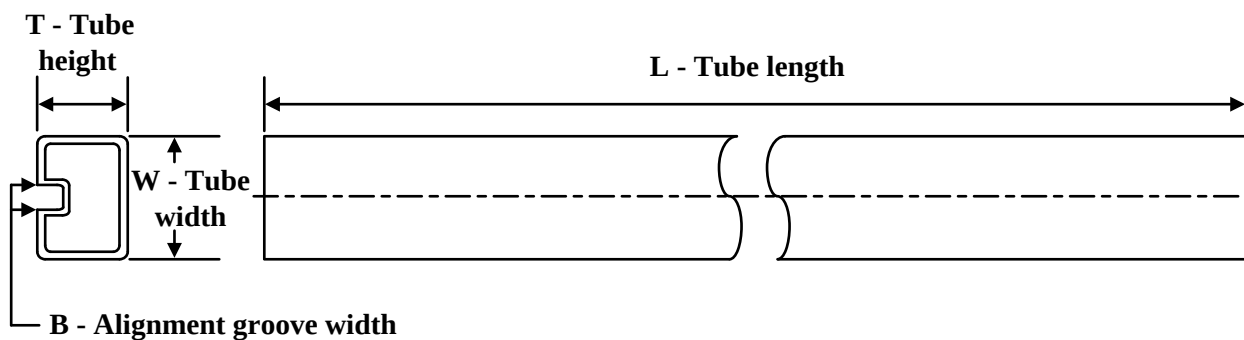


\*All dimensions are nominal

| Device        | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|---------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TL071ACDR     | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL071BCDR     | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL071CDR      | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL071CPSR     | SO           | PS              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL071HIDBVR   | SOT-23       | DBV             | 5    | 3000 | 210.0       | 185.0      | 35.0        |
| TL071HIDBVRG4 | SOT-23       | DBV             | 5    | 3000 | 210.0       | 185.0      | 35.0        |
| TL071HIDCKR   | SC70         | DCK             | 5    | 3000 | 190.0       | 190.0      | 30.0        |
| TL071HIDR     | SOIC         | D               | 8    | 3000 | 353.0       | 353.0      | 32.0        |
| TL071IDR      | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL071IDR1G4   | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072ACDR     | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072ACDR     | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072BCDR     | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072CDR      | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072CDR      | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072CPSR     | SO           | PS              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL072CPWR     | TSSOP        | PW              | 8    | 2000 | 353.0       | 353.0      | 32.0        |
| TL072HIDDFR   | SOT-23-THIN  | DDF             | 8    | 3000 | 210.0       | 185.0      | 35.0        |

| Device      | Package Type | Package Drawing | Pins | SPQ  | Length (mm) | Width (mm) | Height (mm) |
|-------------|--------------|-----------------|------|------|-------------|------------|-------------|
| TL072HIDR   | SOIC         | D               | 8    | 3000 | 353.0       | 353.0      | 32.0        |
| TL072HIPWR  | TSSOP        | PW              | 8    | 3000 | 353.0       | 353.0      | 32.0        |
| TL072IDR    | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL072IDR    | SOIC         | D               | 8    | 2500 | 353.0       | 353.0      | 32.0        |
| TL074ACDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074ACDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074ACNSR  | SOP          | NS              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074BCDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074BCDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074CDBR   | SSOP         | DB              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074CDR    | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074CDR    | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074CDRG4  | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074CDRG4  | SOIC         | D               | 14   | 2500 | 340.5       | 336.1      | 32.0        |
| TL074CNSR   | SOP          | NS              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074CPWR   | TSSOP        | PW              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074CPWR   | TSSOP        | PW              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074HIDR   | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074HIDRG4 | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074HIDYYR | SOT-23-THIN  | DYY             | 14   | 3000 | 336.6       | 336.6      | 31.8        |
| TL074HIPWR  | TSSOP        | PW              | 14   | 2000 | 353.0       | 353.0      | 32.0        |
| TL074IDR    | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |
| TL074IDR    | SOIC         | D               | 14   | 2500 | 353.0       | 353.0      | 32.0        |

## TUBE



\*All dimensions are nominal

| Device      | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|-------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| 81023052A   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| 8102305HA   | U            | CFP          | 10   | 25  | 506.98 | 26.16  | 6220   | NA     |
| 81023062A   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| 8102306DA   | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL071ACP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071ACP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071BCP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071BCP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071CP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071CP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071IP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL071IP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072ACP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072ACP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072ACPS   | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL072ACPS.A | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL072BCP    | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072BCP.A  | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072CP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072CP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072CPS    | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL072CPS.A  | PS           | SOP          | 8    | 80  | 530    | 10.5   | 4000   | 4.1    |
| TL072IP     | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072IP.A   | P            | PDIP         | 8    | 50  | 506    | 13.97  | 11230  | 4.32   |
| TL072MFKB   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL072MFKB.A | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL072MUB    | U            | CFP          | 10   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL072MUB.A  | U            | CFP          | 10   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL074ACN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |

| Device      | Package Name | Package Type | Pins | SPQ | L (mm) | W (mm) | T (μm) | B (mm) |
|-------------|--------------|--------------|------|-----|--------|--------|--------|--------|
| TL074ACN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074ACN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074ACN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074BCN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074BCN    | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074BCN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074BCN.A  | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074CN     | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074CN.A   | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074IN     | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074IN.A   | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074ING4   | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074ING4.A | N            | PDIP         | 14   | 25  | 506    | 13.97  | 11230  | 4.32   |
| TL074MFK    | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL074MFK.A  | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL074MFKB   | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL074MFKB.A | FK           | LCCC         | 20   | 55  | 506.98 | 12.06  | 2030   | NA     |
| TL074MWB    | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |
| TL074MWB.A  | W            | CFP          | 14   | 25  | 506.98 | 26.16  | 6220   | NA     |

**JG0008A**

CERAMIC DUAL IN-LINE PACKAGE



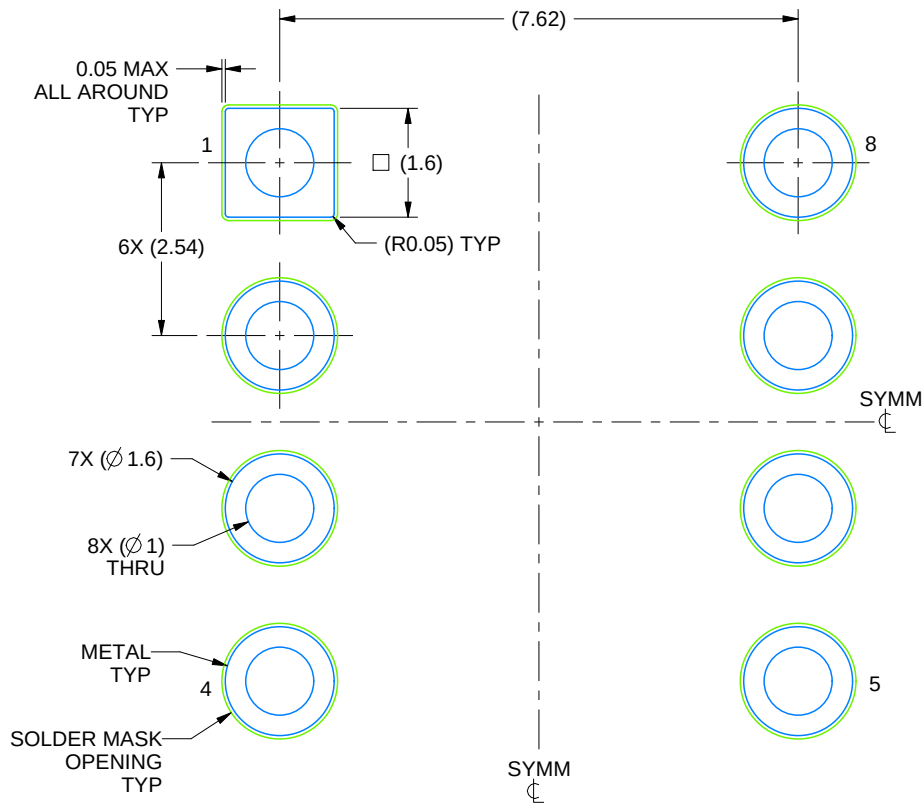
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package can be hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification.
5. Falls within MIL STD 1835 GDIP1-T8

# EXAMPLE BOARD LAYOUT

JG0008A

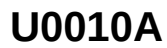
CDIP - 5.08 mm max height

CERAMIC DUAL IN-LINE PACKAGE



LAND PATTERN EXAMPLE  
NON SOLDER MASK DEFINED  
SCALE: 9X

4230036/A 09/2023



**CFP - 2.03 mm max height**

.045 MAX TYP  
 .010 ± .002  
 PIN 1 ID  
 .27 MAX GLASS  
 .005 MIN TYP  
 1  
 10  
 .27 MAX GLASS  
 5  
 6  
 8X .050 ± .005  
 10X .017 ± .002  
 5X .32 ± .01  
 .241 <sup>+0.019</sup>/<sub>-.003</sub>  
 .005 ± .001  
 .067 <sup>+0.013</sup>/<sub>-.012</sub>  
 .045  
 .026



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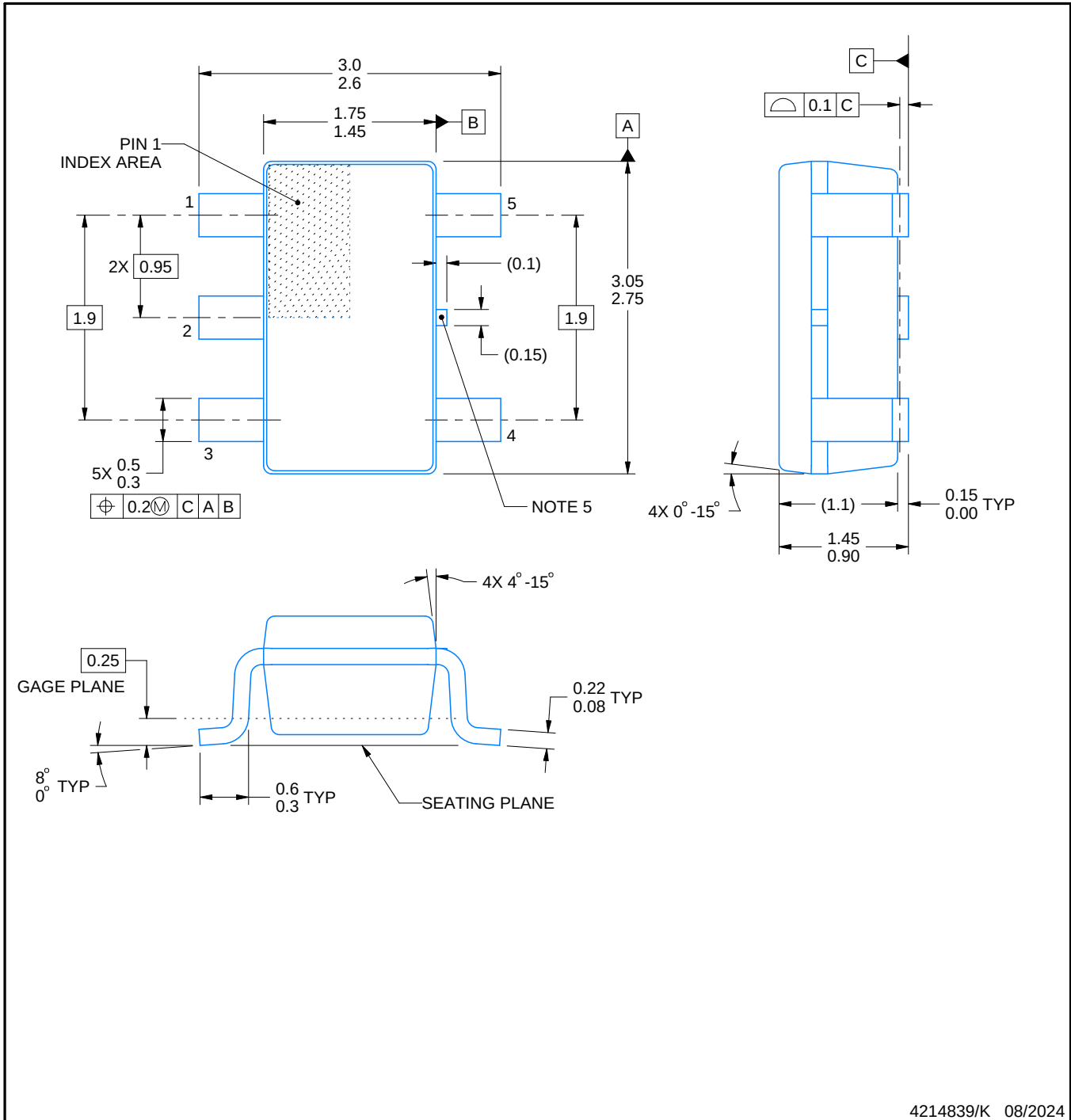


**DBV0005A**

# PACKAGE OUTLINE

**SOT-23 - 1.45 mm max height**

SMALL OUTLINE TRANSISTOR



## NOTES:

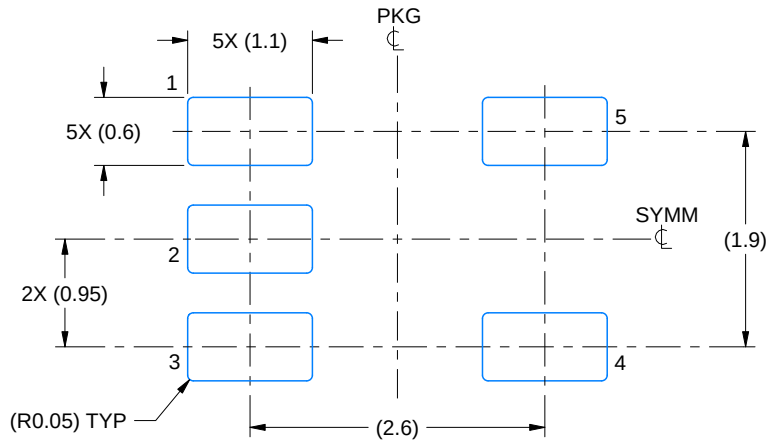
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC MO-178.
4. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25 mm per side.
5. Support pin may differ or may not be present.

# EXAMPLE BOARD LAYOUT

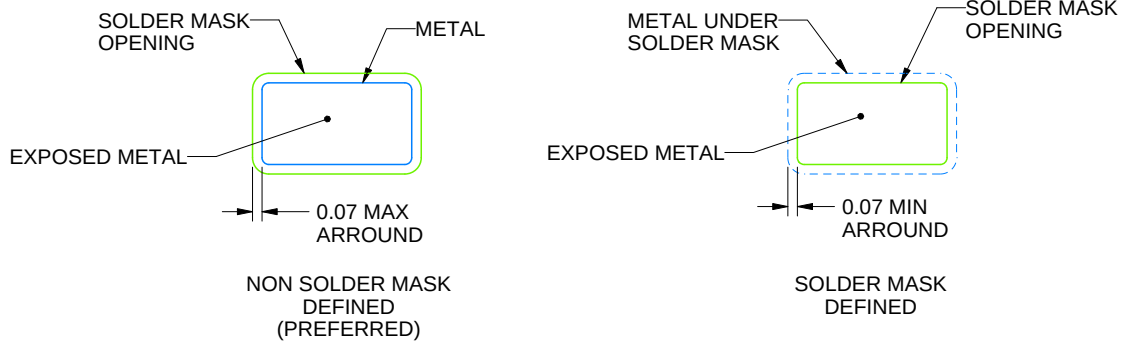
DBV0005A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:15X



SOLDER MASK DETAILS

4214839/K 08/2024

NOTES: (continued)

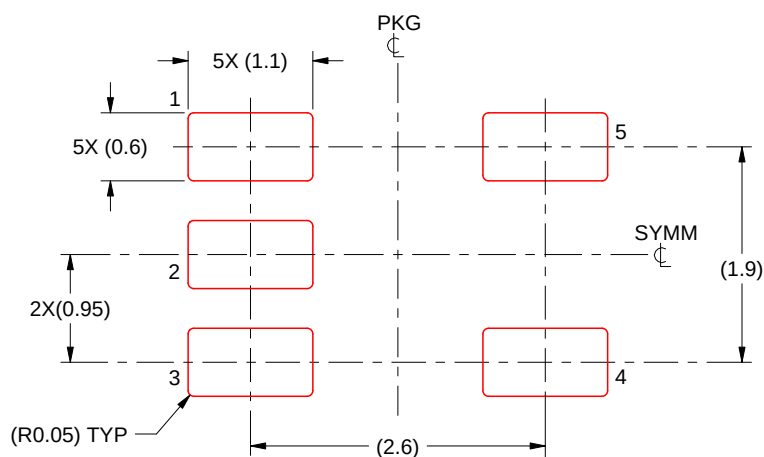
6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

DBV0005A

SOT-23 - 1.45 mm max height

SMALL OUTLINE TRANSISTOR

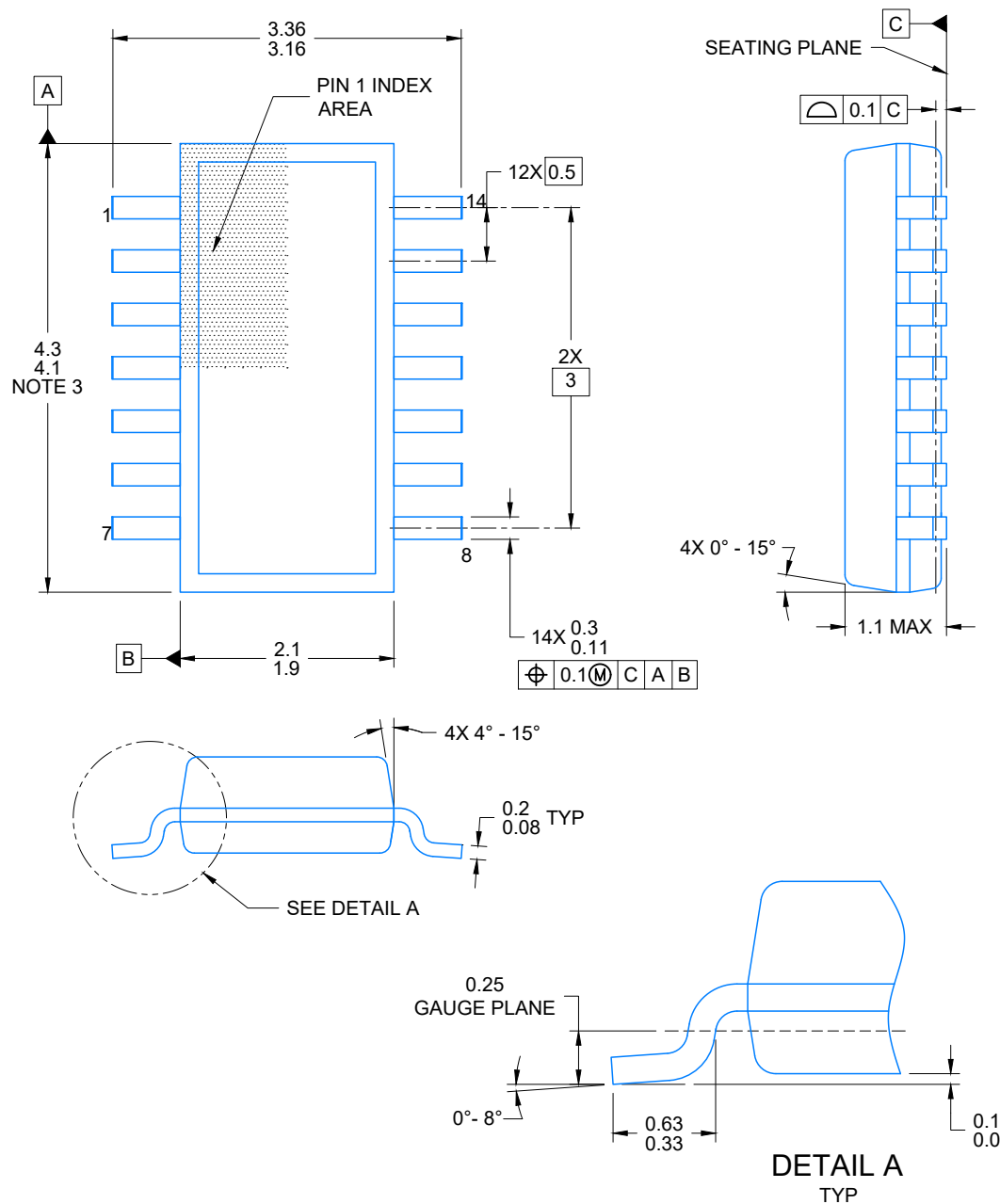


SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:15X

4214839/K 08/2024

NOTES: (continued)

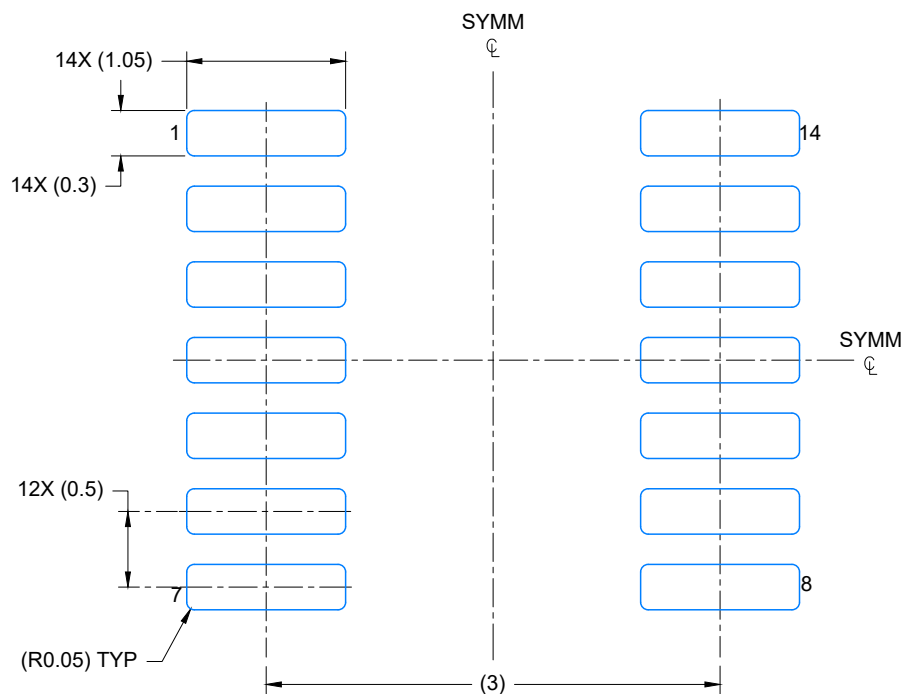
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



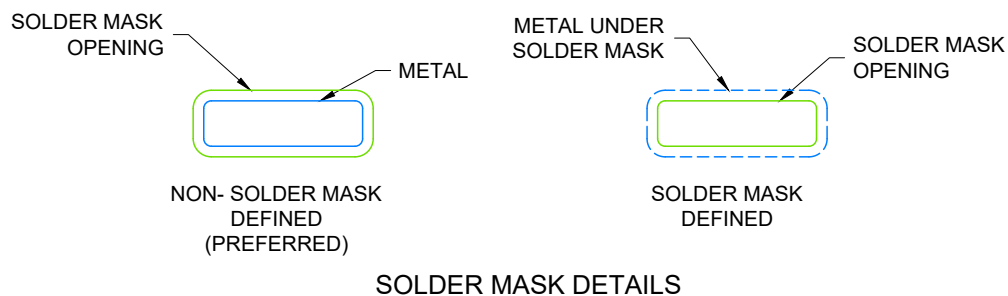
4224643/D 07/2024

NOTES:

1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.50 per side.
5. Reference JEDEC Registration MO-345, Variation AB



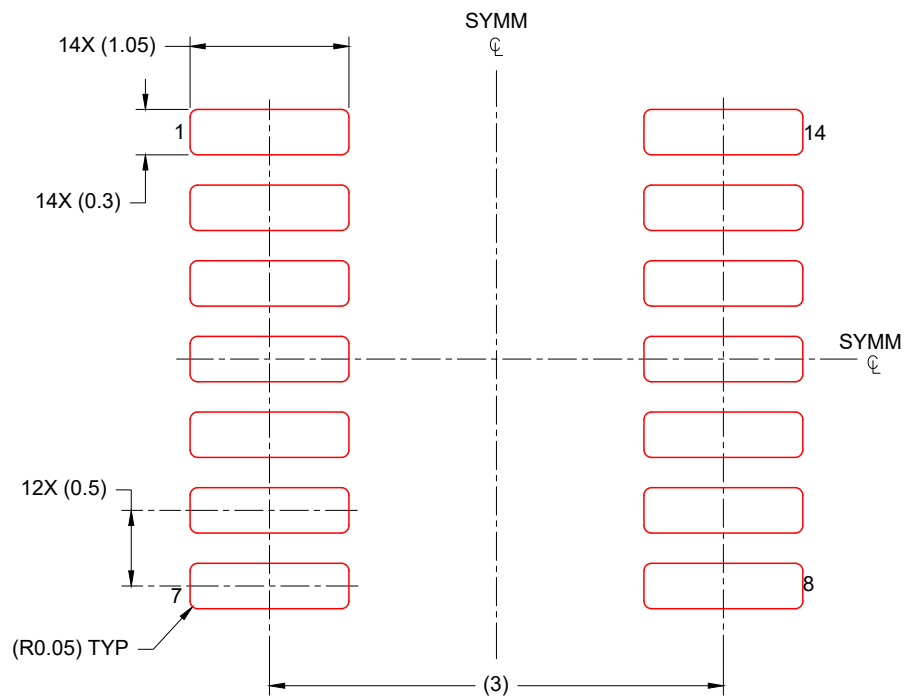
LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 20X



4224643/D 07/2024

## NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

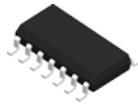


SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 20X

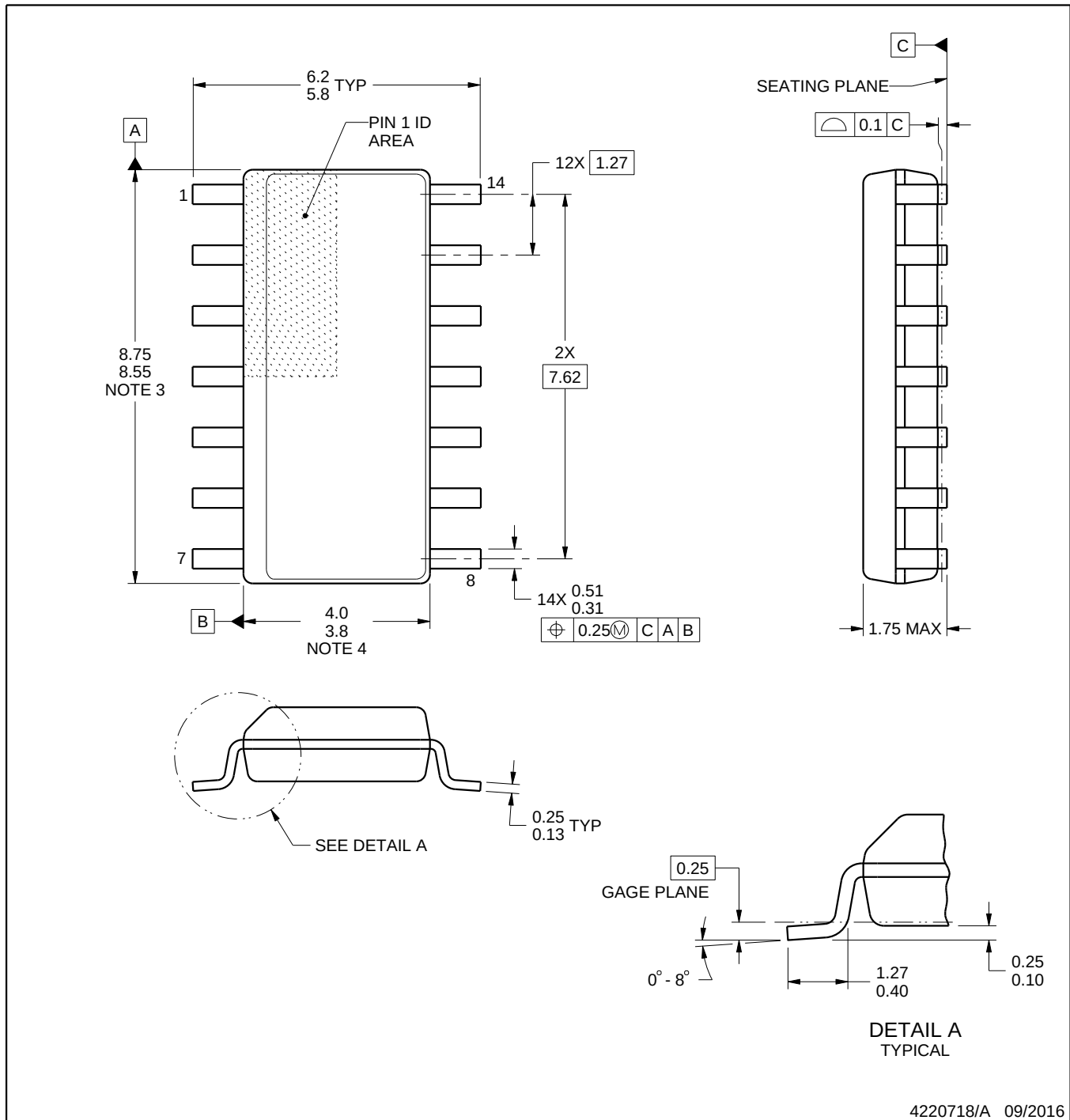
4224643/D 07/2024

## NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

**D0014A****PACKAGE OUTLINE****SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT

**NOTES:**

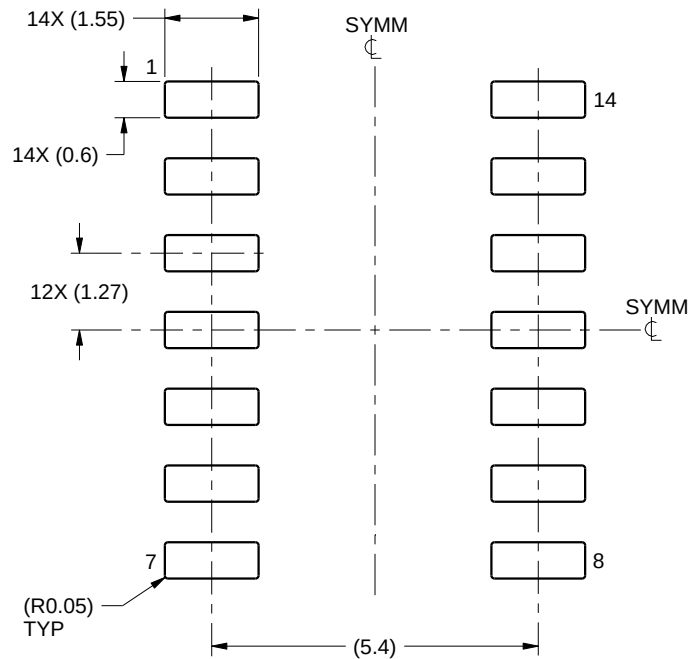
1. All linear dimensions are in millimeters. Dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm, per side.
4. This dimension does not include interlead flash. Interlead flash shall not exceed 0.43 mm, per side.
5. Reference JEDEC registration MS-012, variation AB.

# EXAMPLE BOARD LAYOUT

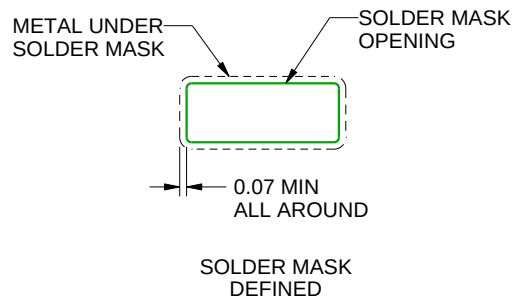
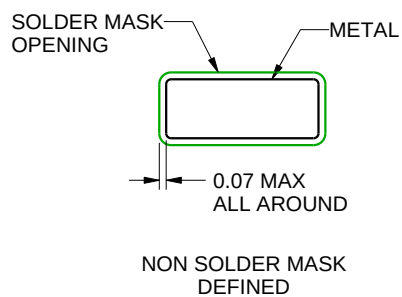
D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



LAND PATTERN EXAMPLE  
SCALE:8X



SOLDER MASK DETAILS

4220718/A 09/2016

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

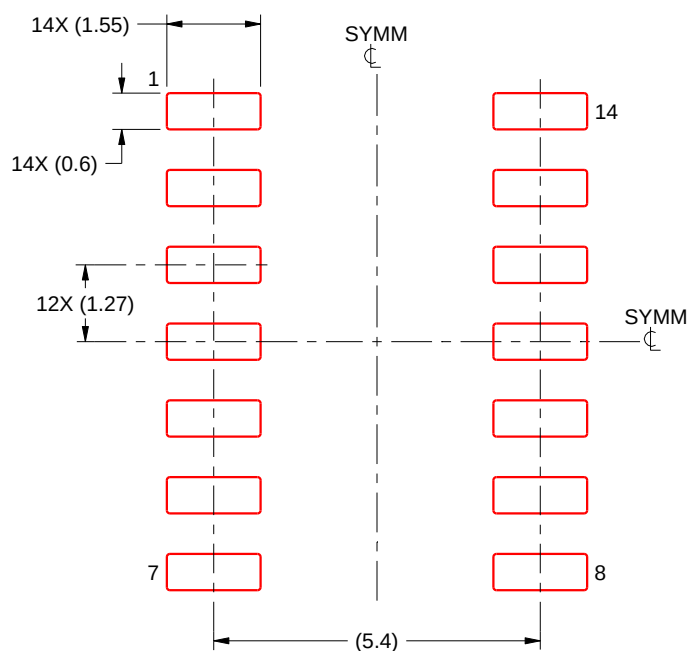
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

D0014A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:8X

4220718/A 09/2016

NOTES: (continued)

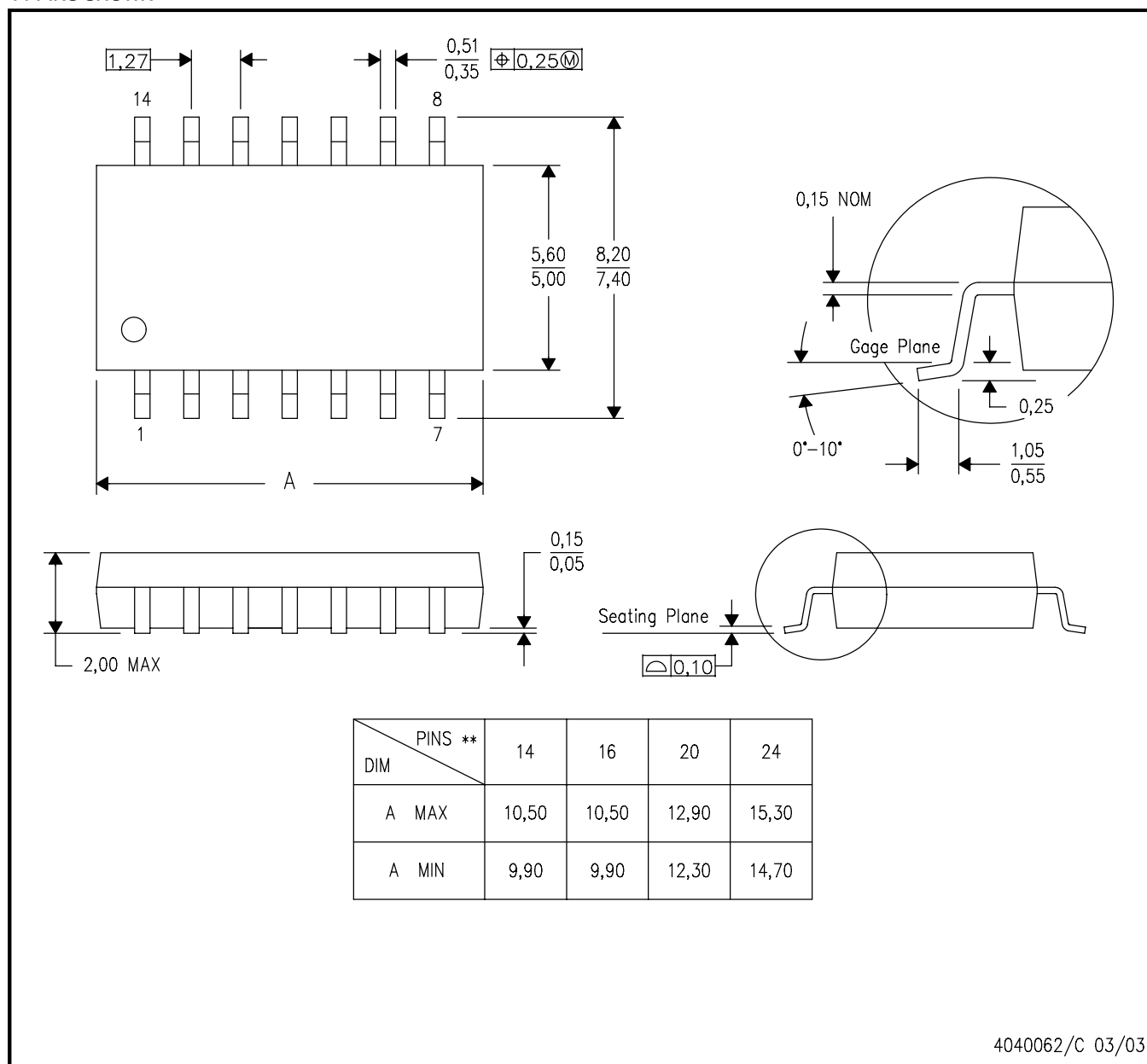
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

# MECHANICAL DATA

NS (R-PDSO-G\*\*)

PLASTIC SMALL-OUTLINE PACKAGE

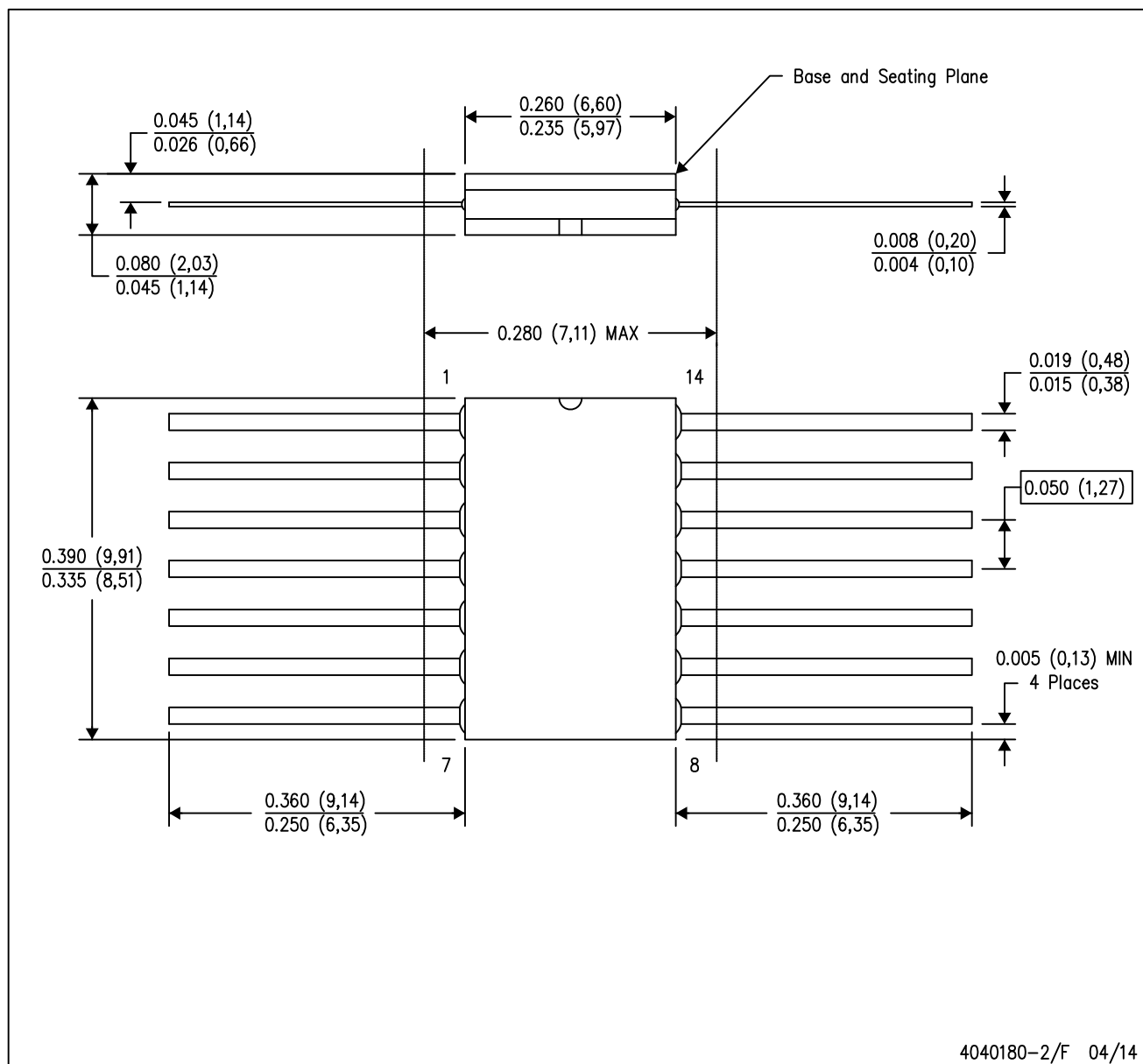
14-PINS SHOWN

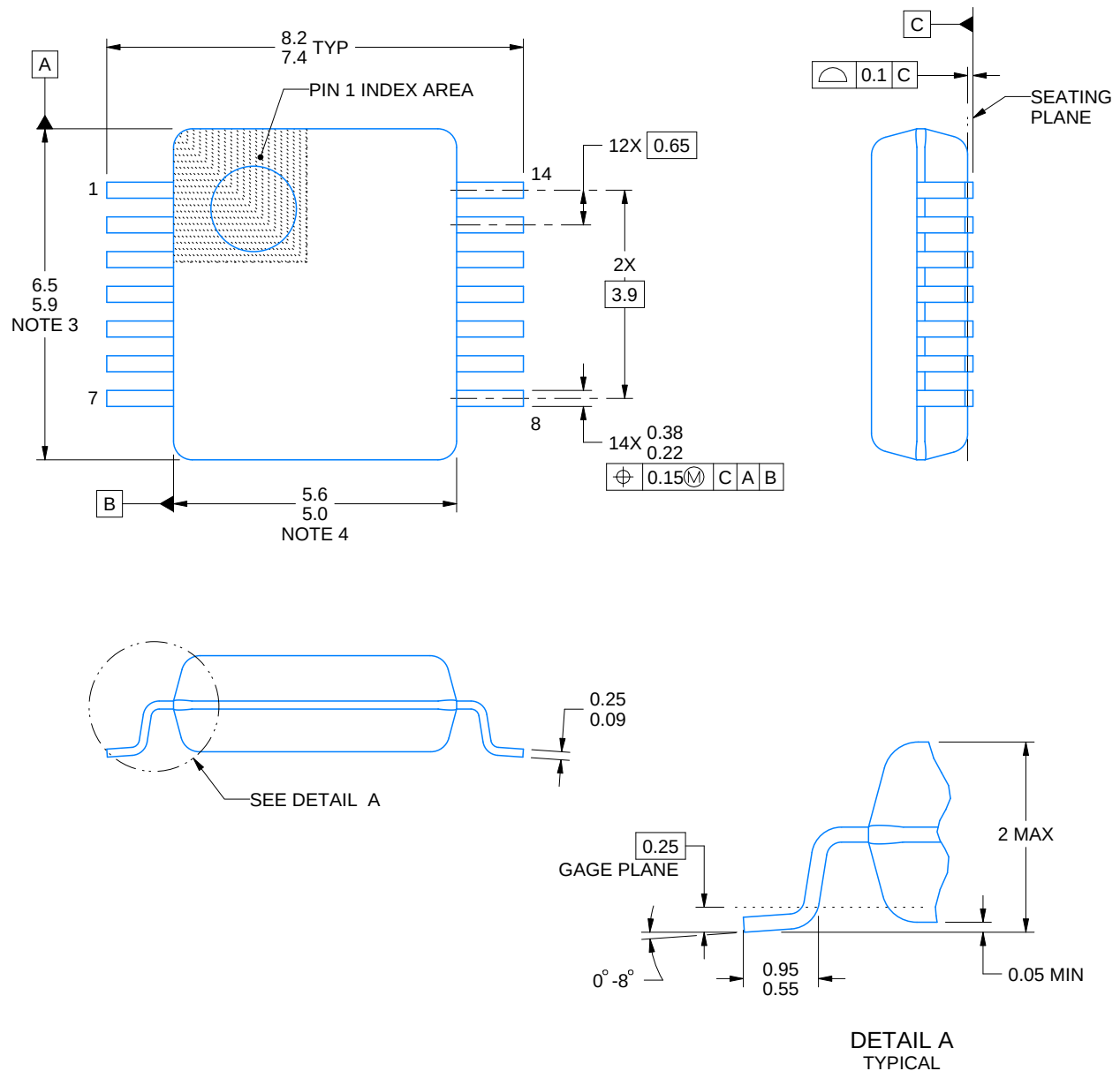
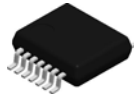


- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

W (R-GDFP-F14)

CERAMIC DUAL FLATPACK





4220762/A 05/2024

## NOTES:

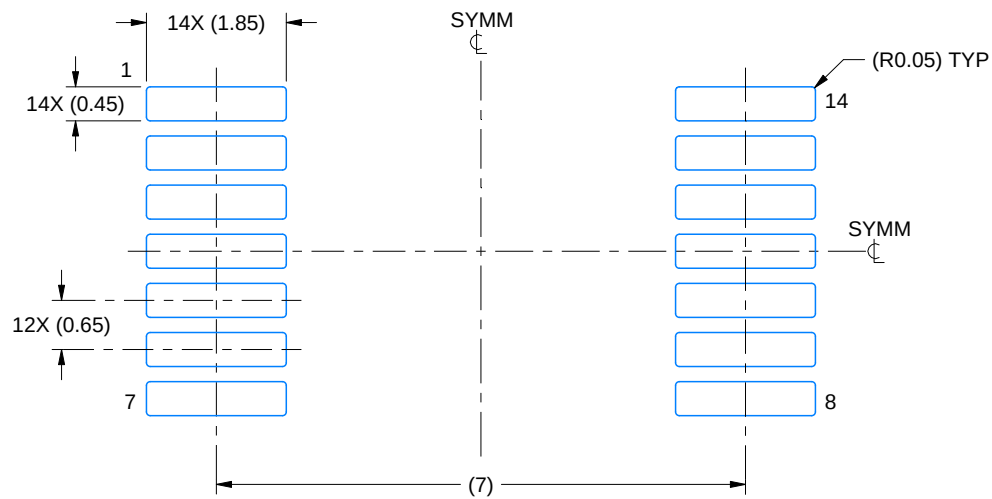
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.
4. Reference JEDEC registration MO-150.

# EXAMPLE BOARD LAYOUT

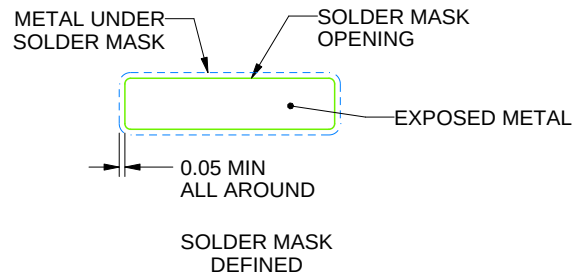
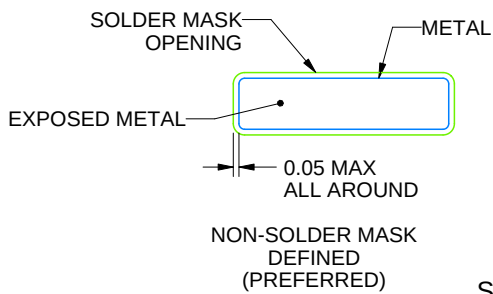
DB0014A

SSOP - 2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220762/A 05/2024

NOTES: (continued)

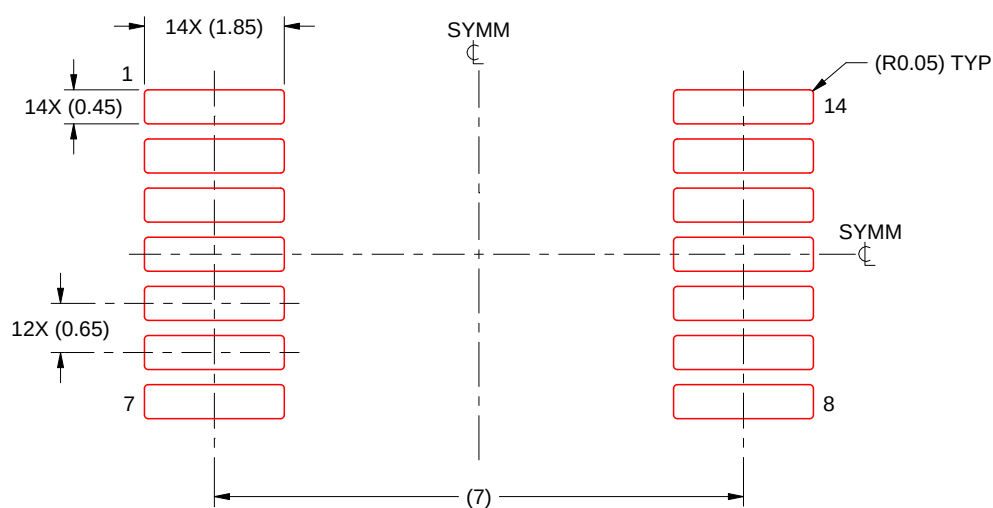
- Publication IPC-7351 may have alternate designs.
- Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

DB0014A

SSOP - 2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220762/A 05/2024

NOTES: (continued)

7. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
8. Board assembly site may have different recommendations for stencil design.

## GENERIC PACKAGE VIEW

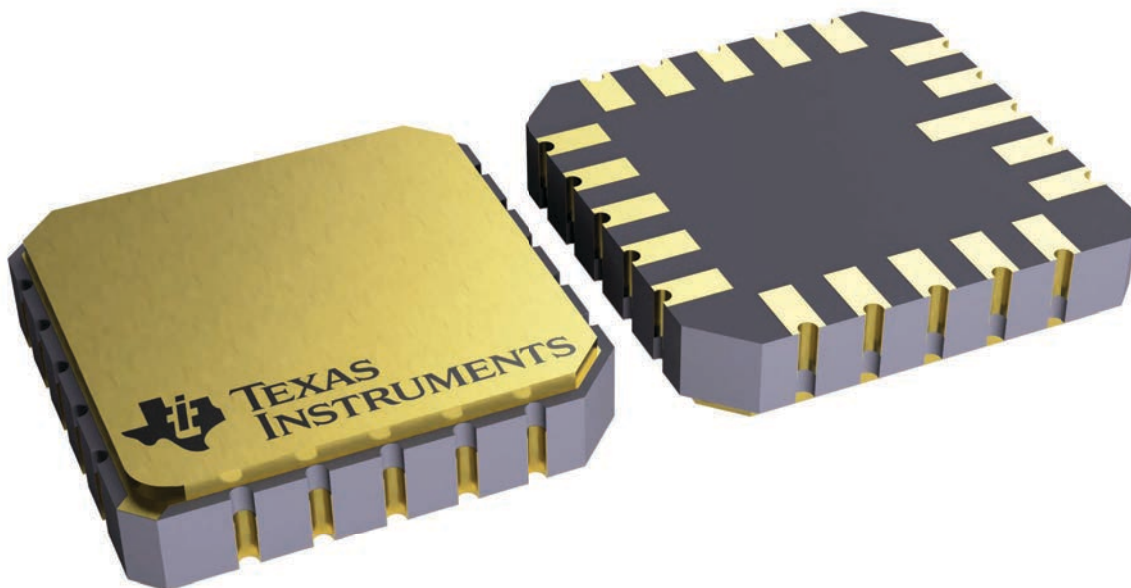
**FK 20**

**LCCC - 2.03 mm max height**

8.89 x 8.89, 1.27 mm pitch

LEADLESS CERAMIC CHIP CARRIER

This image is a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.



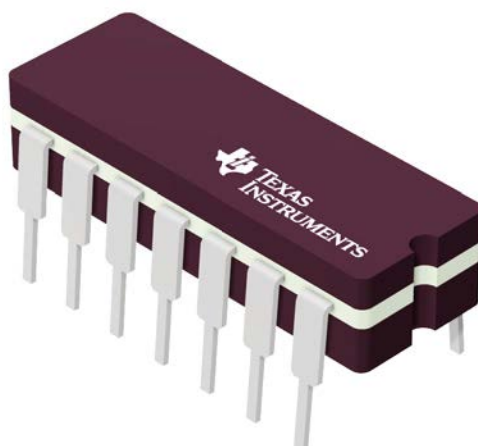
4229370VA\

**J 14**

## GENERIC PACKAGE VIEW

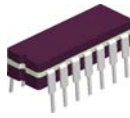
**CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE

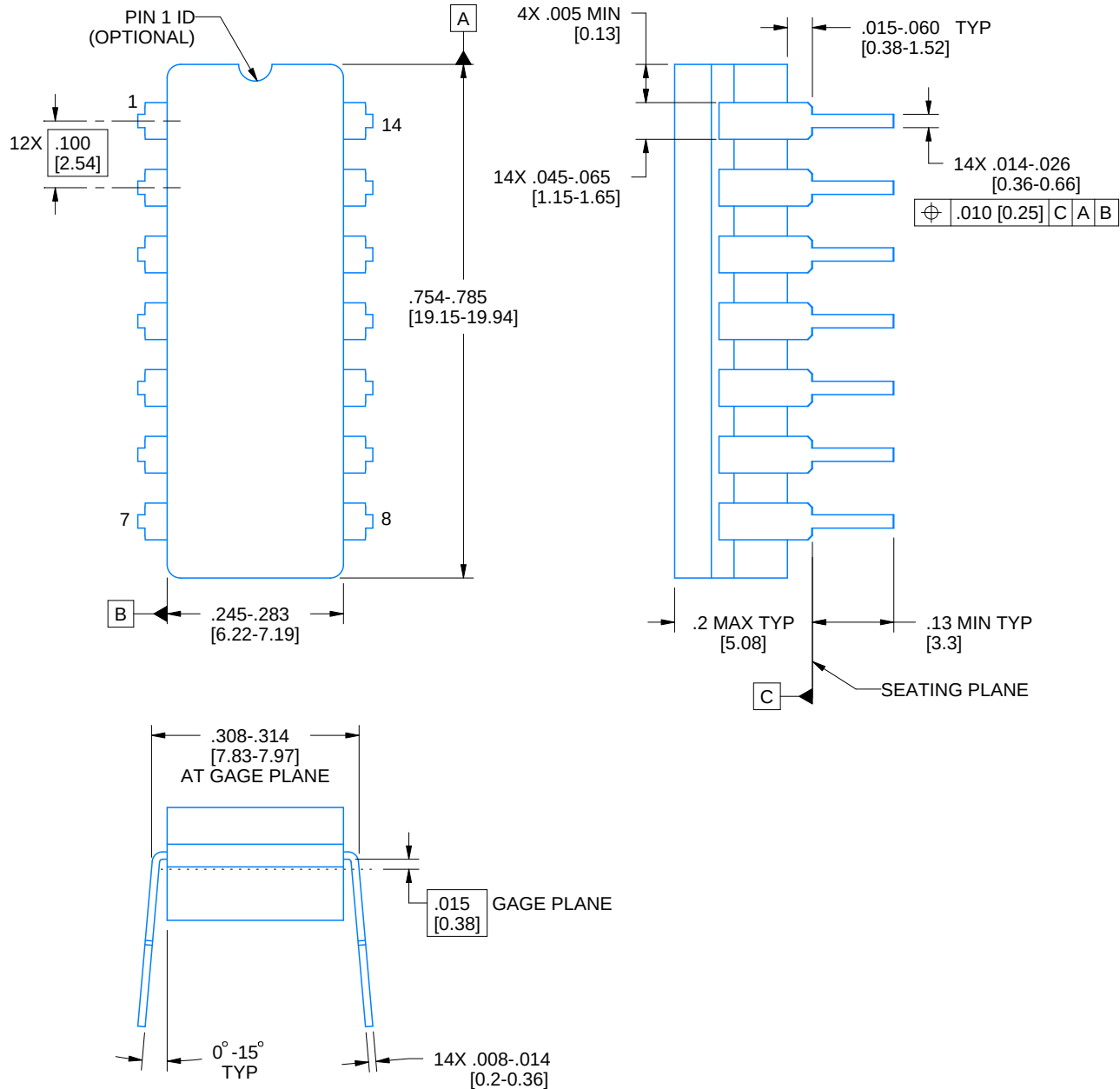


Images above are just a representation of the package family, actual package may vary.  
Refer to the product data sheet for package details.

4040083-5/G

**J0014A****PACKAGE OUTLINE****CDIP - 5.08 mm max height**

CERAMIC DUAL IN LINE PACKAGE



4214771/A 05/2017

**NOTES:**

1. All controlling linear dimensions are in inches. Dimensions in brackets are in millimeters. Any dimension in brackets or parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This package is hermetically sealed with a ceramic lid using glass frit.
4. Index point is provided on cap for terminal identification only and on press ceramic glass frit seal only.
5. Falls within MIL-STD-1835 and GDIP1-T14.



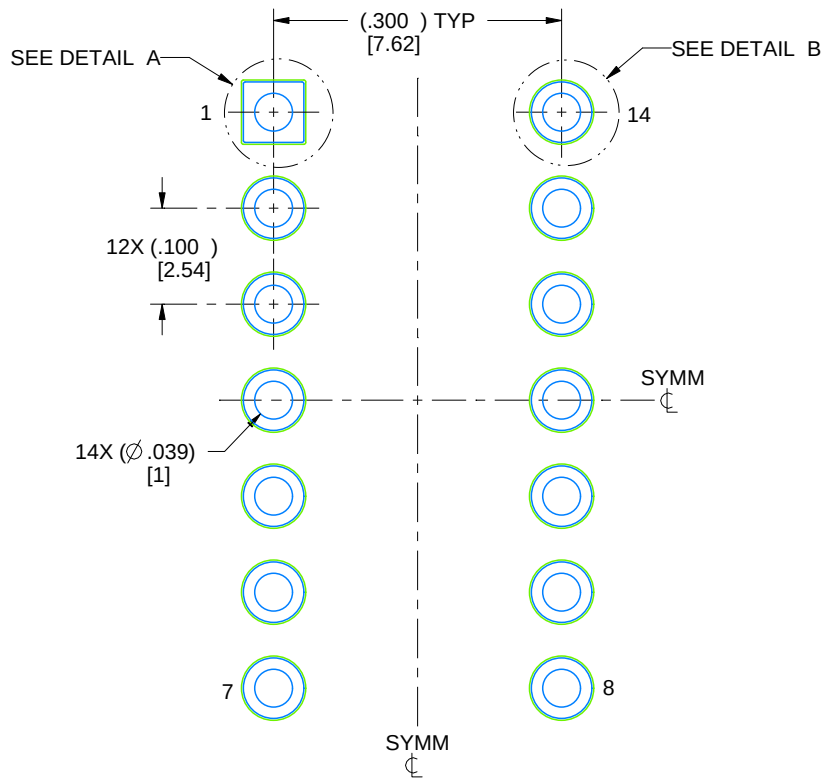
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# EXAMPLE BOARD LAYOUT

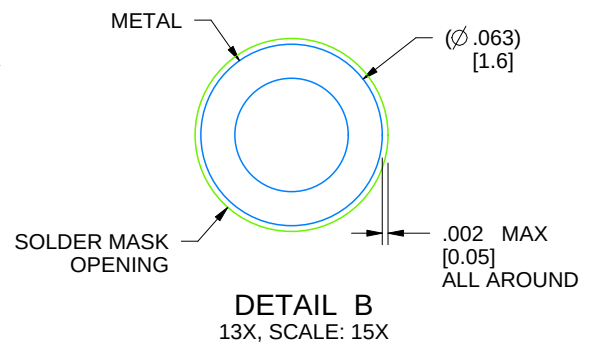
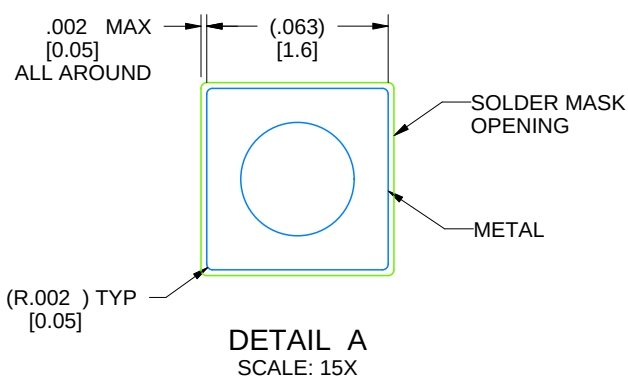
J0014A

CDIP - 5.08 mm max height

CERAMIC DUAL IN LINE PACKAGE



LAND PATTERN EXAMPLE  
NON-SOLDER MASK DEFINED  
SCALE: 5X



4214771/A 05/2017



### SOT - 1.1 max height

Technical drawing of a mechanical part showing three views: front, side, and top.

**Front View Dimensions:**

- Overall width: 2.4 (1.8)
- Overall height: 2.15 (1.85)
- Horizontal hole locations: 1.4 (1.1)
- Vertical hole locations: 1.3 (1.3)
- Hole sizes: 0.15 (0.1)
- Feature: NOTE 4
- Feature: PIN 1 INDEX AREA (hatched area)
- Feature: 2X 0.65
- Feature: 5X 0.33 0.15

**Side View Dimensions:**

- Width: 1.1 MAX
- Thickness: 0.1 C

**Top View Dimensions:**

- Width: 0.46 TYP
- Height: 0.22 TYP
- Feature: 4X 4° -15°
- Feature: 8° 0° TYP
- Feature: 0.22 0.08 TYP
- Feature: GAGE PLANE
- Feature: SEATING PLANE

**Surface Texture Symbols:**

- Front View: 0.1 (0.05) A B
- Top View: 0.1 (0.05) C A B

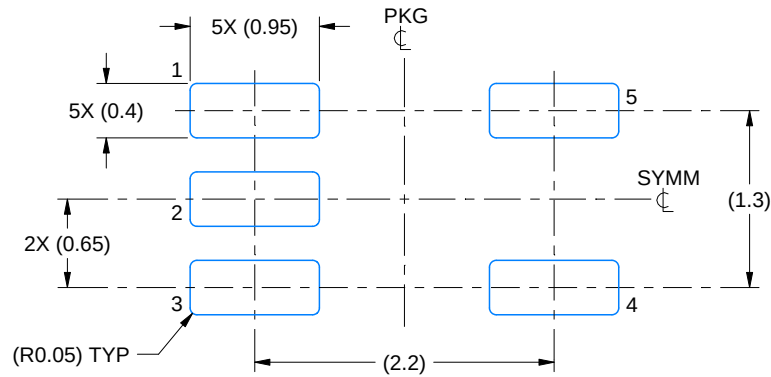
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. Reference JEDEC MO-203.
4. Support pin may differ or may not be present.
5. Lead width does not comply with JEDEC.
6. Body dimensions do not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.25mm per side

# EXAMPLE BOARD LAYOUT

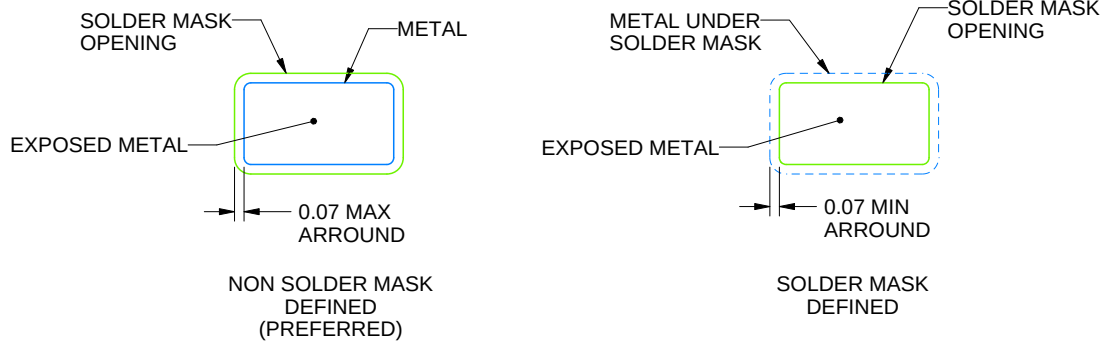
DCK0005A

SOT - 1.1 max height

SMALL OUTLINE TRANSISTOR



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:18X

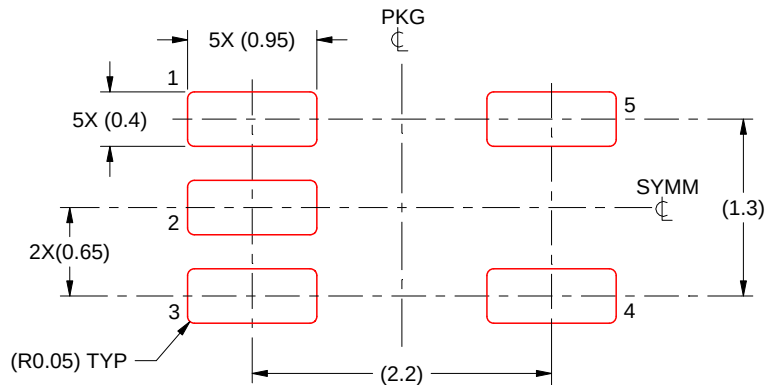


SOLDER MASK DETAILS

4214834/G 11/2024

NOTES: (continued)

7. Publication IPC-7351 may have alternate designs.
8. Solder mask tolerances between and around signal pads can vary based on board fabrication site.



SOLDER PASTE EXAMPLE  
BASED ON 0.125 THICK STENCIL  
SCALE:18X

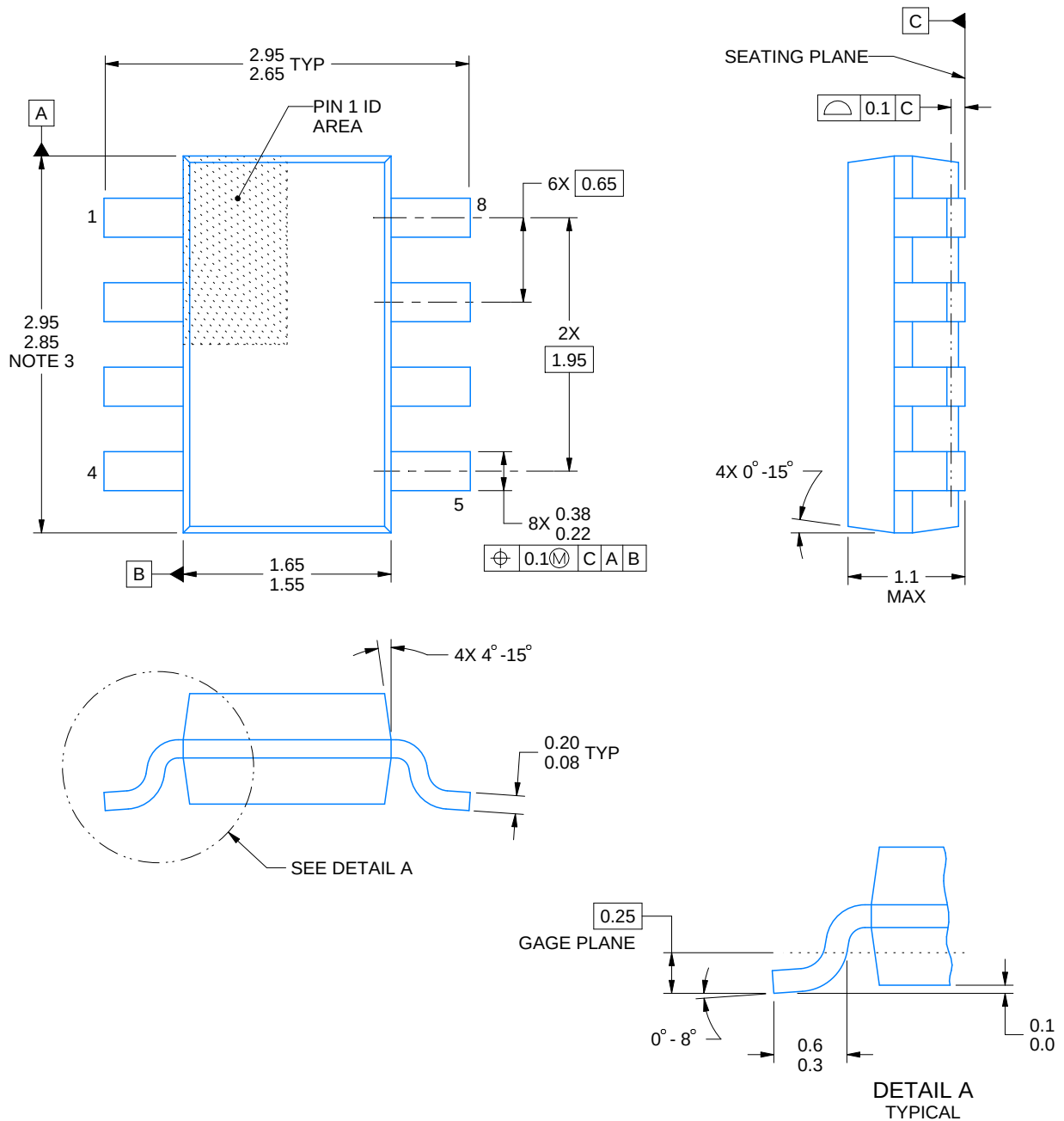
4214834/G 11/2024

NOTES: (continued)

9. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
10. Board assembly site may have different recommendations for stencil design.

**DDF0008A****PACKAGE OUTLINE****SOT-23-THIN - 1.1 mm max height**

PLASTIC SMALL OUTLINE



4222047/E 07/2024

**NOTES:**

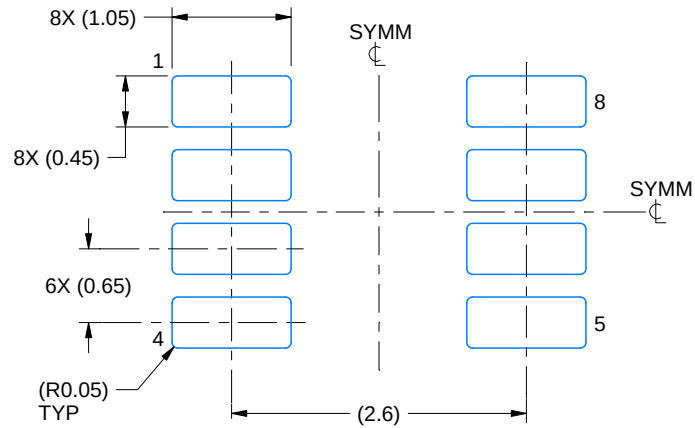
1. All linear dimensions are in millimeters. Any dimensions in parenthesis are for reference only. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed 0.15 mm per side.

# EXAMPLE BOARD LAYOUT

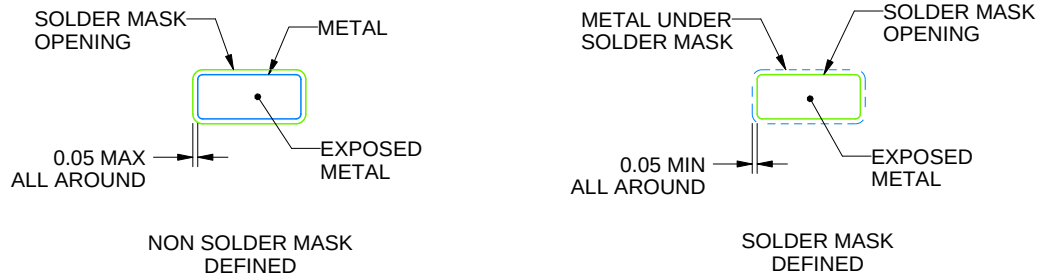
DDF0008A

SOT-23-THIN - 1.1 mm max height

PLASTIC SMALL OUTLINE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE:15X



SOLDER MASK DETAILS

4222047/E 07/2024

NOTES: (continued)

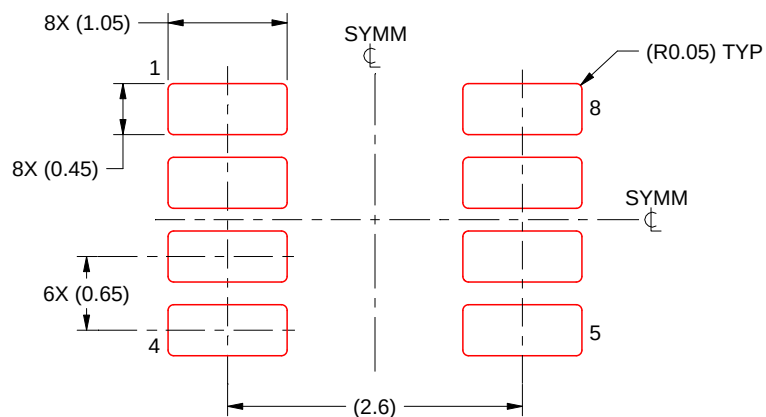
4. Publication IPC-7351 may have alternate designs.
5. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

DDF0008A

SOT-23-THIN - 1.1 mm max height

PLASTIC SMALL OUTLINE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:15X

4222047/E 07/2024

NOTES: (continued)

6. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
7. Board assembly site may have different recommendations for stencil design.



## PACKAGE OUTLINE

**SOIC - 1.75 mm max height**

SMALL OUTLINE INTEGRATED CIRCUIT



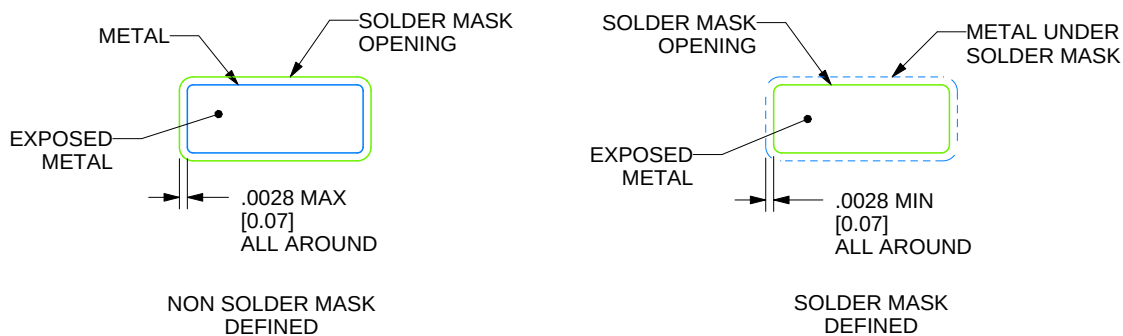
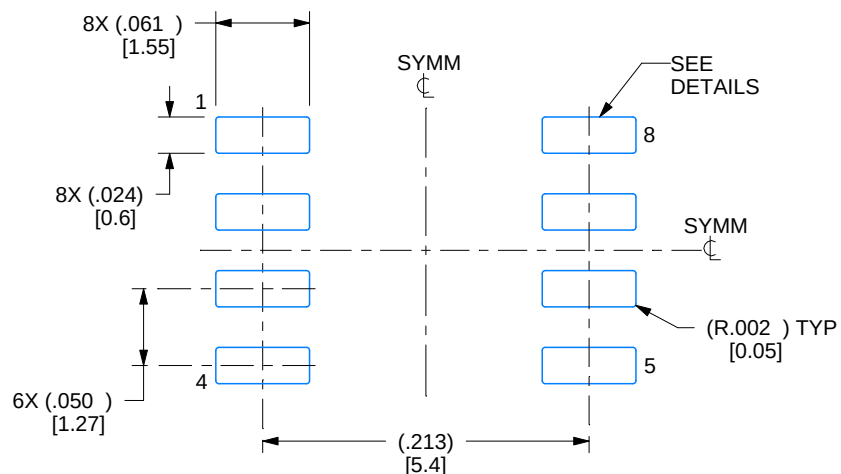
1. Linear dimensions are in inches [millimeters]. Dimensions in parenthesis are for reference only. Controlling dimensions are in inches. Dimensioning and tolerancing per ASME Y14.5M.
2. This drawing is subject to change without notice.
3. This dimension does not include mold flash, protrusions, or gate burrs. Mold flash, protrusions, or gate burrs shall not exceed .006 [0.15] per side.
4. This dimension does not include interlead flash.
5. Reference JEDEC registration MS-012, variation AA.

# EXAMPLE BOARD LAYOUT

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



4214825/C 02/2019

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

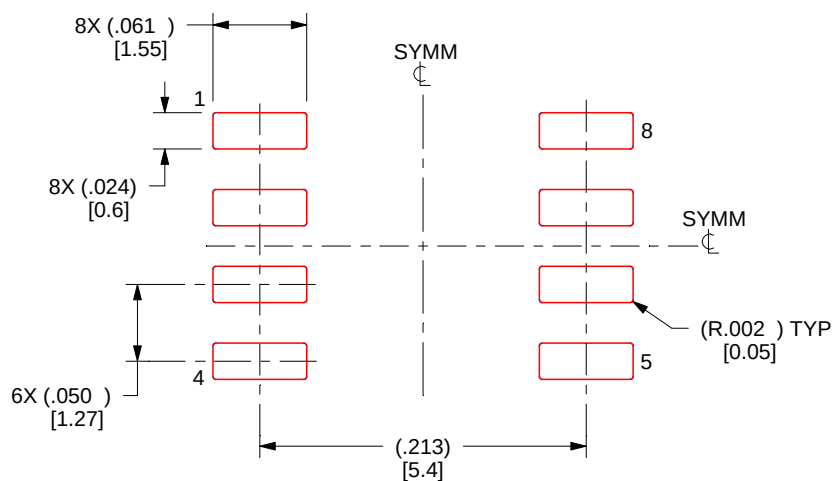
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

D0008A

SOIC - 1.75 mm max height

SMALL OUTLINE INTEGRATED CIRCUIT



SOLDER PASTE EXAMPLE  
BASED ON .005 INCH [0.125 MM] THICK STENCIL  
SCALE:8X

4214825/C 02/2019

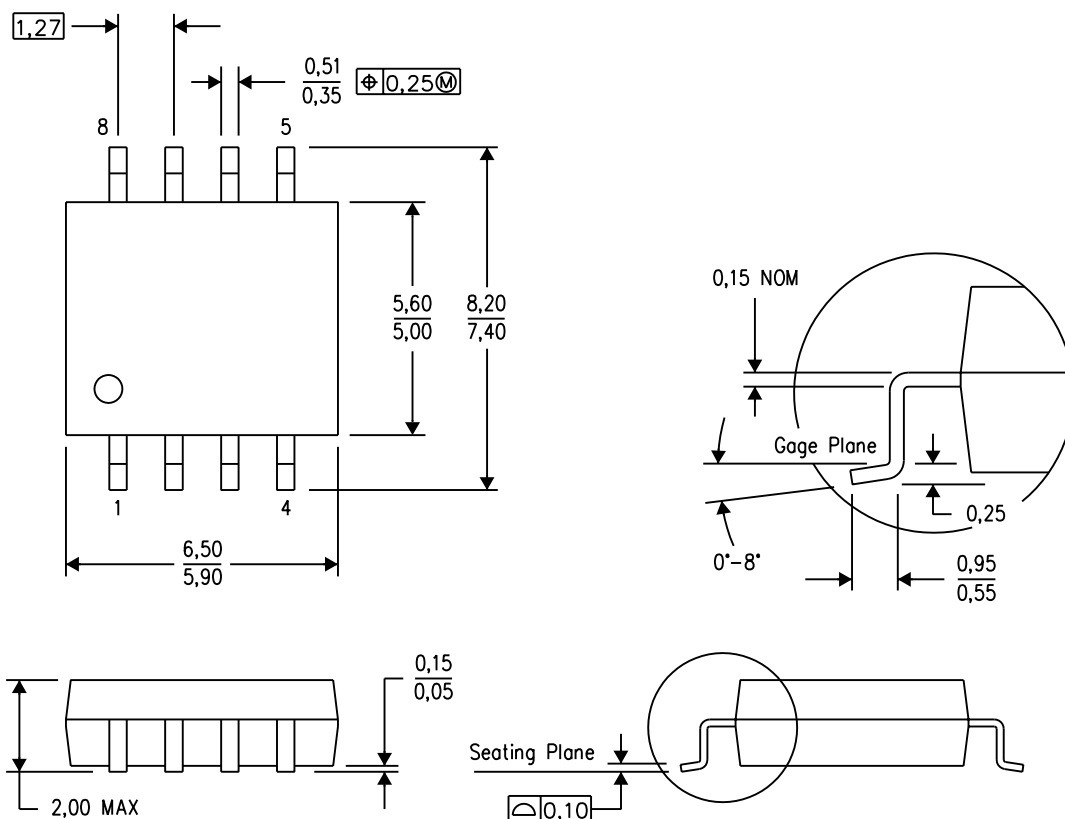
NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

## MECHANICAL DATA

PS (R-PDSO-G8)

PLASTIC SMALL-OUTLINE PACKAGE

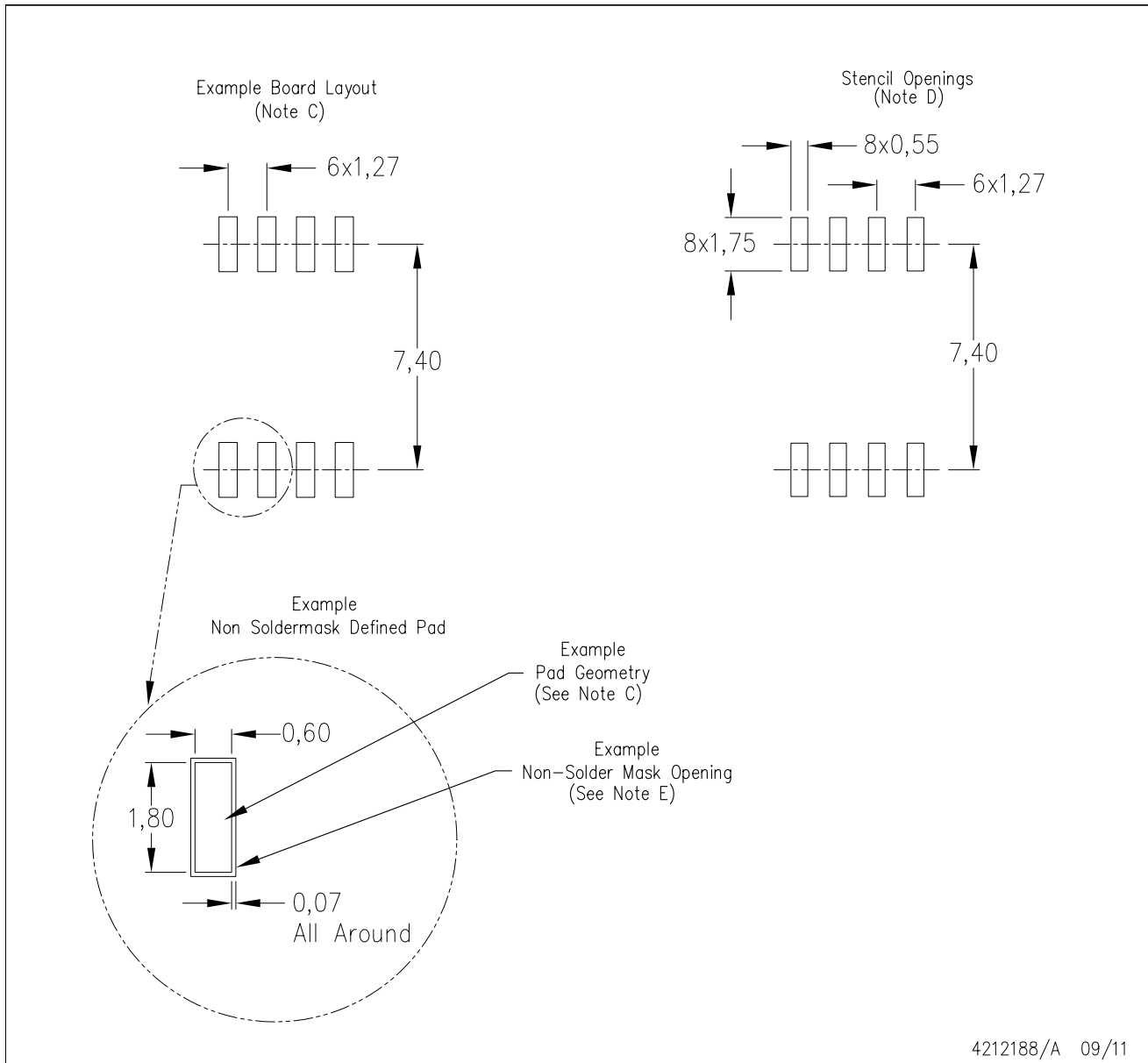


4040063/C 03/03

- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Body dimensions do not include mold flash or protrusion, not to exceed 0,15.

PS (R-PDSO-G8)

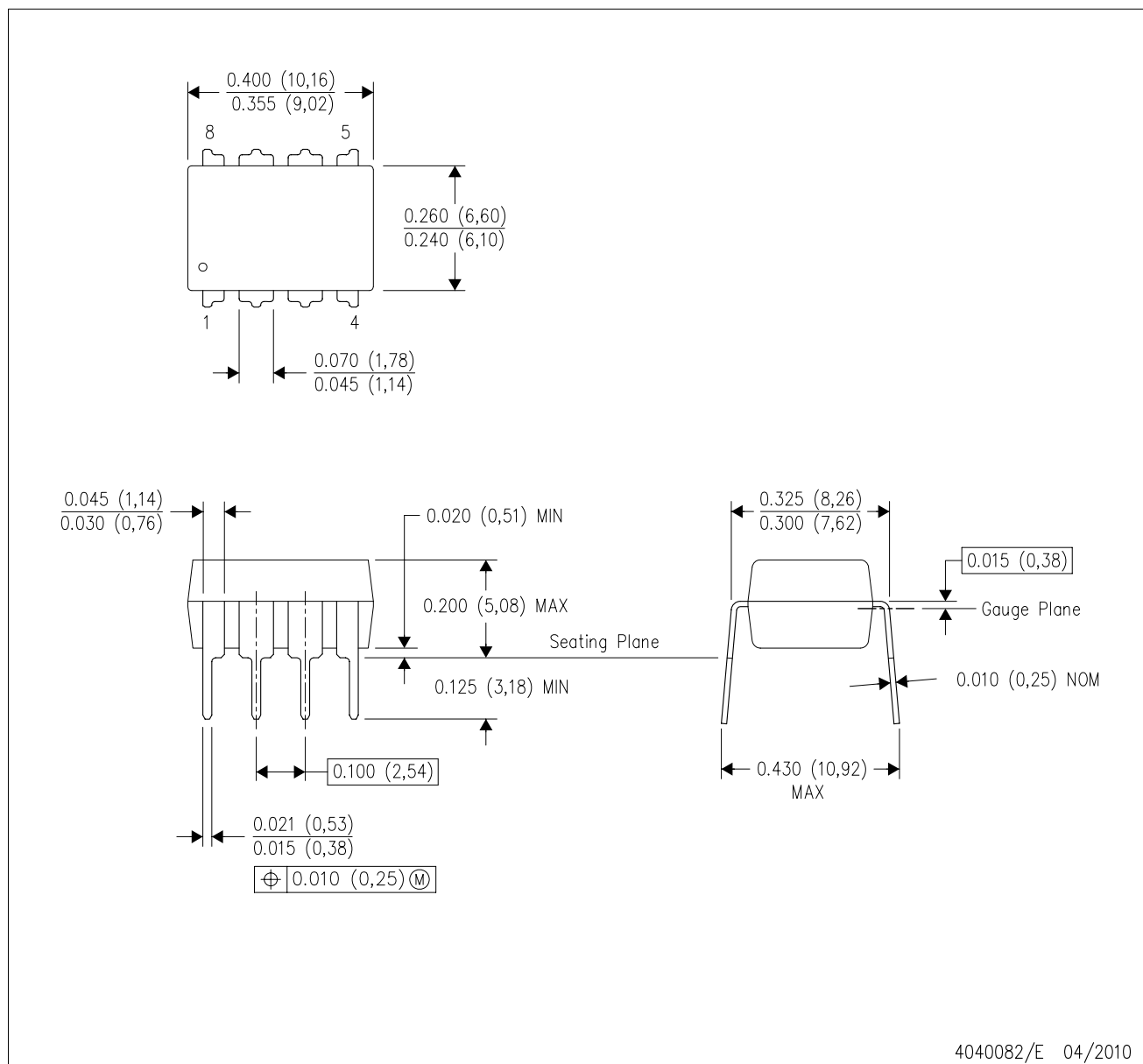
PLASTIC SMALL OUTLINE



- NOTES:
- A. All linear dimensions are in millimeters.
  - B. This drawing is subject to change without notice.
  - C. Publication IPC-7351 is recommended for alternate designs.
  - D. Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC-7525 for other stencil recommendations.
  - E. Customers should contact their board fabrication site for solder mask tolerances between and around signal pads.

P (R-PDIP-T8)

PLASTIC DUAL-IN-LINE PACKAGE

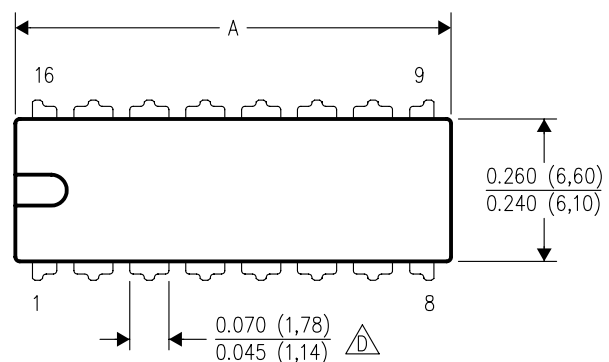


- NOTES:
- A. All linear dimensions are in inches (millimeters).
  - B. This drawing is subject to change without notice.
  - C. Falls within JEDEC MS-001 variation BA.

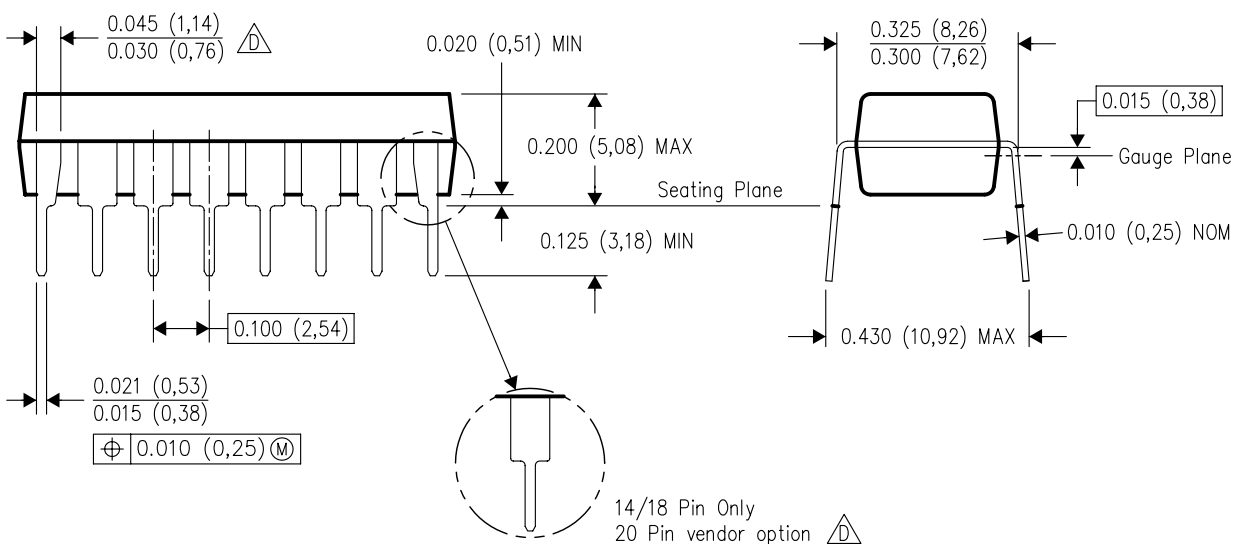
N (R-PDIP-T\*\*)

16 PINS SHOWN

## PLASTIC DUAL-IN-LINE PACKAGE



| PINS **<br>DIM      | 14               | 16               | 18               | 20               |
|---------------------|------------------|------------------|------------------|------------------|
| A MAX               | 0.775<br>(19,69) | 0.775<br>(19,69) | 0.920<br>(23,37) | 1.060<br>(26,92) |
| A MIN               | 0.745<br>(18,92) | 0.745<br>(18,92) | 0.850<br>(21,59) | 0.940<br>(23,88) |
| MS-001<br>VARIATION | AA               | BB               | AC               | AD               |



4040049/E 12/2002

NOTES:

- A. All linear dimensions are in inches (millimeters).  
B. This drawing is subject to change without notice.
-  Falls within JEDEC MS-001, except 18 and 20 pin minimum body length (Dim A).  
 The 20 pin end lead shoulder width is a vendor option, either half or full width.

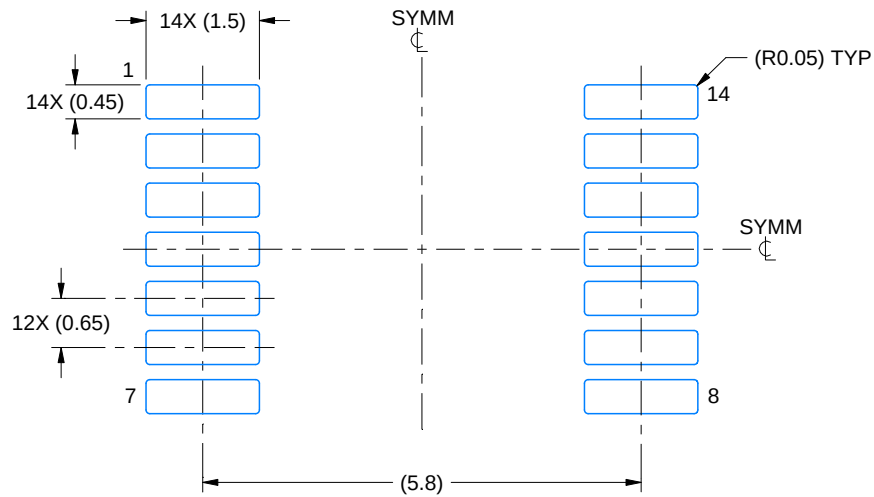


# EXAMPLE BOARD LAYOUT

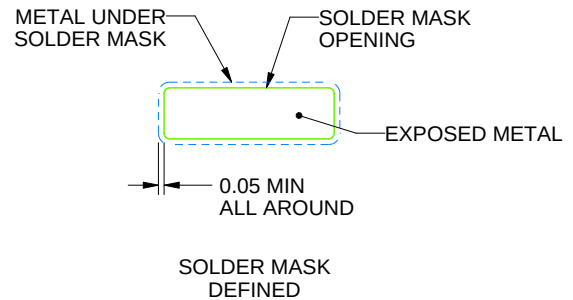
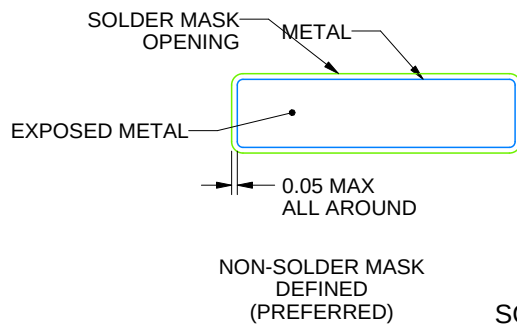
PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
EXPOSED METAL SHOWN  
SCALE: 10X



SOLDER MASK DETAILS

4220202/B 12/2023

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

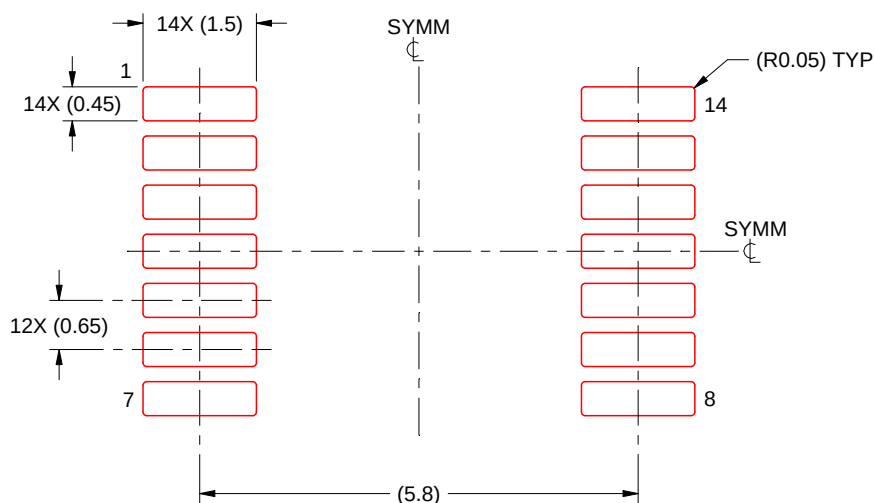
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

# EXAMPLE STENCIL DESIGN

PW0014A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE: 10X

4220202/B 12/2023

NOTES: (continued)

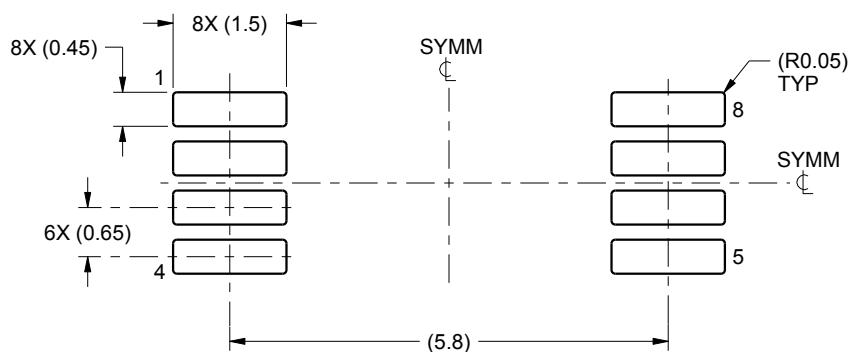
8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.



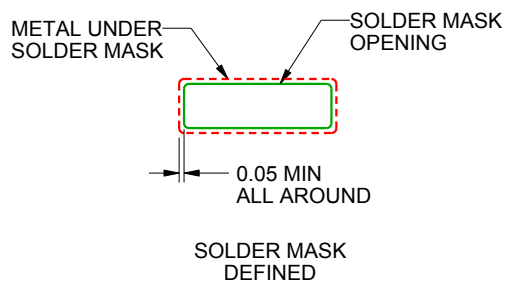
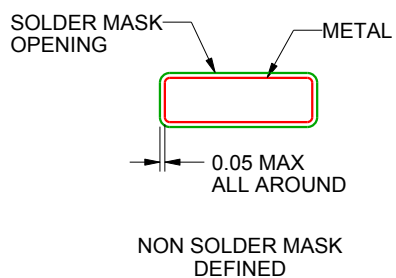
**PW0008A**

## TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



LAND PATTERN EXAMPLE  
SCALE:10X



SOLDER MASK DETAILS  
NOT TO SCALE

4221848/A 02/2015

NOTES: (continued)

6. Publication IPC-7351 may have alternate designs.

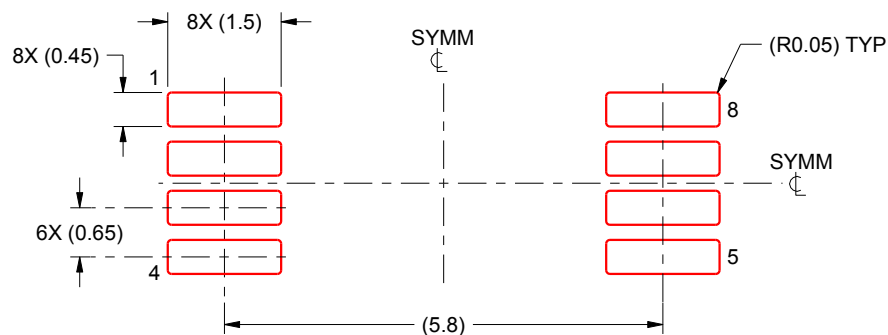
7. Solder mask tolerances between and around signal pads can vary based on board fabrication site.

## EXAMPLE STENCIL DESIGN

PW0008A

TSSOP - 1.2 mm max height

SMALL OUTLINE PACKAGE



SOLDER PASTE EXAMPLE  
BASED ON 0.125 mm THICK STENCIL  
SCALE:10X

4221848/A 02/2015

NOTES: (continued)

8. Laser cutting apertures with trapezoidal walls and rounded corners may offer better paste release. IPC-7525 may have alternate design recommendations.
9. Board assembly site may have different recommendations for stencil design.

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Last updated 10/2025